

MATH 2101 LINEAR ALGEBRA I, FALL SEMESTER 2025
CHAPTER 1: MATRIX ALGEBRA AND DETERMINANTS

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Goal of this section:

- (1) Basic terminology: matrices, entries, columns, rows, etc
- (2) Basic operations: additions, matrix multiplication, matrix addition, transpose
- (3) Properties of matrix multiplications
- (4) Determinants: definition, rules to compute
- (5) Inverse: computing inverse from a determinant, determining invertibility by determinants

1. MATRIX ALGEBRA

1.1. Basic terminology.

Definition 1.1. (1) We use $\mathbb{R}$ to represent the set of real numbers. A **scalar** is usually referred to a number in $\mathbb{R}$ (in this course).

- (2) An $n \times m$ **matrix** A is a rectangular array of the form:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}$$

where a_{ij} are in $\mathbb{R}$. We sometimes write (a_{ij}) to represent an arbitrary matrix. The **(i, j) -entry** of A will be referred to the number a_{ij} .

- (3) For a matrix A , we also write A_{ij} to be the (i, j) -th entry of A .
(4) The **i' -th row** of (a_{ij}) is the $1 \times m$ matrix

$$(a_{i'1} \quad a_{i'2} \quad \cdots \quad a_{i'm})$$

- (5) The **j' -th column** of (a_{ij}) is the $n \times 1$ matrix

$$\begin{pmatrix} a_{1j'} \\ a_{2j'} \\ \vdots \\ a_{nj'} \end{pmatrix}.$$

Example 1.2. Let $A = \begin{pmatrix} 2 & 4 & 1 \\ 0 & 7 & 9 \end{pmatrix}$. Then $A_{13} = 1$. The 2nd column of A is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ and the 1st row of A is $(2 \ 4 \ 1)$.

1.2. Operations of matrices.

(1) **Addition** of two $n \times m$ matrices $(a_{ij}) + (b_{ij})$ is defined as:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1m} + b_{1m} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2m} + b_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} + b_{n1} & a_{n2} + b_{n2} & \dots & a_{nm} + b_{nm} \end{pmatrix}$$

(2) **Scalar multiplication** on a matrix: for a scalar $c \in \mathbb{R}$ and a $n \times m$ matrix (a_{ij}) , $c(a_{ij})$ is defined as:

$$c \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} = \begin{pmatrix} ca_{11} & ca_{12} & \dots & ca_{1m} \\ ca_{21} & ca_{22} & \dots & ca_{2m} \\ \vdots & \vdots & & \vdots \\ ca_{n1} & ca_{n2} & \dots & ca_{nm} \end{pmatrix}$$

(3) **Matrix multiplication**: Let $A = (a_{ij})$ be a $n \times m$ matrix and let $B = (b_{ij})$ be a $m \times p$ matrix. We define the multiplication AB such that the (i, j) -th entry of AB is:

$$(AB)_{ij} = \sum_{x=1}^m a_{ix} b_{xj}$$

Remark 1.3. Let A be a $n \times m$ matrix and let B be a $p \times q$ matrix. If $m \neq p$, then AB is *not* well-defined.

(*Curious students may ask why to define matrix multiplication in such way. This is related to how to the concept of linear transformation, which we will come back later.*)

Example 1.4. (1) $\begin{pmatrix} 1 & 2 & 4 \\ -1 & 7 & 2 \end{pmatrix} + \begin{pmatrix} -1 & 3 & 2 \\ 1 & 3 & 8 \end{pmatrix} = \begin{pmatrix} 0 & 5 & 6 \\ 0 & 10 & 10 \end{pmatrix}$.

(2)

$$(-2) \cdot \begin{pmatrix} 2 & 3 & 0 \\ 1 & 0.5 & 8 \end{pmatrix} = \begin{pmatrix} -4 & -6 & 0 \\ -2 & -1 & 0 \end{pmatrix}$$

(3)

$$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 4 & 3 \\ 0 & 1 \\ 9 & 2 \end{pmatrix} = (1(4) + 2(0) + 3(9) \quad 1(3) + 2(1) + 3(2)) = (31 \quad 11)$$

(4)

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & -5 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & -16 \end{pmatrix}$$

is not well-defined.

1.3. Vectors.

Definition 1.5. (1) A **column vector** is a $n \times 1$ -matrix i.e. of the form

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

(2) A **row vector** is a $1 \times m$ -matrix i.e. of the form

$$(a_1, \dots, a_m).$$

(3) A **vector** simply refers to either a row vector or a column vector.

(4) $\mathbb{R}^n$ refers to the set of all column vectors with n real entries.

(5) We sometimes simply write 0 for the row vector or column vector with all entries to be zero. We shall call such vector to be the **zero vector**.

Exercise: Visualize geometrically the additions and scalar multiplication for vectors in $\mathbb{R}^2$.

1.4. Identity matrix. The **identity matrix** I_n is defined as:

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & \vdots \\ \vdots & \dots & \dots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix},$$

that is

$$(I_n)_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

The main properties of the identity matrix are the following:

Theorem 1.6. *Let A be an $n \times m$ matrix. Then*

(1) $I_n A = A$; and

(2) $A I_m = A$.

Proof. Exercise. □

Indeed, one should do the following more general exercise:

Exercise 1.7. An $n \times n$ matrix A is said to be **diagonal** if for any i, j with $i \neq j$, $A_{ij} = 0$. For example, $\begin{pmatrix} 1 & 0 \\ 0 & 7 \end{pmatrix}$ is diagonal but $\begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$ is not diagonal. The **diagonal entries** of a $n \times n$ matrix (a_{ij}) are referred to the entries $a_{11}, a_{22}, \dots, a_{nn}$.

Let A be a $n \times m$ matrix.

(1) For a diagonal $n \times n$ matrix D , describe the matrix multiplication DA ;

(2) For a diagonal $m \times m$ matrix E , describe the matrix multiplication AE .

1.5. Zero matrices. The $n \times m$ matrix is called the **zero matrix** $0_{n \times m}$ (or sometimes simply 0) if all its entries are zero. The main properties of the zero matrix is the following:

Theorem 1.8. *Let A be an $n \times m$ matrix. Then*

- (1) $0_{p \times n}A = 0_{p \times m}$; and
- (2) $A0_{m \times q} = 0_{n \times q}$; and
- (3) $A + 0 = A$.

Proof. Exercise. □

Remark 1.9. For simplicity, we usually write 0 for different things. One may need a bit thought to identify the meaning of 0 in a content. Can you explain the meaning of of 0 in the following equations? Let A be a $n \times p$ -matrix.

- (1) Then $0_{m \times n}A = 0$.
- (2) Let $c = 0$ be the zero scalar. Then $cA = 0$.
- (3) $A - A = 0$.

1.6. Square matrices.

Definition 1.10. A $n \times m$ matrix is called **square** if $n = m$.

Example 1.11. (1) Identity matrices are square. Diagonal matrices are also square.

(2) The row vector $(1 \ 3)$ is not square.

(3) The matrix $\begin{pmatrix} 1 & 2 \\ 2 & 0 \\ -1 & 1 \end{pmatrix}$ is not square.

1.7. Transpose of a matrix.

Definition 1.12. The **transpose** A^T of an $n \times m$ matrix A is the $m \times n$ matrix obtained from interchanging rows and columns i.e.

$$(A^T)_{ij} = A_{ji}$$

Example 1.13. (1) Let $A = \begin{pmatrix} 3 & 5 & 0 \end{pmatrix}$. Then

$$A^T = \begin{pmatrix} 3 \\ 5 \\ 0 \end{pmatrix}.$$

(2) Let $B = \begin{pmatrix} 4 & 1 \\ 0 & 3 \end{pmatrix}$. Then $B^T = \begin{pmatrix} 4 & 0 \\ 1 & 3 \end{pmatrix}$.

There are some basic properties of the transpose:

Theorem 1.14. (1) (*Transpose commutes with addition*) For $n \times m$ -matrices A and B ,

$$(A + B)^T = A^T + B^T.$$

(2) (*Transpose anti-commutes with matrix multiplication*) For an $n \times m$ -matrix A and a $m \times p$ -matrix B ,

$$(AB)^T = B^T A^T.$$

(3) (*Transpose commutes with scalar multiplication*) For an $n \times m$ -matrix A and a scalar c ,

$$(cA)^T = cA^T.$$

Example 1.15. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and let $B = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -11 \end{pmatrix},$$

and so

$$(AB)^T = \begin{pmatrix} 1 & -11 \end{pmatrix},$$

$$B^T A^T = \begin{pmatrix} -3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -11 \end{pmatrix}.$$

Theorem 1.16. (*Transpose twice*) Let A be an $n \times m$ matrix. Then $(A^T)^T = A$.

Proof.

$$((A^T)^T)_{ij} = (A^T)_{ji} = A_{ij}.$$

□

1.8. Symmetric matrices.

Definition 1.17. A matrix A is said to be *symmetric* if $A^T = A$.

Example 1.18. (1) $\begin{pmatrix} 1 & 3 \\ 3 & -2 \end{pmatrix}$ is symmetric.

(2) $\begin{pmatrix} 1 & 3 \\ -3 & -2 \end{pmatrix}$ is not symmetric.

(3) Any diagonal matrices are symmetric.

Exercise 1.19. Show that if a matrix is symmetric, then A is square.

Exercise 1.20. Write down the form of symmetric 2×2 matrices and the form of symmetric 3×3 matrices.

1.9. Upper triangular matrices.

- Definition 1.21.** (1) A square matrix (a_{ij}) is said to be an **upper triangular matrix** if for any $i > j$, $a_{ij} = 0$.
 (2) A square matrix (a_{ij}) is said to be a **lower triangular matrix** if for any $i < j$, $a_{ij} = 0$.

Example 1.22. (1) The matrices $\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 3 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & -2 \end{pmatrix}$ are upper triangular.

(2) The matrices $\begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 \end{pmatrix}$ are lower triangular matrices.

(3) The matrix $\begin{pmatrix} 1 & 2 \\ -1 & -2 \end{pmatrix}$ is neither upper triangular nor lower triangular.

Exercise 1.23. Show that a square matrix A is both upper triangular and lower triangular if and only if A is diagonal.

2. MORE ON MATRIX MULTIPLICATIONS

2.1. The slogan in matrix multiplication. Let $A = (a_{ij})$ be a $n \times m$ matrix and let $B = (b_{ij})$ be a $m \times p$ matrix. Then one may say that: *the (i, j) -th entry of AB is obtained by the i -th row of A times the j -th column of B .*

The meaning of the above slogan is that:

$$(AB)_{ij} = (a_{i1} \quad \dots \quad a_{im}) \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{mj} \end{pmatrix}.$$

2.2. Associativity. Here are some properties of the matrix multiplication:

Theorem 2.1. (*Associativity*) *The matrix multiplication is an associative operation, meaning that: Let A be a $n \times m$ matrix. Let B be a $m \times p$ matrix. Let C be a $p \times q$ matrix. Then*

$$A(BC) = (AB)C.$$

Proof. Write $A = (a_{ij})$, $B = (b_{jk})$ and $C = (c_{kl})$.

Compute $(AB)C$:

$$(AB)_{ij} = \sum_{x=1}^m a_{ix} b_{xj},$$

$$((AB)C)_{il} = \sum_{y=1}^p (AB)_{iy} C_{yl} = \sum_{y=1}^p \sum_{x=1}^m a_{ix} b_{xy} c_{yl}$$

Compute $A(BC)$:

$$(BC)_{kl} = \sum_{y=1}^p b_{ky} c_{yl},$$

$$A(BC)_{il} = \sum_{x=1}^m a_{ix} (BC)_{xl} = \sum_{x=1}^m \sum_{y=1}^p a_{ix} b_{xy} c_{yl}.$$

Thus, we have two expressions coincide. □

2.3. Distributive.

Theorem 2.2. (*Distributive property*) For a $n \times m$ matrix A and $m \times p$ matrices B, C , we have:

$$A(B + C) = AB + AC.$$

Exercise 2.3. Verify that $A(B + C) = AB + AC$ by using definitions.

2.4. Non-commutativity. For an $n \times m$ matrix A and a $m \times n$ matrix B , we *do not always* have $AB = BA$.

Example 2.4. Let $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 1+ab & a \\ b & 1 \end{pmatrix}$$

and

$$BA = \begin{pmatrix} 1 & a \\ b & 1+ab \end{pmatrix}.$$

So $AB \neq BA$ when $a \neq 0$ and $b \neq 0$.

Exercise 2.5. Let A and B be two $n \times n$ symmetric matrices. Show that $AB = BA$ if and only if AB is also symmetric..

2.5. Block multiplication. To motivate the term 'block multiplication', it is better to illustrate first with an example:

Example 2.6. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$. Let

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, C = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}.$$

Let

$$D = \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Suppose we have already known that

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad AC = \begin{pmatrix} 2 & -2 \\ -1 & -6 \end{pmatrix}.$$

Can you tell the answer the matrix multiplication:

$$AD$$

without direct computations?

Indeed, the answer is

$$AD = \begin{pmatrix} 1 & 2 & 2 & -2 \\ 3 & -1 & -1 & -6 \end{pmatrix}!$$

This is done by investigating how each entry is obtained in the matrix multiplication by using the above slogan!

The above example suggests that one can partition a matrix into blocks suitably in order to carry out the matrix multiplication.

Proposition 2.7. (*Block multiplication*) Let A be an $n \times m$ -matrix. Let $B_1, B_2, \dots, B_r$ be $m \times p_1, m \times p_2, \dots, m \times p_r$ matrices respectively. Let

$$C = (B_1 \ B_2 \ \dots \ B_r)$$

which is a $m \times (p_1 + \dots + p_r)$ -matrix. Then

$$AC = (AB_1 \ AB_2 \ \dots \ AB_r).$$

The following is a particular case that we shall use few times later:

Corollary 2.8. Let A be an $n \times m$ matrix. Let $v_1, \dots, v_k$ be column vectors in $\mathbb{R}^m$. Let

$$B = (v_1 \ v_2 \ \dots \ v_k),$$

which is a $m \times k$ matrix. Then

$$AB = (Av_1 \ Av_2 \ \dots \ Av_k).$$

We can also consider some other variations of block matrix multiplication:

Exercise 2.9. Let A_1, A_2 be 2×2 matrices, B_1, B_2 be 2×3 matrices, C_1, C_2 be 3×2 matrices and D_1, D_2 be 3×3 matrices. Write the multiplication of two block matrices

$$\begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \begin{pmatrix} A_2 & B_2 \\ C_2 & D_2 \end{pmatrix}$$

in the form of one matrix.

Exercise 2.10. Formulate a general version of block multiplication.

3. DETERMINANT OF A MATRIX

The determinant will be defined for square matrices.

3.1. Definition.

Definition 3.1. (2×2 matrix) The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$.

Example 3.2. $\det \begin{pmatrix} 3 & 4 \\ 2 & 9 \end{pmatrix} = 27 - 8 = 19$

Definition 3.3. ($n \times n$ matrix) Let $A = (a_{ij})$ be an $n \times n$ matrix.

- (1) Define the **(i, j) -minor**, denoted by $\tilde{A}_{ij}$, of A to be the matrix obtained by deleting the i -th row and j -th column.
- (2) Define the **(i, j) -cofactor** C_{ij} of A to be the scalar $(-1)^{i+j} \det \tilde{A}_{ij}$
- (3) Fix certain j^* . Define *recursively* the **determinant**, denoted by $\det A$, of A to be a scalar

$$(3.1) \quad \det(A) = \sum_{i=1}^n a_{ij^*} C_{ij^*} = \sum_{i=1}^n (-1)^{i+j^*} a_{ij^*} \det(\tilde{A}_{ij^*})$$

or fixing certain i^* , define $\det A$ as:

$$(3.2) \quad \det(A) = \sum_{j=1}^n a_{i^*j} C_{i^*j} = \sum_{j=1}^n (-1)^{i^*+j} a_{i^*j} \det(\tilde{A}_{i^*j}).$$

The above definition is *independent of a choice of a row or a column*. We shall call (3.1) (respectively (3.2)) to be the **cofactor expansion** of $\det(A)$ along j^* -th-column (respectively along i^* -th-row).

Example 3.4. Let $A = \begin{pmatrix} 3 & 4 & 1 \\ 2 & 9 & 1 \\ 0 & -1 & -5 \end{pmatrix}$. The minors for the second column is:

$$\tilde{A}_{12} = \begin{pmatrix} 2 & 1 \\ 0 & -5 \end{pmatrix}, \quad \tilde{A}_{22} = \begin{pmatrix} 3 & 1 \\ 0 & -5 \end{pmatrix}, \quad \tilde{A}_{32} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

Then

$$\begin{aligned} \det(A) &= (-1)^{1+2}(4)\det \begin{pmatrix} 2 & 1 \\ 0 & -5 \end{pmatrix} + (-1)^{2+2}(9)\det \begin{pmatrix} 3 & 1 \\ 0 & -5 \end{pmatrix} + (-1)^{2+3}(-1)\det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \\ &= (-1)(4)(-10) + (9)(-15) - 1(-1) \end{aligned}$$

Exercise 3.5. Find

$$\det \begin{pmatrix} 3 & 4 & 0 & 0 \\ 2 & 9 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & -1 & 0 & -5 \end{pmatrix}$$

by using the cofactor expansion.

Exercise 3.6. (a) Write down the determinant of a 3×3 -matrix:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}.$$

View the formula by a 'diagonal multiplication'.

(b) Do you think a similar 'diagonal multiplication' can hold for 4×4 -matrices?

(Hint: consider

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Exercise 3.7. Let A be a 3×3 matrix. Prove that there exists $t \in \mathbb{R}$ such that $\det(A - tI_3) = 0$. (Hint: Consider $\det(A - xI_3)$ as a polynomial in x and solve for x .)

3.2. Rules and properties for computing determinants.

Theorem 3.8. (*Property 1: switching rows*) Let A be an $n \times n$ matrix. If B is obtained from A by switching two rows, then

$$\det(A) = -\det(B).$$

Proof. (Sketch of the proof) First consider that B is obtained from A by switching the i -th and $(i + 1)$ -th rows.

Relation of the (i, j) -minor of A and the $(i + 1, j)$ -minor of B

$$(3.3) \quad \tilde{A}_{ij} = \tilde{B}_{i+1,j}$$

Compute $\det(B)$ by the cofactor expansion along the $(i + 1)$ -th row:

$$(3.4) \quad \det(B) = \sum_{j=1}^n (-1)^{(i+1)+j} b_{i+1,j} \det \tilde{B}_{i+1,j} = (-1) \sum_{j=1}^n (-1)^{i+j} b_{i+1,j} \det \tilde{B}_{i+1,j}.$$

Compute $\det(A)$ by the cofactor expansion along the i -th row:

$$(3.5) \quad \det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det \tilde{A}_{ij}.$$

Combining (3.3), (3.4) and (3.5), we have

$$\det(A) = -\det(B).$$

The general case follows from switching two consecutive rows multiple times. $\square$

Example 3.9. $\det \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = -\det \begin{pmatrix} 3 & -1 \\ 2 & 1 \end{pmatrix}.$

Corollary 3.10. (*Property 2: zero determinant for matrix with two identical rows*) Let A be an $n \times n$ matrix. If A has two identical rows, then $\det(A) = 0$.

Proof. By using the previous theorem, we have $\det(A) = -\det(A)$. Hence $2\det(A) = 0$ and so $\det(A) = 0$. $\square$

Example 3.11. $\det \begin{pmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 4 & 5 \end{pmatrix} = 0$.

Theorem 3.12. (*Property 3: Adding a scalar multiple on a row vector*) Let $v_1, \dots, v_n$ be row vectors with n -entries and let u be another row vector with n entries. Let $c \in \mathbb{R}$. Then

$$\det \begin{pmatrix} v_1 \\ \vdots \\ v_{r-1} \\ v_r + cu \\ v_{r+1} \\ \vdots \\ v_n \end{pmatrix} = \det \begin{pmatrix} v_1 \\ \vdots \\ v_{r-1} \\ v_r \\ v_{r+1} \\ \vdots \\ v_n \end{pmatrix} + c \det \begin{pmatrix} v_1 \\ \vdots \\ v_{r-1} \\ u \\ v_{r+1} \\ \vdots \\ v_n \end{pmatrix}$$

Proof. (Sketch) One uses the cofactor expansion along the r th row. $\square$

Example 3.13.

$$\begin{aligned} \det \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} &= \det \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} + 0 \\ &= \det \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} + \det \begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix} \\ &= \det \begin{pmatrix} 2 & 3 \\ 1+2 & 4+3 \end{pmatrix} \\ &= \det \begin{pmatrix} 2 & 3 \\ 3 & 7 \end{pmatrix} \end{aligned}$$

Exercise 3.14. Try to use Corollary 3.10 and Theorem 3.12 to show

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix} = 0.$$

Corollary 3.15. Let A be an $n \times n$ matrix.

- (1) If one row of A is multiplied by a scalar c to obtain A' , then $\det(A') = c \cdot \det(A)$.
- (2) If one row of A is scalar multiple of another row of A , then $\det(A) = 0$.

Theorem 3.16. (*Property 4: Taking transpose*) Let A be an $n \times n$ matrix. Then $\det(A^T) = \det(A)$.

Proof. Write $A = (a_{ij})$. By the cofactor expansion for $\det(A)$ along the first row,

$$(3.6) \quad \det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(\tilde{A}_{1j})$$

Write $A^T = (b_{ij})$. By the cofactor expansion for $\det(A^T)$ and we have:

$$(3.7) \quad \det(A^T) = \sum_{i=1}^n (-1)^{i+1} b_{i1} \det(\tilde{A}^T_{i1}).$$

By the definition of minors,

$$(\tilde{A}_{1x})^T = \tilde{A}^T_{x1}.$$

Now, by the mathematical induction on n , we have that

$$(3.8) \quad \det(\tilde{A}_{1x}) = \det((\tilde{A}_{1x})^T) = \det((\tilde{A}^T)_{x1}).$$

By the definition of transpose,

$$(3.9) \quad a_{1x} = b_{x1}$$

Combining (3.6), (3.7), (3.8) and (3.9), we have:

$$\det(A) = \det(A^T).$$

□

Recall that transpose takes a row to a column and so one may apply Theorem 3.16 to deduce the column version of Properties 1-3.

Theorem 3.17. (*Column version of properties 1-3*) Let A be an $n \times n$ matrix. Then

- (1) If two columns of A are switched to obtain B , then $\det(B) = -\det(A)$.
- (2) If two columns of A are identical, or more generally one column of A is a scalar multiple of another column, then $\det(A) = 0$.
- (3) Let $v_1, \dots, v_n$ be column vectors in $\mathbb{R}^n$ and let u be another column vector in $\mathbb{R}^n$. Let $c \in \mathbb{R}$. Then

$$\begin{aligned} & \det \begin{pmatrix} v_1 & \dots & v_{r-1} & v_r + cu & v_{r+1} & \dots & v_n \end{pmatrix} \\ = & \det \begin{pmatrix} v_1 & \dots & v_{r-1} & v_r & v_{r+1} & \dots & v_n \end{pmatrix} + c \cdot \det \begin{pmatrix} v_1 & \dots & v_{r-1} & u & v_{r+1} & \dots & v_n \end{pmatrix} \end{aligned}$$

Theorem 3.18. (*Property 5: Scalar multiplication*) For an $n \times n$ matrix A and a scalar c ,

$$\det(cA) = c^n \det(A).$$

Proof. We write

$$A = \begin{pmatrix} v_1 & v_2 & \dots & v_n \end{pmatrix},$$

for column vectors $v_1, \dots, v_n$ in $\mathbb{R}^n$.

Then,

$$cA = \begin{pmatrix} cv_1 & cv_2 & \dots & cv_n \end{pmatrix}.$$

Applying Theorem 3.17(3) n -times,

$$\begin{aligned} \det(cA) &= \det(\begin{pmatrix} cv_1 & cv_2 & \dots & cv_n \end{pmatrix}) \\ &= c \det(\begin{pmatrix} v_1 & v_2 & \dots & v_n \end{pmatrix}) \\ &= \vdots & = c^n \det(\begin{pmatrix} v_1 & v_2 & \dots & v_n \end{pmatrix}) \\ &= c^n \det(A) \end{aligned}$$

□

Theorem 3.19. (*Property 6: Multiplicative property*) Let A, B be two $n \times n$ -matrices. Then $\det(AB) = \det(A) \cdot \det(B)$.

We shall not discuss a proof here. But one may readily verifies the theorem for some examples.

Exercise 3.20. An $n \times n$ matrix A is said to be *orthogonal* if $AA^T = I_n$. Prove that if Q is orthogonal, then $\det(Q) = \pm 1$.

3.3. Simplifying computation of determinants by using rules. We can write a matrix A in the form of rows:

$$A = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix},$$

where $u_1, \dots, u_n$ are row vectors with n entries. Suppose B is obtained from a matrix A by subtracting the i -th row of A by the c -multiple of the j -th row of A .

Then

$$\begin{aligned}
 \det(B) &= \det \begin{pmatrix} u_1 \\ \vdots \\ u_i - c \cdot u_j \\ \vdots \\ u_n \end{pmatrix} \\
 &= \det \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} - c \cdot \det \begin{pmatrix} u_1 \\ \vdots \\ u_j \\ \vdots \\ u_j \\ \vdots \\ u_n \end{pmatrix} \quad \text{by Theorem 3.12} \\
 &= \det(A) - 0 \quad \text{by Corollary 3.10} \\
 &= \det(A)
 \end{aligned}$$

Such rule is useful in simplifying computing some determinants.

Example 3.21.

$$\begin{aligned}
 \det \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 3 & -1 \\ 3 & 1 & -2 & 1 \\ -1 & 1 & 1 & 2 \end{pmatrix} &= \det \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 1 & -5 & -2 \\ 0 & 1 & 2 & 3 \end{pmatrix} \\
 &= (1) \det \begin{pmatrix} 1 & 1 & -3 \\ 1 & -5 & -2 \\ 1 & 2 & 3 \end{pmatrix} \quad \text{by cofactor expansion along 1st column} \\
 &= \det \begin{pmatrix} 1 & 1 & -3 \\ 0 & -6 & 1 \\ 0 & 1 & 6 \end{pmatrix} \\
 &= (1) \det \begin{pmatrix} -6 & 1 \\ 1 & 6 \end{pmatrix} \quad \text{by cofactor expansion along 1st column} \\
 &= -36 - 1 \\
 &= -37
 \end{aligned}$$

4. INVERSE

4.1. Definition.

Definition 4.1. An $n \times n$ matrix A is said to be **invertible** (or non-singular) if there exists a matrix B such that $AB = BA = I_n$. We say B to be the **inverse** of A . We shall denote such matrix by A^{-1} .

Remark 4.2. The inverse of an invertible matrix A is unique if it exists, which means that there is *at most one* matrix B such that $AB = BA = I_n$.

4.2. Relation to determinants and a formula for the inverse.

Theorem 4.3. (*Determinant of an inverse*) Let A be an invertible $n \times n$ -matrix. Then $\det(A) \neq 0$. Moreover, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof.

$$\begin{aligned} AA^{-1} = I_n &\xrightarrow{\text{taking det}} \det(AA^{-1}) = \det(I_n) \xrightarrow{\det(I_n)=1} \det(AA^{-1}) = 1 \\ &\xrightarrow{\text{use } \det(AB) = \det(A)\det(B)} \det(A)\det(A^{-1}) = 1 \end{aligned}$$

Hence, $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$. □

Definition 4.4. Let A be an $n \times n$ matrix. Define the **adjugate** of A as:

$$\text{adj}(A) = \begin{pmatrix} (-1)^{1+1}\det(\tilde{A}_{11}) & \dots & (-1)^{1+n}\det(\tilde{A}_{1n}) \\ \vdots & & \vdots \\ (-1)^{n+1}\det(\tilde{A}_{n1}) & \dots & (-1)^{n+n}\det(\tilde{A}_{nn}) \end{pmatrix}^T$$

Theorem 4.5. (*Formula for the inverse*) Let A be an $n \times n$ matrix. If $\det(A) \neq 0$, then

$$\frac{1}{\det A} \cdot \text{adj}(A)$$

is the inverse of A .

Proof. We compute $A(\text{adj}(A))$.

Compute the (i, i) -entry on $A(\text{adj}(A))$

The i -th row of A is

$$(a_{i1} \quad a_{i2} \quad \dots \quad a_{in})$$

and the i -th column of $\text{adj}(A)$ is

$$\begin{pmatrix} (-1)^{i+1}\det\tilde{A}_{i1} \\ \vdots \\ (-1)^{i+n}\det\tilde{A}_{in} \end{pmatrix}.$$

According to Section 2.1, the **(i, i) -th entry** of $A(\text{adj}(A))$ is

$$(4.10) \quad \sum_{x=1}^n (-1)^{i+x} a_{ix} \det \tilde{A}_{ix}.$$

Note that the term (4.10) is the **cofactor expansion for $\det(A)$ along the i -th row**, and hence is equal to **$\det(A)$** . Thus, all the diagonal entries in $A(\text{adj}(A))$ are $\det(A)$.

Compute the (i, i') -entry of $A\text{adj}(A)$ with $i \neq i'$

Again, according to Section 2.1, the (i, i') entry of $A \operatorname{adj}(\tilde{A})$ is:

$$(4.11) \quad \sum_{x=1}^n (-1)^{i'+x} a_{ix} \det \tilde{A}_{i'x}$$

Let B be the matrix obtained by replacing the i' -th row of A with the i -th row of A . However, the term in (4.11) is simply the cofactor expansion for $\det(B)$ along the i' -th row of B , and so is $\det(B)$. Since B has two identical rows, $\det(B) = 0$ by Corollary 3.10. In other words, all the (i, i') -entry ($i \neq i'$) of $A \operatorname{adj}(A)$ is zero.

We combine the above two cases to have:

$$A(\operatorname{adj}(A)) = \det(A)I_n.$$

When $\det(A) \neq 0$, we have:

$$A \left(\frac{1}{\det(A)} \operatorname{adj}(A) \right) = I_n.$$

□

Exercise 4.6. Suppose the 5-th column of a 6×6 -matrix A is

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix}$$

Find

$$2\det(\tilde{A}_{13}) - 4\det(\tilde{A}_{23}) + 6\det(\tilde{A}_{33}) - 8\det(\tilde{A}_{43}) + 10\det(\tilde{A}_{53}) - 12\det(\tilde{A}_{63})$$

Example 4.7. Find the inverse of the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

First, $\det(A) = 4 - 6 = -2$. Next we have:

$$\operatorname{adj}(A) = \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}^T = \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}.$$

$$\text{Hence, } A^{-1} = \frac{1}{\det(A)} \cdot \operatorname{adj}(A) = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

4.3. Criteria for the invertibility of a matrix. Combining Theorems 4.3 and 4.5, one can check the invertibility of a matrix as follows:

Theorem 4.8. (*Invertibility by computing determinant*) Let A be an $n \times n$ matrix. Then A is invertible if and only if $\det(A) \neq 0$.

Example 4.9. Determine if the following matrices are invertible. If it is invertible, find the inverse.

- $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 5 & 7 \end{pmatrix}$
- $B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 5 & 7 \end{pmatrix}$
- $C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$.

Solution:

- Compute $\det(A) = \det(B) = 0$ and so A and B are not invertible.
- Compute $\det(C) = 1 \neq 0$ and so C is invertible. Then

$$\text{adj}(C) = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}^T, C^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & -3 \\ 0 & -1 & 2 \end{pmatrix}.$$

4.4. Inverse with other operations.

Theorem 4.10. Let A, B be invertible $n \times n$ matrices. Let $c \in \mathbb{R}$. Then

- (1) (*Inverse anti-commutes with multiplication*) $(AB)^{-1} = B^{-1}A^{-1}$;
- (2) (*Inverse commutes with scalar multiplication*) $(cA)^{-1} = c^{-1}A^{-1}$;
- (3) (*Inverse commutes with transpose*) $(A^T)^{-1} = (A^{-1})^T$;
- (4) (*Inverse of inverse is identity*) $(A^{-1})^{-1} = A$.

Proof. We only check for the left inverse.

- (1) $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_n B = B^{-1}B = I_n$
- (2) $(c^{-1}A^{-1})(cA) = c^{-1}cA^{-1}A = I_n$
- (3) $(A^{-1})^T A^T = (AA^{-1})^T = I_n$
- (4) $AA^{-1} = I_n$

□

MATH 2101 LINEAR ALGEBRA I, FALL SEMESTER 2025
CHAPTER 2: SYSTEM OF LINEAR EQUATIONS

K.Y. CHAN

Goal of this section:

- (1) Terminologies for system of linear equations, solving by inverses, Cramer's rule
- (2) Elementary row operations and matrices, row echelon forms and reduced ones
- (3) Jordan-Gaussian eliminations
- (4) Number of solutions for a system of linear equations
- (5) Use row operations to find an inverse

1. SYSTEM OF LINEAR EQUATIONS AND SOME BASIC METHODS

1.1. **Basic terminologies for system of linear equations.**

Definition 1.1. (1) A **system of n linear equations in m variables** is a collection of equations:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m &= b_2 \\&\vdots \\a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m &= b_n\end{aligned}$$

where all $a_{ij} \in \mathbb{R}$ and $b_i \in \mathbb{R}$, and x_j are unknowns (called *variables*).

(2) Let

$$(1.1) \quad \textcolor{brown}{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{n,m} \end{pmatrix}, \quad \textcolor{red}{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

Let

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

whose entries are unknowns. We can rewrite (1.1) as a matrix equation:

$$(1.2) \quad Ax = b$$

We also refer such matrix equation to be a system of n linear equations in m -variables. We shall call A to be the **coefficient matrix**, and call the $n \times (m + 1)$ -matrix $(A|b)$ to be the **augmented matrix**.

- (3) A **solution** to (1.2) is a column vector v in $\mathbb{R}^m$ such that $Av = b$. The solution set refers to the set $\{v \in \mathbb{R}^m : Av = b\}$.

1.2. Method of using matrix inversions. Let A be a $n \times m$ matrix. Suppose the following two conditions hold:

- (1) A is a square matrix i.e. $m = n$; and
- (2) A is invertible.

To solve the following system of linear equations:

$$(1.3) \quad Ax = b,$$

we can multiply A^{-1} on both sides to obtain:

$$x = A^{-1}b.$$

Thus to solve (1.3), one carries out the following two steps:

- (1) find A^{-1} ;
- (2) find $A^{-1}b$.

In particular, we have:

Theorem 1.2. (*Unique solution under invertibility*) Let A be an invertible $n \times n$ matrices and let $b \in \mathbb{R}^n$. Then the system of linear equations $Ax = b$ always has a **unique solution**.

Example 1.3. Solve the following system of linear equations by using A^{-1} :

$$x_1 + 2x_2 - x_3 = 5$$

$$x_1 + x_2 + x_3 = 2$$

$$2x_1 - 2x_2 + x_3 = 0$$

Let $A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1 & 1 \\ 2 & -2 & 1 \end{pmatrix}$. Then we have

$$\det(A) = 9, \quad \text{adj}(A) = \begin{pmatrix} 3 & 1 & -4 \\ 0 & 3 & 6 \\ 3 & -2 & -1 \end{pmatrix}^T = \begin{pmatrix} 3 & 0 & 3 \\ 1 & 3 & -2 \\ -4 & 6 & -1 \end{pmatrix}.$$

Hence, $A^{-1} = \frac{1}{9} \begin{pmatrix} 3 & 0 & 3 \\ 1 & 3 & -2 \\ -4 & 6 & -1 \end{pmatrix}$, and so the solution is given by:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A^{-1} \begin{pmatrix} 5 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{15}{9} \\ \frac{11}{9} \\ -\frac{8}{9} \end{pmatrix}.$$

Exercise 1.4. Given a system of linear equations:

$$2x_1 + cx_2 = 2$$

$$3x_1 + dx_2 = 3,$$

where c, d are some real numbers. Find all c, d such that the system has a unique solution.

1.3. Cramer's rule. Cramer's rule is a simpler formula that lighten some computations in matrix inversion:

Theorem 1.5. (*Cramer's rule*) Let A be an $n \times n$ invertible matrix and let b be a column vector in $\mathbb{R}^n$. The solution of the following system of linear equations

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = b$$

is given by:

$$x_i = \frac{\det(A_i)}{\det(A)},$$

where A_i is the matrix obtained by replacing the i -th column of A by b .

Proof. Recall from Chapter 1 that we have the following fomrula for the inverse:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Compute entries in $\text{adj}(A)b$:

By the **definition of matrix multiplication** in Chapter 1,

$$(\text{adj}(A)b)_{i1} = (-1)^{i+1}b_1\det(\tilde{A}_{1i}) + \dots + (-1)^{i+n}b_n\det(\tilde{A}_{ni}).$$

Use the **cofactor expansion** for $\det(A_i)$ along the i -th column
 $\det(A_i)$ is preciesely the term $(-1)^{i+1}b_1\det(\tilde{A}_{1i}) + \dots + (-1)^{i+n}b_n\det(\tilde{A}_{ni})$.

Thus, combining above equations,

$$x_i = (A^{-1}b)_{i1} = \left(\frac{1}{\det A} \text{adj}(A)b\right)_{i1} = \frac{\det(A_i)}{\det(A)}.$$

□

Example 1.6. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and let $b = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$. Solve $Ax = b$ by Cramer's rule.

Solution: $\det(A) = -3$. We also have:

$$\det \begin{pmatrix} 3 & 2 \\ 3 & 1 \end{pmatrix} = -3, \det \begin{pmatrix} 1 & 3 \\ 2 & 3 \end{pmatrix} = -3$$

Hence, by Cramer's rule,

$$x = \frac{1}{-3} \begin{pmatrix} -3 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

2. ROW OPERATIONS

2.1. Motivating Row operations.

Example 2.1. Solve

$$x_1 + x_2 = 1$$

$$x_2 = 2$$

The system is quite straightforward for solving since we can substitute $x_2 = 2$ to the first equation and so get $x_1 = -1$. In the form of matrix equation, the augmented matrix takes the form:

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 2 \end{array} \right).$$

In general, a system of linear equations may not take such simple form. Our goal is to first introduce some operations that help one systematically transform matrices into such simple form.

2.2. Elementary row operations on system of linear equations.

Definition 2.2. Let A be an $n \times m$ matrix. Any one of the following three operations on rows is called an **elementary row operations**:

- (I) **interchange any two rows of A** ;
- (II) **multiply any row of A by a *non-zero* scalar**;
- (III) **add a scalar multiple of a row of A to another row of A** .

Example 2.3. Let $A = \begin{pmatrix} 1 & 2 & -1 & 5 \\ 0 & 7 & 1 & 2 \\ 9 & 2 & 0 & 4 \end{pmatrix}$.

- (1) Interchange the first and third rows:

$$\begin{pmatrix} 1 & 2 & -1 & 5 \\ 0 & 7 & 1 & 2 \\ 9 & 2 & 0 & 4 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 9 & 2 & 0 & 4 \\ 0 & 7 & 1 & 2 \\ 1 & 2 & -1 & 5 \end{pmatrix}$$

- (2) Multiply the second row by the scalar $-\sqrt{2}$:

$$\begin{pmatrix} 1 & 2 & -1 & 5 \\ 0 & 7 & 1 & 2 \\ 9 & 2 & 0 & 4 \end{pmatrix} \xrightarrow{(-\sqrt{2})R_2 \rightarrow R_2} \begin{pmatrix} 1 & 2 & -1 & 5 \\ 0 & -7\sqrt{2} & -\sqrt{2} & -2\sqrt{2} \\ 9 & 2 & 0 & 4 \end{pmatrix}$$

- (3) Add the scalar multiple 2 of the first row to the third row:

$$\begin{pmatrix} 1 & 2 & -1 & 5 \\ 0 & 7 & 1 & 2 \\ 9 & 2 & 0 & 4 \end{pmatrix} \xrightarrow{2R_1 + R_3 \rightarrow R_3} \begin{pmatrix} 1 & 2 & -1 & 5 \\ 0 & 7 & 1 & 2 \\ 11 & 6 & -2 & 14 \end{pmatrix}$$

2.3. Elementary matrices.

Definition 2.4. An **elementary matrix** is a square matrix obtained by performing *precisely one* elementary operation on I_n .

Example 2.5. (1) $I_4 \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ is an elementary matrix by interchanging R_1 and R_3 .

(2) $I_2 \xrightarrow{2R_2 \rightarrow R_2} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ is an elementary matrix obtained by multiplying 2 on R_2 .

(3) $I_3 \xrightarrow{-2R_1 + R_3 \rightarrow R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$ is an elementary matrix obtained by adding the scalar multiple -2 of R_1 to R_3 .

We have the following general description of elementary matrices:

Example 2.6. (Non-examples) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ are not elementary matrices.

Theorem 2.7. (*Elementary row operations via multiplication of elementary matrices*) Let A be an $n \times m$ matrix.

- (1) Suppose E is an elementary matrix. Then EA is the matrix obtained from A by the corresponding row elementary operation.
- (2) Conversely, if B is a matrix obtained from A by a row elementary operation, then $B = EA$ for the elementary matrix E corresponding to that operation.

Example 2.8. Let $E = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, which corresponds to the row elementary operation for adding the scalar multiple 2 of the second row to the first row. Let $A = \begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix}$. Then

$$EA = \begin{pmatrix} u_1 + 2v_1 & u_2 + 2v_2 \\ v_1 & v_2 \end{pmatrix}$$

Exercise 2.9. Find the elementary matrices corresponding to the following row operations on a 3×3 -matrix:

- multiplication of the third row by -3 ;
- interchanging the first and second rows;
- adding 2-times of the second row to the third row.

Exercise 2.10. While matrices in Example 2.6 are not elementary, the multiplication between elementary matrices gives those matrices. Describe how those corresponding row operations work.

2.4. More properties of elementary matrices.

Theorem 2.11. (*Invertibility of elementary matrices*) Elementary matrices are invertible. Moreover, the inverse of an elementary matrix is still an elementary matrix of the same type.

Proof. By Theorem 2.7, it suffices to find an elementary row operation taking the matrix back to the original one. This will be left for an exercise. $\square$

Exercise 2.12. Determine if the followings *always* give an elementary matrix. If yes, give a proof. If no, give a counterexample.

- Addition of two elementary matrices
- Multiplication of two elementary matrices
- Transpose of an elementary matrix

2.5. Reduced Row Echelon form.

Definition 2.13. A matrix is said to be in a row echelon form if the following conditions are satisfied:

- (1) All zero rows (i.e. rows with all zero entries), if any, must be at the bottom.
- (2) The leading entry (meaning the leftmost non-zero entry) in a non-zero row is on the right of the leading entry of every row above.
- (3) The leading entry is equal to 1, called a leading one.

Example 2.14. Which of the following matrix is in a row echelon form?

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -1 & 3 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, I_n, 0_{n \times m}$$

Which entries are leading ones?

Definition 2.15. A matrix is said to be in a **reduced row echelon form** if it is in a row echelon form with the following further property:

- (4) each column with a leading entry has zero in all other entries.

Example 2.16. All 2×2 matrices in a reduced row echelon form:

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & a \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Example 2.17. The matrices $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ are in reduced row echelon forms.

Example 2.18. The following matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

are in row echelon forms, but not in reduced row echelon forms.

Exercise 2.19. Write down all reduced row echelon forms for 3×2 matrices.

Theorem 2.20. (*Invertibility of a square matrix in reduced row echelon form*) Let A be an $n \times n$ matrix in a reduced row echelon form. Then the following statements are equivalent:

- (1) A has n leading ones;
- (2) A is invertible;
- (3) $A = I_n$.

Proof. (3) $\Rightarrow$ (1) and (3) $\Rightarrow$ (2) are clear.

(1) $\Rightarrow$ (3). Since there are n rows, (1) implies that **each row must have a leading one**. Then the leading ones occupy all the diagonal entries. By **Definition 2.15(4)**, all non-diagonal entries are then zero. Thus A has to be I_n .

(2) $\Rightarrow$ (1). Since A is invertible, there is **no zero row in A** . In other words, each row has a leading one and so A has n leading ones.

(1) $\Rightarrow$ (2) follows from (1) $\Rightarrow$ (3) and (3) $\Rightarrow$ (2).

(2) $\Rightarrow$ (3) follows from (2) $\Rightarrow$ (1) and (1) $\Rightarrow$ (3). □

3. JORDAN-GAUSSIAN ELIMINATIONS

3.1. Row equivalence.

Definition 3.1. For two $n \times m$ matrices A and B , we say A is **row equivalent** to B if there exists a sequence of elementary matrices $E_1, \dots, E_k$ such that $E_k \dots E_1 A = B$.

By Theorem 2.7, A is row equivalent to B if and only if B is obtained from A by a sequence of elementary row operations in Section 2.2.

Exercise 3.2. (1) Is the row equivalence a *symmetric* relation?

(2) Is the row equivalence a *transitive* relation?

Theorem 3.3. (*Solution set under row equivalence*) Given two systems of linear equations $Ax = b$ and $A'x = b'$, if the augmented matrices $(A|b)$ and $(A'|b')$ are row equivalent, then the two systems have the same solution set.

Proof. By definition, there exists elementary matrices $E_1, \dots, E_r$ such that

$$E_r \dots E_1 (A|b) = (A'|b')$$

By the block multiplication in the previous chapter,

$$(E_r \dots E_1 A | E_r \dots E_1 b) = (A' | b').$$

Hence, $A' = E_r \dots E_1 A$ and $b' = E_r \dots E_1 b$.

Prove $Av = b \Rightarrow A'v = b'$

Let the size of A is $n \times m$. If $v \in \mathbb{R}^m$ satisfies $Av = b$, then

$$E_r \dots E_1 Av = E_r \dots E_1 b$$

and so $A'v = b'$.

Prove $A'v = b' \Rightarrow Av = b$

If $v \in \mathbb{R}^m$ satisfies $A'v = b'$, then

$$E_r \dots E_1 Av = E_r \dots E_1 b.$$

Since $E_1, \dots, E_r$ are invertible by Theorem 2.11, $Av = b$.

□

3.2. Jordan-Gaussian eliminations. Recall from Section 2.1 that we would like to transform a matrix to a simple form for solving equations. More precisely, we want a systematic way to find a matrix in a *reduced row echelon form* that is *row equivalent* to a given matrix. Such procedure is called the **Jordan-Gaussian eliminations**.

Example 3.4. Find a matrix in the reduced row echelon form which is row equivalent to the following augmented matrix:

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 2 \\ 2 & 2 & 4 & 6 \\ 1 & 0 & 1 & 1 \end{array}\right)$$

The Jordan-Gaussian elimination is to carry out the following steps:

- (1) Find the leftmost non-zero column. Interchange the **first row** with a row containing a **non-zero entry** in that column e.g.

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 2 \\ 2 & 2 & 4 & 6 \\ 1 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 1 & 1 \end{array}\right)$$

- (2) Change **such non-zero entry** to 1 by a scalar multiplication on the (new) first row e.g.

$$\left(\begin{array}{ccc|c} 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 1 & 1 \end{array}\right) \xrightarrow{\frac{1}{2}R_1 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 1 & 1 \end{array}\right)$$

- (3) Add a scalar multiple of the first row to other rows so that other entries below the leading one in that column are zero.

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 1 & 1 \end{array}\right) \xrightarrow{-R_1 + R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & -1 & -2 \end{array}\right)$$

- (4) Repeat a similar process of Steps (1)-(3) for the second row, third row, and so on

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & -1 & -2 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -1 & -1 & -2 \\ 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{-R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

Steps (1) to (4) give a systematic way to transform A to a row echelon form

- (5) Add a scalar multiple of the last non-zero row with a non-zero entry to rows above that so that the column with the **leading one** of that row has only one non-zero entry.

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{-R_3 + R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{-2R_3 + R_1 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

- (6) Repeat the process for **preceding rows**.

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{-R_2 + R_1 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

Steps (5) and (6) are to transform the matrix to the reduced row echelon form.

Definition 3.5. A matrix B is said to be the **reduced row echelon form (RREF)** of a matrix A if B is obtained by the Jordan-Gaussian elimination on A . Alternatively, B is the RREF of A if B is in reduced row echelon form and is row equivalent to A .

Exercise 3.6. Let A be an $n \times n$ matrix. Then A is invertible if and only if the reduced row echelon form of A is the identity matrix.

Definition 3.7. Let A be an $n \times m$ -matrix and let $b \in \mathbb{R}^n$. Given a system of linear equations $Ax = b$, we obtain an augmented matrix $(A|b)$. The reduced row echelon form of $(A|b)$ takes the form

$$(A'|b').$$

A variable corresponding to a leading entry of $(A'|b')$ is called a **leading variable**. A variable which does not correspond to a leading entry of $(A'|b')$ is called a **free variable**.

Suppose the system has a solution (see more discussions on Theorem 4.4 and Section 4.1). To find the solution for $Ax = b$, we solve for $A'x = b'$ by Theorem 3.3. Now we set the free variables (from $(A'|b')$) to be any arbitrary values, and then solve for the leading variables in terms of those free variables.

Example 3.8. Solve the system of linear equations:

$$\begin{aligned} x_3 &= 2 \\ 2x_1 + 2x_2 + 4x_3 &= 6 \\ x_1 + x_3 &= 1 \end{aligned}$$

by using the reduced row echelon form in Example 3.4.

Sol: The augmented matrix from the system is:

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 2 \\ 2 & 2 & 4 & 6 \\ 1 & 0 & 1 & 1 \end{array} \right).$$

By Example 3.4 (and Theorem 3.3), we solve the system with the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array} \right).$$

Thus, $x_1 = -1, x_2 = 0, x_3 = 2$, and the solution set is $\{(-1, 0, 2)\}$.

Example 3.9. Solve the following system of linear equations:

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 &= -1 \\ 2x_1 + 4x_2 - x_3 + 8x_4 &= -3 \\ x_2 + x_4 &= 2 \end{aligned}$$

Then the augmented matrix takes the form:

$$\begin{pmatrix} 1 & 2 & -1 & 4 & | & -1 \\ 2 & 4 & -1 & 8 & | & -3 \\ 0 & 1 & 0 & 1 & | & 2 \end{pmatrix} \xrightarrow{-2R_1+R_2 \rightarrow R_2} \begin{pmatrix} 1 & 2 & -1 & 4 & | & -1 \\ 0 & 0 & 1 & 0 & | & -1 \\ 0 & 1 & 0 & 1 & | & 2 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & -1 & 4 & | & -1 \\ 0 & 1 & 0 & 1 & | & 2 \\ 0 & 0 & 1 & 0 & | & -1 \end{pmatrix}$$

$$\xrightarrow{-R_3+R_1 \rightarrow R_1} \begin{pmatrix} 1 & 2 & 0 & 4 & | & -2 \\ 0 & 1 & 0 & 1 & | & 2 \\ 0 & 0 & 1 & 0 & | & -1 \end{pmatrix} \xrightarrow{-2R_2+R_1 \rightarrow R_1} \begin{pmatrix} 1 & 0 & 0 & 2 & | & -6 \\ 0 & 1 & 0 & 1 & | & 2 \\ 0 & 0 & 1 & 0 & | & -1 \end{pmatrix}$$

Hence, x_4 is a free variable and we set $x_4 = t$. Then, the solution set is

$$(3.4) \quad \{(-2t - 6, -t + 2, -1)^T \in \mathbb{R}^4 : t \in \mathbb{R}\}.$$

We shall sometimes call (3.4) to be the **general solution set** of the system.

Example 3.10. Let $Ax = b$ be a system of linear equations. Suppose the reduced row echelon form of $(A|b)$ takes the form:

$$\begin{pmatrix} 0 & 1 & 2 & 0 & 3 & | & 2 \\ 0 & 0 & 0 & 1 & 2 & | & 3 \end{pmatrix}.$$

Then, x_2, x_4 are leading variables and x_1, x_3, x_5 are free variables. Set $x_1 = s, x_3 = t, x_5 = u$. We have $x_2 = 2 - 2t - 3u$ and $x_4 = 3 - 2u$.

Exercise 3.11. Let A and A' be two matrices in reduced row echelon forms. If A and A' are row equivalent, then $A = A'$. This exercise implies that the reduced row echelon form of a matrix is unique.

4. NUMBER OF SOLUTIONS FOR A SYSTEM OF LINEAR EQUATIONS

4.1. Consistency and inconsistency of a system.

Definition 4.1. A system of linear equations is said to be **consistent** if the solution set to that system is non-empty. Otherwise, we say that the system of linear equations is **inconsistent**.

Example 4.2. The equation $0x_1 + 0x_2 = 1$ has no solution! This is the first example of inconsistent system of linear equations.

Example 4.3. The system of linear equations:

$$2x_1 + x_2 = 1$$

$$4x_1 + 2x_2 = 3$$

has no solution. Hence, the system is also inconsistent.

Theorem 4.4. (*Criteria of consistency in terms of the reduced row echelon form*)
A system of linear equations $Ax = b$ is inconsistent if and only if the reduced row echelon form of the augmented matrix $(A|b)$ has a row which has only one non-zero entry in the last column (i.e. a leading one is in the last column).

Proof. (Main idea of proof) Let $(A'|b')$ be the reduced row echelon form of $(A|b)$. By Theorem 3.3, it suffices to solve the system $A'x = b'$. If $(A'|b')$ has a row with a leading one in the last column, the system has an equation of the form

$$(4.5) \quad 0x_1 + 0x_2 + \dots + 0x_m = 1.$$

There is **no solution for the equation (4.5)** and so the system $A'x = b'$ is inconsistent. $\square$

Example 4.5. If the reduced row echelon form of $(A|b)$ takes the form

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right),$$

then the system $Ax = b$ is inconsistent.

Example 4.6. Solve the following system of linear equations:

$$\begin{aligned} x_1 - 2x_2 - x_3 &= -2 \\ 2x_1 - 3x_2 + x_3 &= 0 \\ 3x_1 - 4x_2 - 2x_3 &= -6 \\ x_1 + 5x_3 &= 4 \end{aligned}$$

Solution: This gives that

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 2 & -3 & 1 & 0 \\ 3 & -4 & -2 & -6 \\ 1 & 0 & 5 & 4 \end{array} \right) \xrightarrow{-2R_1+R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 3 & -4 & -2 & -6 \\ 1 & 0 & 5 & 4 \end{array} \right) \\ & \xrightarrow{-3R_1+R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 5 & 4 \end{array} \right) \xrightarrow{-R_1+R_4 \rightarrow R_4} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & 6 & 6 \end{array} \right) \\ & \xrightarrow{-2R_2+R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -5 & -5 \\ 0 & 2 & 6 & 6 \end{array} \right) \xrightarrow{-2R_2+R_4 \rightarrow R_4} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & -2 \end{array} \right) \end{aligned}$$

From this, we know the reduced row echelon form has a row with only zeros in the last column. Hence, the system does not have a solution.

If we adjust the previous equation slightly, we can obtain a system of linear equations with a solution:

Example 4.7. Solve the following system of linear equations:

$$x_1 - 2x_2 - x_3 = -2$$

$$2x_1 - 3x_2 + x_3 = 0$$

$$3x_1 - 4x_2 - 2x_3 = -3$$

$$x_1 + 5x_3 = 6$$

We first obtain an augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 2 & -3 & 1 & 0 \\ 3 & -4 & -2 & -3 \\ 1 & 0 & 5 & 6 \end{array} \right)$$

This gives that

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 2 & -3 & 1 & 0 \\ 3 & -4 & -2 & -3 \\ 1 & 0 & 5 & 6 \end{array} \right) \xrightarrow{-2R_1+R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 3 & -4 & -2 & -3 \\ 1 & 0 & 5 & 6 \end{array} \right) \xrightarrow{-3R_1+R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 5 & 6 \end{array} \right) \\ & \xrightarrow{-R_1+R_4 \rightarrow R_4} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & 6 & 8 \end{array} \right) \xrightarrow{-2R_2+R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -5 & -5 \\ 0 & 2 & 6 & 8 \end{array} \right) \\ & \xrightarrow{-2R_2+R_4 \rightarrow R_4} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{\frac{1}{-5}R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{-3R_3+R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & -2 & -1 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \\ & \xrightarrow{R_3+R_1 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & -2 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{2R_2+R_1 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

From this, we can read off the solution: $x_1 = x_2 = x_3 = 1$.

Exercise 4.8. In Theorem 1.2, we showed that $Ax = b$ has a unique solution if A is a square invertible matrix. Explain this fact using the reduced row echelon form.

4.2. Infinitely many solutions.

Exercise 4.9. Show that if a system has more than one solution, then the system must have infinitely many solutions.

The above exercise shows that a system of linear equations can only be in one of the following three possibilities:

- (1) has no solution (is inconsistent);
- (2) has a unique solution;

(3) has infinitely many solutions.

As a consequence of the above criteria, we have:

Theorem 4.10. (*Criteria for infinitely many solutions*) A system of linear equations $Ax = b$ has infinitely many solutions if and only if the system is consistent and has a free variable.

Definition 4.11. Let A be an $n \times m$ matrix. A system of linear equations $Ax = b$ is called **homogeneous** if $b = 0_{n \times 1}$. A homogeneous system of linear equations is always consistent since

$$x = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

is always a solution. The zero vector is called the **trivial solution**.

Exercise 4.12. Use Theorem 4.4 to show that a homogeneous system of linear equations is always consistent.

Theorem 4.13. (*homogeneous system with more variables than equations*) Let $Ax = 0$ be a homogeneous system of linear equations of n equations and m variables. If $m > n$, then there is an infinite number of solutions.

Proof. The augmented matrix of the system is $(A \mid 0_{n \times 1})$, and its reduced row echelon form is of the form $(A' \mid 0)$ for some $n \times m$ matrix A' . Then the system has a free variable, and so has infinitely many solutions by Theorem 4.10. $\square$

Example 4.14. The equation $x_1 + 3x_2 - 2x_3 = 0$ has the solution set:

$$\{(-3s + 2t, s, t)^T : s, t \in \mathbb{R}\}.$$

In general, we have the following:

Theorem 4.15. (*Infinite number of solutions for consistent system with more variables than equations*) Let $Ax = b$ be a system of n linear equations and m variables. Suppose $m > n$ and the system is consistent. Then,

- (1) There is an infinite number of solutions for the system.
- (2) Moreover, let x_p satisfy $Ax_p = b$. The solution set for the system $Ax = b$ is

$$\{x_p + y \in \mathbb{R}^m : Ay = 0\}.$$

Proof. Reduction to (2): Note that (1) will follow from (2) and Theorem 4.13 and so it suffices to prove (2).

We now prove(2). To this end, let

$$\mathcal{S} = \{x \in \mathbb{R}^m : Ax = b\}.$$

Prove $\mathcal{S} \supset \{x_p + y : Ay = 0\}$

Let $x' \in \{x_p + y : Ay = 0\}$ and so $x' = x_p + y$ for some $Ay = 0$. Then we have that $Ax' = A(x_p + y) = Ax_p + Ay = b + 0 = b$. So, $x' \in \mathcal{S}$.

Prove $\mathcal{S} \subset \{x_p + y : Ay = 0\}$

Let $x' \in \mathcal{S}$ and so $Ax' = b$. Then $A(x' - x_p) = b - b = 0$. Hence,

$$x' = x_p + (x' - x_p) \in \{x_p + y \in \mathbb{R}^m : Ay = 0\}.$$

This shows that $x' \in \{x_p + y : Ay = 0\}$. □

Example 4.16. Let A be an $n \times 4$ matrix. Suppose the solution set of the system of linear equations $Ax = 0$ is given by $\{(2t, s, 3t + 2s, t - s)^T : t, s \in \mathbb{R}\}$. If $(0, 1, 0, 0)^T$ is a solution to $Ax = b$, find the solution set of $Ax = b$.

Solution: The solution set is

$$\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 2t \\ s \\ 3t + 2s \\ t - s \end{pmatrix} : t, s \in \mathbb{R} \right\}.$$

4.3. More on invertibility of a matrix on solving a system of linear equations.

Theorem 4.17. (*Homogeneous system with number of variables equal to number of equations*) Let A be an $n \times n$ matrix. Then the following statements are equivalent:

- (1) the system $Ax = 0$ has a non-trivial solution;
- (2) the system $Ax = 0$ has infinitely many solutions;
- (3) A is not invertible.

Proof. (1) $\Rightarrow$ (3)

Suppose A is invertible to arrive a contradiction. Then, by Theorem 1.2, the system has unique solution and so only has the trivial solution. This contradicts (1).

(2) $\Rightarrow$ (1)

This is clear.

(3) $\Rightarrow$ (1)

If A is not invertible, then the reduced row echelon form of A is also not invertible (work out Exercise 3.6). Thus the reduced row echelon form of $(A|0)$ has less leading entries than the number of columns, by Theorem 2.20. In other words, the system

has free variables. Hence, the solution has a non-trivial solution.

(1) $\Rightarrow$ (2)

This is Theorem 4.10.

(2) $\Rightarrow$ (3) This follows from $(2) \Rightarrow (1) \Rightarrow (3)$.

(3) $\Rightarrow$ (2) This follows from $(3) \Rightarrow (1) \Rightarrow (2)$.

□

Example 4.18. Determine if the following system of linear equations has infinitely many solutions:

$$3x_1 - 4x_2 + 5x_3 = 0$$

$$2x_1 - x_2 + 4x_3 = 0$$

$$-x_1 - x_2 + 2x_3 = 0$$

Exercise 4.19. (Homogenous system with number of variables less than number of equations) Let $Ax = 0$ be a system of linear equations with m variables and n equations. Suppose $m < n$. Determine if the following situation could happen:

- (1) the system has a unique solution
- (2) the system has infinitely many solutions
- (3) the system has no solution.

If such situation could happen, give an example. If such situation could not happen, explain why.

5. INVERSES VIA ELEMENTARY ROW OPERATIONS

5.1. Inverse and row operations: theoretical aspects.

Theorem 5.1. (Row echelon form of an invertible matrix) Let A be an $n \times n$ matrix. Then A is invertible if and only if the reduced row echelon form of A is I_n .

Proof. If direction Suppose A is invertible. Then the reduced row echelon form of A is also invertible by Exercise 3.6. By Theorem 2.20, the reduced row echelon form is I_n .

Only if direction Suppose the reduced row echelon form of A is I_n and so is invertible. Then, by Exercise 3.6 again, A is invertible.

□

MATH 2101 LINEAR ALGEBRA I, FALL SEMESTER 2024
CHAPTER 3: VECTOR SPACES (PART 1)

K.Y. CHAN

Goals:

- (1) Definition of vector spaces, examples, vector subspaces
- (2) Construct vector subspaces from spanning sets
- (3) Linear independence
- (4) Basis, dimensions, construction of a basis

1. VECTOR SPACES

1.1. Abstract Vector spaces. A vector space is a main object in the linear algebra which abstractizes properties in $\mathbb{R}^n$.

Definition 1.1. A **vector space** V over $\mathbb{R}$ consists of a set with two operations, called **additions** $+: V \times V \rightarrow V$ and **scalar multiplications** $\mathbb{R} \times V \rightarrow V$ as follows

- (1) (addition) for $x, y \in V$, a unique element $x + y$ is defined in V :
- (2) (scalar multiplication) for $x \in V$ and $a \in \mathbb{R}$, a unique element ax is defined in V .

such that

- (1) (**Commutativity of addition**) For any $x, y \in V$, $x + y = y + x$
- (2) (**Associativity of addition**) For any $x, y, z \in V$, $(x + y) + z = x + (y + z)$
- (3) (**Existence of zero element in V**) There exists an element in V , denoted by **0**, such that $x + 0 = 0 + x = x$.
- (4) (**Existence of additive inverse**) For each $x \in V$, there exists an element $y \in V$ such that $x + y = 0$.
- (5) (**Multiplication by the scalar 1**) For any $x \in V$, $1x = x$.
- (6) (**Associativity of scalar multiplications**) For any $a, b \in \mathbb{R}$ and $x \in V$, $a(bx) = (ab)x$.
- (7) (**Distributive property of scalar multiplication**) For any $a, b \in \mathbb{R}$ and $x \in V$, $(a + b)x = ax + bx$.
- (8) (**Second distributive property of scalar multiplication**) For any $a \in \mathbb{R}$ and any $x, y \in V$, $a(x + y) = ax + ay$.

The zero element 0 is sometimes also called the **zero vector**.

Exercise 1.2. Let V be a vector space. Show that $0v$ is always the zero vector 0. (Hint: consider $0v + 0v$) Also show that $(-1)v$ is the additive inverse of v . (Later on, we simply write $-v$ for the additive inverse of v .)

Example 1.3. (Column vectors) The set $\mathbb{R}^n$ of column vectors with the addition and the scalar multiplication defined as the usual ones: for $(x_1, \dots, x_n)^T, (y_1, \dots, y_n)^T \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

$$(x_1, \dots, x_n)^T + (y_1, \dots, y_n)^T = (x_1 + y_1, \dots, x_n + y_n)^T,$$

$$c(x_1, \dots, x_n)^T = (cx_1, \dots, cx_n)^T.$$

The above example is the typical example in the course which we shall use a lot in the remaining of the course. Later on, we shall also regard $\mathbb{R}^n$ as a vector space (not just a set) with the usual addition and multiplication.

Example 1.4. ($n \times m$ Matrices) The set of $n \times m$ matrices with the matrix addition and the scalar multiplication. We shall denote the vector space by **Mat** $_{n \times m}$. What is the zero vector in that vector space?

Example 1.5. (Functions) Let V be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Define the two operations as follows:

- (1) (Addition) for $f_1, f_2 \in V$, $f_1 + f_2$ is a function from $\mathbb{R}$ to $\mathbb{R}$ given by $(f_1 + f_2)(x) = f_1(x) + f_2(x)$ for any $x \in \mathbb{R}$;
- (2) (Scalar multiplication) for $f \in V$ and a scalar c , $(cf)(x) = c(f(x))$ for any $x \in \mathbb{R}$.

What is the zero vector in V ?

Exercise 1.6. Let

$$f_1(x) = \begin{cases} 1 & \text{for } x \geq 1 \\ 0 & \text{for } x < 1 \end{cases}$$

and

$$f_2(x) = \begin{cases} 0 & \text{for } x \geq 2 \\ 2 & \text{for } x < 2 \end{cases}$$

Describe the function $f_1 + f_2$ and $2f_1$.

1.2. Vector subspaces.

Definition 1.7. A non-empty subset W of a vector space V is called a **vector subspace** of V if it is closed under addition and scalar multiplication i.e. for $v_1, v_2 \in W$,

- (1) (**Closed under addition**) $v_1 + v_2 \in W$;

- (2) (**Closed under Scalar multiplication**) for any scalar c and any $v \in W$, $cv \in W$.

Theorem 1.8. (*Subspaces as vector spaces*) Let W be a vector subspace of a vector space V . Then W is also a vector space, with the addition and scalar multiplication from V .

Proof. Note that Definition 1.7(1) and (2) guarantees that the addition and the scalar multiplication are **well-defined operations** on W . Checking axioms in Definition 1.7 is straightforward. $\square$

Since a vector subspace is still a vector space, we shall sometimes also talk about properties such as bases and dimensions for vector subspaces later.

Example 1.9. A vector space V is always a vector subspace of V itself!

Example 1.10. (**Zero vector space**) Let W be the subset of V containing only the zero vector i.e. $W = \{0\}$. Then W is a vector subspace of V . We call W to be the **zero subspace** of V .

Exercise 1.11. Let W be a vector subspace of $\mathbb{R}^n$. Show that $0 \in W$.

Example 1.12. The set $W = \{(x, y, z)^T \in \mathbb{R}^3 : x + 2y - 3z = 0\}$ is a vector subspace of $\mathbb{R}^3$.

Sol: Let $v_1 = (x_1, y_1, z_1)^T, v_2 = (x_2, y_2, z_2)^T \in \mathbb{R}^3$. Let c be a scalar.

- (1) (Check addition) Note that $v_1 + v_2 = (x_1 + x_2, y_1 + y_2, z_1 + z_2)^T$. Then $(x_1 + x_2) + 2(y_1 + y_2) - 3(z_1 + z_2) = (x_1 + 2y_1 - 3z_1) + (x_2 + 2y_2 - 3z_2) = 0 + 0 = 0$. Hence, $v_1 + v_2 \in W$.
- (2) (Check scalar multiplication) Note that $cv_1 = (cx_1, cy_1, cz_1)^T$. Then $(cx_1) + 2(cy_1) - 3(cz_1) = c(x_1 + 2y_1 - 3z_1) = 0$. Hence, $cv_1 \in W$.

Exercise 1.13. Let v be a non-zero vector in $\mathbb{R}^n$.

- (1) Show that $\{cv : c \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^n$.
- (2) Visualize the above subspace geometrically.

Exercise 1.14. Verify that the set of all diagonal $n \times n$ matrices is a vector subspace of the space of all $n \times n$ matrices.

Example 1.15. Let $\mathcal{P}$ be the set of polynomials in the vector space V of functions from $\mathbb{R}$ to $\mathbb{R}$ (See Example 1.5). Then $\mathcal{P}$ is a vector subspace of V .

Example 1.16. (**Examples of non-vector subspaces**)

- (1) The set $\{(1, 2)^T, (3, 4)^T\}$ with only two non-zero points is not a vector subspace of $\mathbb{R}^2$, since the multiple 2 of $(1, 2)^T$ is not in the set!

- (2) Exercise: Let W be a *finite* subset of a vector space V consisting of at least one *non-zero* vector. Show that W is not a vector subspace of V

Example 1.17. Determine if the following subsets are subspaces:

- (1) $S_1 = \{(1, 2)^T, (2, 3)^T\}$ in $\mathbb{R}^2$;
- (2) $S_2 = \{(2t, 3t, 4t)^T \in \mathbb{R}^3 : t \in \mathbb{R}\}$ in $\mathbb{R}^3$;
- (3) $S_3 = \mathbb{R}^3 \setminus \{(2t, 3t, 4t)^T : t \in \mathbb{R}\}$ in $\mathbb{R}^3$;
- (4) $S_4 = \{(1 + 2t, 2t, 4t)^T \in \mathbb{R}^3 : t \in \mathbb{R}\} \cup \{(0, 0, 0)^T\}$ in $\mathbb{R}^3$;
- (5) $S_5 = \{(0, 0, 0, 0)^T\}$ in $\mathbb{R}^4$;
- (6) $S_6 = \{(x_1, x_2, x_3)^T \in \mathbb{R}^3 : x_1^3 + x_2^3 + x_3^3 = 0\}$ in $\mathbb{R}^3$.

2. SPANNING SETS

2.1. Some motivations.

Example 2.1. If we start with a *fixed* vector $v \in \mathbb{R}^n$, we can form a vector subspace of $\mathbb{R}^n$ by considering the set:

$$\{cv : c \in \mathbb{R}\}.$$

For example, let $v = (3, 4, 5)^T$ and then $\{(3c, 4c, 5c)^T : c \in \mathbb{R}\}$ is a vector subspace.

Example 2.2. How about if we add one more vector $(2, -1, 3)^T$? How to form a vector subspace of $\mathbb{R}^3$ containing $(3, 4, 5)^T$ and $(2, -1, 3)^T$?

2.2. Linear combinations.

Definition 2.3. Let S be a non-empty subset of a vector space V . A vector v in V is called a **linear combination** of vectors in S if there exists a *finite* number of vectors $u_1, \dots, u_n$ in S and scalars $a_1, \dots, a_n$ in $\mathbb{R}$ such that

$$v = a_1 u_1 + \dots + a_n u_n$$

We also say that v is a **linear combination** of vectors $u_1, \dots, u_n$.

Example 2.4. (1) In $\mathbb{R}^3$, $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ is a linear combination of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

since

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = (1) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

(2) In $\text{Mat}_{2 \times 2}$, $\begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}$ is a linear combination of $\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 2 & -1 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 2 & 1 \end{pmatrix}$

since

$$\begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 2 & -1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 2 & 1 \end{pmatrix}.$$

(3) In $\mathcal{P}$, $1 + 2x - 6x^5$ is a linear combination of $1 + 3x$ and $x + 6x^5$ since

$$1 + 2x - 6x^5 = (1 + 3x) - (x + 6x^5).$$

Example 2.5. (Linear combinations by solving system of linear equations) Express the vector $(1, 3)^T$ in linear combinations of $(2, 3)^T$ and $(3, 5)^T$.

Solution: We have to solve $x_1, x_2 \in \mathbb{R}$ for $(1, 3)^T = x_1(2, 3)^T + x_2(3, 5)^T$ i.e.

$$\begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Let $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$. Then $A^{-1} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}$. We then have:

$$A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}.$$

Example 2.6. Let $u_1 = (1, 1, 0, 0)^T$, $u_2 = (0, 1, 1, 0)^T$, $u_3 = (0, 0, 1, 1)^T$. Show that $(1, 2, 3, 4)^T$ cannot be written as a linear combination of u_1, u_2 and u_3 .

Solution: We have to show the system of linear equations: $x_1u_1 + x_2u_2 + x_3u_3 = (1, 2, 3, 4)^T$ does not have a solution. We write the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{array} \right)$$

By the Jordan-Gaussian eliminations, the row echelon form takes the form:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 2 \end{array} \right)$$

The last row implies there is no solution.

Example 2.7. Express $(0, -1, 4)^T$ as a linear combination of $(1, 2, -1)^T$ and $(2, 3, 2)^T$.

Solution: We have to solve x_1, x_2 such that

$$x_1(1, 2, -1)^T + x_2(2, 3, 2)^T = (0, -1, 4)^T.$$

Then, we have to solve the system of linear equations with augmented matrix

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 2 & 3 & -1 \\ -1 & 2 & 4 \end{array} \right).$$

By the Jordan-Gaussian eliminations (exercise: fill out the details),

$$\left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right).$$

2.3. Spanning sets.

Definition 2.8. • Let S be a non-empty subset of a vector space V . The **span** of S , denoted by $\text{span}(S)$, is the set exactly consisting of all linear combinations of the vectors in S .

- Define $\text{span}(\emptyset) = \{0\}$.

Example 2.9. Let $S = \{(1, 0, 0)^T, (0, 1, 0)^T\}$ be a subset in $\mathbb{R}^3$. Then

$$\text{span}(S) = \{(x, y, 0)^T \in \mathbb{R}^3 : x, y \in \mathbb{R}\}.$$

Example 2.10. Let v be a non-zero vector in $\mathbb{R}^n$ and let $S = \{v\}$. Then

$$\text{span}(S) = \{av : a \in \mathbb{R}\}$$

Geometrically, $\text{span}(S)$ is a line in $\mathbb{R}^n$ passing through the origin.

Exercise 2.11. Let $n \geq 2$. Let v be a non-zero vector in $\mathbb{R}^n$. Show that $\text{span}(\{v\}) \neq \mathbb{R}^n$.

Theorem 2.12. (*Vector subspaces from span of vectors*) The span of any subset S of V is a vector subspace of V .

Proof. Case 1: $S = \emptyset$. It follows from that $\text{span}(S)$ is a zero subspace.

Case 2: $S \neq \emptyset$. For any $v, u \in \text{span}(S)$, there exist $a_1, \dots, a_r \in \mathbb{R}$ and $v_1, \dots, v_r \in S$ such that

$$v = a_1 v_1 + \dots + a_r v_r,$$

and similarly there exist $b_1, \dots, b_s \in \mathbb{R}$ and $u_1, \dots, u_s \in S$ such that

$$u = b_1 u_1 + \dots + b_s u_s.$$

Check $v + u \in \text{span}(S)$

$$v + u = a_1 v_1 + \dots + a_r v_r + b_1 u_1 + \dots + b_s u_s$$

and so is a linear combination of vectors in S . Hence, $v + u \in \text{span}(S)$.

Check $cv \in \text{span}(S)$

For $c \in \mathbb{R}$,

$$cv = c(a_1 v_1) + \dots + c(a_r v_r) = (ca_1) v_1 + \dots + (ca_r) v_r.$$

Then, cv is also a linear combination of vectors in S and so is also in $\text{span}(S)$. $\square$

The intuitive meaning of Theorem 2.12 is that $\text{span}(S)$ is the minimal vector space containing all vectors in S .

Exercise 2.13. (**Column spaces**) Let A be an $n \times m$ matrix and let $v_1, \dots, v_m$ be the columns of A . Define

$$\text{col}(A) = \text{span}(\{v_1, \dots, v_m\}),$$

called the **column space of A** . Show that

$$\text{col}(A) = \{Ax \in \mathbb{R}^n : x \in \mathbb{R}^m\}.$$

Exercise 2.14. Let S be a subset of a vector space V . If $v \in \text{span}(S)$, show that $\text{span}(S \cup \{v\}) = \text{span}(S)$.

2.4. Spanning sets.

Definition 2.15. A subset S of a vector space V is said to **span** V if $\text{span}(S) = V$. We also say that the vectors in S **span** V , or S is a **spanning set** of V .

Example 2.16. Let $\mathbf{e}_i \in \mathbb{R}^n$ be the column vector with the $(i, 1)$ -entry to be 1 and zero elsewhere. For example, in $\mathbb{R}^3$, $\mathbf{e}_1 = (1, 0, 0)^T$, $\mathbf{e}_2 = (0, 1, 0)^T$ and $\mathbf{e}_3 = (0, 0, 1)^T$. Then $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ spans $\mathbb{R}^n$.

Example 2.17. Show that $(1, 2, 0)^T, (0, 1, 2)^T, (0, 1, 1)^T$ span $\mathbb{R}^3$.

Solution: For any $(u_1, u_2, u_3)^T \in \mathbb{R}^3$, we consider the following system of linear equations:

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}.$$

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 0 & 2 & 1 \end{pmatrix}$. Then

$$\det(A) = (1)\det \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = -1 \neq 0.$$

Hence, A is invertible. Thus, we can find

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A^{-1} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}.$$

In other words, any vector in $\mathbb{R}^3$ is a linear combination of that three vectors.

3. LINEAR DEPENDENCE AND LINEAR INDEPENDENCE

3.1. Linearly independent and linearly dependent.

Definition 3.1. • A subset S of a vector space V is called **linearly dependent** if there exists a finite number of distinct vectors $v_1, \dots, v_n$ in S and scalars $a_1, \dots, a_n$ in $\mathbb{R}$ *not all zero* such that

$$a_1v_1 + \dots + a_nv_n = 0.$$

- A subset S of a vector space V is said to be **linearly independent** if S is not linearly dependent. Equivalently, S is linearly independent if for any $v_1, \dots, v_r \in S$, the only $a_1, \dots, a_r \in \mathbb{R}$ satisfying

$$a_1 v_1 + \dots + a_r v_r = 0$$

are the trivial ones i.e. $a_1 = \dots = a_r = 0$.

Example 3.2. Let $u_1 = (1, 0, -1)^T$, $u_2 = (0, 1, -1)^T$ and $u_3 = (1, 0, -2)^T$. Then u_1, u_2, u_3 are linearly dependent since $u_1 + u_2 + (-1)u_3 = 0$.

Example 3.3. (**Linearly dependence between two non-zero vectors**) Let u, v be two non-zero vectors in a vector space V . Then u and v are linearly dependent if and only if $u = av$ for some non-zero scalar $a \in \mathbb{R}$.

Example 3.4. Let S be a subset of a vector space V . If $0 \in S$, then S is always linearly dependent.

Exercise 3.5. Let $S_1 \subset S_2 \subset \mathbb{R}^n$. Prove the following statements:

- If S_2 is linearly independent, then S_1 is also linearly independent.
- If S_1 is linearly dependent, then S_2 is also linearly dependent.

Example 3.6. Show that the vectors $(1, 0, 0, 2)^T, (0, 1, 0, 1)^T, (0, 0, 1, 0)^T, (0, 0, 1, 1)^T$ form a set of linearly independent vectors in $\mathbb{R}^4$.

Solution: We have to solve x_1, x_2, x_3, x_4 such that

$$x_1(1, 0, 0, 2)^T + x_2(0, 1, 0, 1)^T + x_3(0, 0, 1, 0)^T + x_4(0, 0, 1, 1)^T = (0, 0, 0, 0)^T.$$

The corresponding augmented matrix is:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 \end{array} \right)$$

The Jordan-Gaussian eliminations (here I skip the details) take to the following reduced row echelon form:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$$

and so $x_1 = x_2 = x_3 = x_4 = 0$ is the only solution.

Example 3.7. Let V be the vector space of all functions from $\mathbb{R}$ to $\mathbb{R}$. (Recall that the addition and the scalar multiplication are defined in Example 1.5.) Let $f_1(x) = x$ and $f_2(x) = x^2$. Show that $\{f_1, f_2\}$ is a linearly independent set.

Solution: Let $a_1, a_2 \in \mathbb{R}$ such that $a_1 f_1 + a_2 f_2 = 0$. (What is 0 in that vector space?)

- If we substitute $x = 1$, we then have $a_1 + a_2 = 0$.
- If we substitute $x = 2$, we then have $2a_1 + 4a_2 = 0$.

Solving the equations, $a_1 = a_2 = 0$ is the only solution. Hence, $\{f_1, f_2\}$ is a linearly independent set.

4. BASIS AND DIMENSIONS

4.1. Basis.

Definition 4.1. A **basis** for a vector space V is a *linearly independent* subset of V which also *spans* for V .

Example 4.2. Let $\mathbf{e}_1 = (1, 0, \dots, 0)^T$, $\mathbf{e}_2 = (0, 1, 0, \dots, 0)^T$, $\dots$, $\mathbf{e}_n = (0, \dots, 0, 1)^T$. The set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$. We shall call $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ be the **standard basis** for $\mathbb{R}^n$.

Example 4.3. Let $v_1 = (1, 2)^T$ and $v_2 = (2, 3)^T$.

- We first show that $\{v_1, v_2\}$ is linearly independent: For any $a_1, a_2 \in \mathbb{R}$ with

$$a_1 v_1 + a_2 v_2 = (0, 0)^T,$$

we have

$$\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

and so

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

- Show $\{v_1, v_2\}$ spans $\mathbb{R}^2$: Exercise or one may use Theorem 5.2 below.

Example 4.4. (**Basis for a spanning set**)

- (1) Let $S = \{(1, 2, 0)^T, (2, 3, 0)^T\}$. Note that S is a linearly independent set. Then S is a basis for $\text{span}(S)$.
- (2) Let S be a set of *linearly independent* vectors in a vector space V . Then S is a basis for $\text{span}(S)$.
- (3) If S is a (not necessarily linearly independent) subset of vectors in V , then we shall discuss how to find a basis in Section 5.2.

Exercise 4.5. Let β be a basis for a vector space V . Prove the following statements:

- (1) If S is a proper subset of β , then $\text{span}(S) \neq V$.
- (2) If β is a proper subset of another set S' , then S' is not linearly independent.

Theorem 4.6. (*Expressing vectors as a unique linear combination of a basis*) Let β be a subset of a vector space V . Then β is a basis for V if and only if for each vector $u \in V$, u can be **uniquely** written as a **linear combination** of vectors in β i.e.

there exist unique non-zero scalars $a_1, \dots, a_n$ in $\mathbb{R}$ and unique vectors $v_1, \dots, v_n$ in β such that

$$u = a_1v_1 + \dots + a_nv_n$$

Proof. For simplicity, we only consider β is finite and $\beta = \{v_1, \dots, v_n\}$. We prove the 'only if' direction.

Existence of a linear combination from spanning set

Since β is a spanning set of B , there exists integers $a_1, \dots, a_n$ in $\mathbb{R}$ such that

$$u = a_1v_1 + \dots + a_nv_n.$$

Uniqueness from linear independence

Suppose $u = b_1v_1 + \dots + b_nv_n$ for another linear combination of vectors in β . Then,

$$a_1v_1 + \dots + a_nv_n = b_1v_1 + \dots + b_nv_n$$

and so

$$(a_1 - b_1)v_1 + \dots + (a_n - b_n)v_n = 0.$$

Since $v_1, \dots, v_n$ are linearly independent, the only solution is

$$a_1 - b_1 = a_2 - b_2 = \dots = a_n - b_n = 0.$$

Hence, $a_1 = b_1, \dots, a_n = b_n$. This shows the uniqueness. $\square$

4.2. Dimensions.

Exercise 4.7. Explain why the vector space $\mathcal{P}$ in Example 1.15 does not have a *finite* basis.

From now on, we shall assume vector spaces have *finite* bases unless otherwise specified. Vector spaces of an infinite basis are usually studied with some topology, which is an advanced subject of functional analysis.

Theorem 4.8. (*Number of linearly independent vectors bounded by dimension*) Let β be a basis for a vector space V . If β' is a set of linearly independent vectors in V , then $|\beta'| \leq |\beta|$. (Here $|\cdot|$ means the the number of elements in a finite set.)

Proof. Let $n = |\beta|$ and $m = |\beta'|$. Let $\beta = \{v_1, \dots, v_n\}$. Let $\beta' = \{u_1, \dots, u_m\}$.

Step 1: Writing u_i as linear combinations of vectors in β

Since β is a basis for V , β spans V . Hence, we can write as:

$$u_j = a_{1j}v_1 + \dots + a_{nj}v_n$$

Step 2: Set-up equations for β'

Now, we consider the equation:

$$x_1u_1 + \dots + x_mu_m = 0$$

We can rewrite as:

$$\sum_{i=1}^n (a_{i1}x_1 + a_{i2}x_2 + \dots + a_{im}x_m)v_i = 0$$

Step 3: Obtain a system from linearly independence on β

Since $v_1, \dots, v_n$ are linearly independent, we obtain the following **system of linear equations**:

$$\begin{aligned} a_{11}x_1 + \dots + a_{1m}x_m &= 0 \\ &\vdots \\ a_{n1}x_1 + \dots + a_{nm}x_m &= 0 \end{aligned}$$

Step 4: Solve the system

Suppose $m > n$. Chapter 2 discussed that **the above system has non-trivial solutions**. This contradicts that $u_1, \dots, u_m$ are linearly independent. $\square$

Exercise 4.9. Let β_1 and β_2 be any two bases for a vector space V . Show that β_1 and β_2 contain the *same number* of vectors.

Definition 4.10. The number of vectors in a basis for a vector space V is called the **dimension** of V , denoted by **$\dim(V)$** . The above discussion implies that the definition of the dimension is independent of the choice of a basis.

Example 4.11. (**Dimension of $\mathbb{R}^n$**) The dimension of $\mathbb{R}^n$ is n since the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$ has n vectors.

Example 4.12. Let S be a finite set of linearly independent vectors in a vector space V . Then $\dim(\text{span}(S)) = |S|$.

Exercise 4.13. Let W be a vector subspace of a vector space V . Then

$$\dim(W) \leq \dim(V).$$

5. GENERAL CONSTRUCTIONS OF BASIS

5.1. Adding vectors (Extension). In general, one can find a basis for a vector space by adding vectors one-by-one using the following lemma:

Theorem 5.1. (*Construct a larger linearly independent set by adding a vector not in the span*) Let S be a linearly independent set of vectors in V . Let $v \in V$. If $v \notin \text{span}(S)$, then $S \cup \{v\}$ is also a set of linearly independent vectors.

Proof. Set-up an equation for linear independence

Let $S = \{v_1, \dots, v_m\}$. We consider the following equation:

$$(5.1) \quad x_1v_1 + x_2v_2 + \dots + x_mv_m + x_{m+1}v = 0$$

We shall solve by dividing into two cases.

Case 1: Suppose $x_{m+1} \neq 0$. Then,

$$v = -\frac{x_1}{x_{m+1}}v_1 - \dots - \frac{x_m}{x_{m+1}}v_m.$$

Then, v is a linear combination of vectors in S and so $v \in \text{span}(S)$. This gives a contradiction i.e. Case 1 could not happen.

Case 2: Suppose $x_{m+1} = 0$. This reduces (5.1) to:

$$x_1v_1 + x_2v_2 + \dots + x_mv_m = 0.$$

Since $v_1, \dots, v_m$ are linearly independent, we have $x_1 = x_2 = \dots = x_m = 0$.

In other words, $x_1 = \dots = x_m$ is the only solution for (5.1) and so $S \cup \{v\}$ is linearly independent. $\square$

Theorem 5.2. (*dim(V)-number of linearly independent vectors must form a basis*)
Let $m = \dim(V)$. If S is a linearly independent set of m vectors, then S is a basis for V .

Proof. Suppose $\text{span}(S) \neq V$. Then there exists $v \in V$ such that $v \notin \text{span}(S)$. By Theorem 5.1, $S \cup \{v\}$ is linearly independent. However, this contradicts to Theorem 4.8. $\square$

Example 5.3. Construct a basis for $\mathbb{R}^3$, which contains the linearly independent vectors $(1, 2, 0)^T, (1, 1, 1)^T$.

Solution: Let $v = (0, 0, 1)^T$. Since there is no solution for

$$x_1(1, 2, 0)^T + x_2(1, 1, 1)^T = (0, 0, 1)^T,$$

we conclude that $v \notin \text{span}\{(1, 2, 0)^T, (1, 1, 1)^T\}$. Then $\{(1, 2, 0)^T, (1, 1, 1)^T, (0, 0, 1)^T\}$ is a set of linearly independent vectors and so, by Theorem 5.2, it forms a basis.

Example 5.4. By Example 3.6 and Theorem 5.2, $(1, 0, 0, 2)^T, (0, 1, 0, 1)^T, (0, 0, 1, 0)^T, (0, 0, 1, 1)^T$ forms a basis for $\mathbb{R}^4$.

5.2. Removing vectors (Reduction).

Theorem 5.5. (*Construct a smaller spanning set by removing a vector*) Let S be a spanning set of a vector space V . If S is linearly dependent, then there exists $v \in S$ such that $\text{span}(S - \{v\}) = V$.

Proof. Step 1: Equation from linearly dependence

Let $S = \{v_1, \dots, v_r\}$. Since S is linearly dependent, there exists scalars $a_1, \dots, a_r$ not all zero such that

$$a_1 v_1 + \dots + a_r v_r = 0.$$

Step 2: Rewrite as a linear dependence of certain v_i We have $a_i \neq 0$ for some i . Rewrite the above equation as:

$$a_i v_i = -a_1 v_1 - \dots - a_{i-1} v_{i-1} - a_{i+1} v_{i+1} - \dots - a_r v_r$$

and so

$$v_i = -\frac{a_1}{a_i} v_1 - \dots - \frac{a_{i-1}}{a_i} v_{i-1} - \frac{a_{i+1}}{a_i} v_{i+1} - \dots - \frac{a_r}{a_i} v_r.$$

This implies $v_i \in \text{span}(S - \{v_i\})$ and so, by Exercise 2.14,

$$\text{span}(S) = \text{span}(S - \{v_i\}).$$

□

Exercise 5.6. Let S be a finite subset of non-zero vectors in V . Prove that there exists a subset β of S such that β is a basis for $\text{span}(S)$. (Hint: Prove by a back induction on the cardinality of S .)

Exercise 5.7. Let S be a finite subset of vectors in V . Prove that $\dim(\text{span}(S)) \leq |S|$.

Theorem 5.8. (*spanning set of $\dim(V)$ -vectors must form a basis*) Let $m = \dim(V)$. If S is a spanning set of V of m vectors, then S is a basis for V .

Proof. Suppose S is not a basis for V . Then there exists a subset S' of S such that S' is a basis for V . Then $|S'| < |S| = m$. But, by the definition of dimension, $m = \dim(V) = |S'|$. This gives a contradiction. □

Example 5.9. By Example 2.17 and Theorem 5.8, $(1, 2, 0)^T, (0, 1, 2)^T, (0, 1, 1)^T$ is a basis for $\mathbb{R}^3$.

5.3. Comparing linearly independence, spanning set and dimensions.

Exercise 5.10. Let V be a vector space with $\dim(V) = n$. Let S be a finite subset of V with k elements. Circle the best words in the table for the descriptions of S :

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	$ S < n$	$ S = n$	$ S > n$
linearly independent	must be / must not be/ may or may not be	must be / must not be/ may or may not be	must be / must not be/ may or may not be
spanning set for V	must be / must not be/ may or may not be	must be / must not be/ may or may not be	must be / must not be/ may or may not be
basis for V	must be / must not be/ may or may not be	must be / must not be/ may or may not be	must be / must not be/ may or may not be

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CHAPTER 3: VECTOR SPACES (PART 2)

K.Y. CHAN

Goals:

- (1) Column spaces and how to find a basis
- (2) Dimension of column space = dimension of row space
- (3) Null spaces and how to find a basis

1. COLUMN AND ROW SPACES

1.1. Column space of a matrix. Let A be an $n \times m$ matrix. Write

$$A = (v_1 \ \dots \ v_m),$$

where $v_1, \dots, v_m$ are column vectors in $\mathbb{R}^n$.

Recall that the *column space* of A is defined as:

$$\text{col}(A) = \text{span}\{v_1, \dots, v_m\}.$$

It is shown in a previous exercise that

$$\text{col}(A) = \{Ax : x \in \mathbb{R}^m\} \subset \mathbb{R}^n.$$

Exercise 1.1. (Rephrasing $\text{col}(A)$ in the content of a system of linear equations)

The space $\text{col}(A)$ is the set precisely containing all vectors b such that the system $Ax = b$ is consistent/inconsistent*. (*Choose the appropriate word)

Exercise 1.2. Let $v_1, \dots, v_m$ be column vectors in $\mathbb{R}^m$. Let

$$A = (v_1 \ \dots \ v_m)$$

Show that the following statements are equivalent (some are explained before in Chapter 2):

- (1) $\det(A) \neq 0$;
- (2) A is invertible;
- (3) for any $b \in \mathbb{R}^m$, $Ax = b$ has unique solution;
- (4) $\{v_1, \dots, v_m\}$ is a basis for $\mathbb{R}^m$;
- (5) $\{v_1, \dots, v_m\}$ is linearly independent.

Example 1.3. Determine if the set $\{(1, 0, -1)^T, (2, 3, 1)^T, (0, 4, 3)^T\}$ is a basis for $\mathbb{R}^3$.

Solution: Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ -1 & 1 & 3 \end{pmatrix}$. Then

$$\det(A) = (-1)^{1+1}(1)\det\begin{pmatrix} 3 & 1 \\ 4 & 3 \end{pmatrix} + (-1)^{1+3}(-1)\det\begin{pmatrix} 0 & 3 \\ -1 & 1 \end{pmatrix} = 2,$$

and so the vectors form a basis.

1.2. Linearly independent under invertible matrix.

Theorem 1.4. (*Multiplication by an invertible matrix preserves linear independence*) Let P be an invertible $n \times n$ -matrix. Then $\{v_1, \dots, v_k\}$ is a set of linearly independent vectors in $\mathbb{R}^n$ if and only if $\{Pv_1, \dots, Pv_k\}$ is a set of linearly independent vectors in $\mathbb{R}^n$.

Proof. Only if direction:

Suppose $\{v_1, \dots, v_k\}$ is linearly independent. Consider the equation:

$$x_1(Pv_1) + x_2(Pv_2) + \dots + x_k(Pv_k) = 0.$$

for unknowns $x_1, \dots, x_k$ in $\mathbb{R}$. Since P is invertible, multiplying P^{-1} gives

$$x_1v_1 + x_2v_2 + \dots + x_kv_k = 0$$

and so the linear independence of $\{v_1, \dots, v_k\}$ gives $x_1 = \dots = x_k = 0$. Hence, $\{Pv_1, \dots, Pv_k\}$ is linearly independent.

If direction:

Suppose $\{Pv_1, \dots, Pv_k\}$ is linearly independent. Since P is invertible, P^{-1} is also invertible. Then, from the proved only if direction, we also have:

$$\{P^{-1}(Pv_1), P^{-1}(Pv_2), \dots, P^{-1}(Pv_r)\} = \{v_1, \dots, v_r\}$$

is also linearly independent. □

1.3. Finding a basis for $\text{col}(A)$. How to find a basis for $\text{col}(A)$? Let $v_1, \dots, v_m$ be all columns of A . We have discussed before that:

$$\text{span}(\{v_1, \dots, v_m\}) = V.$$

Thus, by previous discussion, there is a subset of $\{v_1, \dots, v_m\}$ that forms a basis for V . We shall provide a systematic method (Theorem 1.7) to that.

Theorem 1.5. (*Multiplication by an invertible matrix preserves dimension*) Let A be an $n \times m$ matrix. Let P be an invertible $n \times n$ matrix. Then $\dim(\text{col}(A)) = \dim(\text{col}(PA))$.

Proof. Let $A = (v_1 \dots v_m)$. Then we can find vectors from $v_1, \dots, v_m$ form a basis for $\text{col}(A)$, say $v_{i_1}, \dots, v_{i_t}$. We want to show that

- $\{Pv_{i_1}, \dots, Pv_{i_t}\}$ is a basis for $\text{col}(PA)$.

Check linearly independence: This follows from Theorem 1.4.

Check spanning set: Since $PA = (Pv_1 \ Pv_2 \ \dots \ Pv_m)$, we also have

$$\text{span}(\{Pv_1, \dots, Pv_m\}) = \text{col}(PA)$$

by definition. By writing each v_j in terms of a linear combination of $v_{i_1}, \dots, v_{i_t}$, each Pv_j can also be expressed in terms of a linear combination of $Pv_{i_1}, \dots, Pv_{i_t}$. Hence,

$$\text{span}(\{Pv_{i_1}, \dots, Pv_{i_t}\}) = \text{span}(\{Pv_1, \dots, Pv_m\}) = \text{col}(PA).$$

Hence, $\{Pv_{i_1}, \dots, Pv_{i_t}\}$ forms a basis. Thus $\dim(\text{col}(A)) = \dim(\text{col}(PA))$. $\square$

Exercise 1.6. Let A be an $n \times m$ matrix. Let P be any $n \times n$ matrix. Show that $\dim(\text{col}(A)) \geq \dim(\text{col}(PA))$.

We now describe how to find a basis for a column space:

Theorem 1.7. (*Determining a basis for $\text{col}(A)$ from leading ones in RREF of A*)
 Let A be a $n \times m$ matrix. Let R be the reduced row echelon form of A . Then the columns in A corresponding to the leading ones in R form a basis for $\text{col}(A)$.

Proof. Step 1: RREF of A Let $A = (v_1 \ v_2 \ \dots \ v_m)$. Let E be the product of the sequences of elementary matrices taking to R . In other words,

$$R = EA = (Ev_1 \ Ev_2 \ \dots \ Ev_m).$$

We pick the columns in R with leading ones $Ev_{i_1}, \dots, Ev_{i_k}$.

Step 2: Find the dimension of $\text{col}(R)$

By looking at the leading ones, indeed we have that $Ev_{i_1}, \dots, Ev_{i_k}$ span $\text{col}(R)$. With Step 1, $\dim \text{col}(R) = k$.

Step 3: Determine a basis for $\text{col}(A)$

- By Theorem 1.4 and the fact that E is invertible, $\{v_{i_1}, \dots, v_{i_k}\}$ is linearly independent.
- By Theorem 1.5, $\dim(\text{col}(R)) = \dim(\text{col}(A))$.

It is shown in the previous part that $\dim(\text{col}(A))$ linearly independent vectors form a basis for $\text{col}(A)$, and so $\{v_{i_1}, \dots, v_{i_k}\}$ forms a basis for $\text{col}(A)$. $\square$

Corollary 1.8. *Let A be an $n \times m$ matrix. Then $\dim(\text{col}(A))$ is equal to the number of leading ones in the RREF of A .*

Example 1.9. Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 4 & 2 & 1 \end{pmatrix}$$

Find a basis for $\text{col}(A)$.

Sol: By using Jordan-Gaussian eliminations,

$$\begin{pmatrix} 2 & 1 & 0 \\ 4 & 2 & 1 \end{pmatrix} \xrightarrow{-2R_1+R_2 \rightarrow R_2} \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1 \rightarrow R_1} \begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The leading ones are in the first and third columns. Hence,

$$\begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

form a basis for $\text{col}(A)$.

Exercise 1.10. Find a basis for

$$\text{span}\left(\left\{\begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 5 \\ 4 \\ 9 \\ -1 \end{pmatrix}\right\}\right)$$

which is a subset of that containing the four vectors in the span.

2. DIMENSION OF COLUMN SPACE = DIMENSION OF ROW SPACE

2.1. Row spaces.

Definition 2.1. Let A be an $n \times m$ matrix. Define $\text{row}(A)$, called the row space of A , to be the subspace of $\mathbb{R}^m$ spanned by the rows of A .

Exercise 2.2. Let A be an $n \times m$ matrix. Let P be an invertible $n \times n$ matrix. Then $\text{row}(PA) = \text{row}(A)$ and in particular, $\dim(\text{row}(PA)) = \dim(\text{row}(A))$.

Exercise 2.3. (Basis for the row space of a matrix in RREF) Let A be an $n \times m$ matrix in a reduced row echelon form. Prove that the non-zero rows in A form a basis for $\text{row}(A)$.

Exercise 2.4. (Finding a basis for $\text{row}(A)$) Let A be an $n \times m$ matrix. Let $\{w_1, \dots, w_k\}$ be the non-zero rows in the reduced row echelon form of A . Then the previous two exercise implies that $\{w_1, \dots, w_k\}$ forms a basis for $\text{row}(A)$.

Exercise 2.5. Let A be any $n \times m$ matrix. Let P be any $n \times n$ matrix. Show that $\dim(\text{row}(A)) \geq \dim(\text{row}(PA))$.

2.2. Rank of a matrix. One fundamental result in linear algebra is the following:

Theorem 2.6. (*row dimension=column dimension*) Let A be an $n \times m$ matrix. Then $\dim(\text{row}(A)) = \dim(\text{col}(A))$.

Proof. **Step 1: Reduction to RREF**

Let R be the reduced row echelon form of A . Since R is obtained by a sequence of (invertible) elementary matrices, we have, by Theorem 1.5 and Exercise 2.2,

$$\dim(\text{row}(A)) = \dim(\text{row}(R))$$

and

$$\dim(\text{col}(A)) = \dim(\text{col}(R)).$$

Step 2: Determine the dimensions by the number of leading ones

- By Theorem 1.7, $\dim(\text{col}(R))$ is equal to the number of leading ones in R .
- By Exercise 2.4, $\dim(\text{row}(R))$ is equal to the number of leading ones in R .

Hence $\dim(\text{col}(A)) = \dim(\text{row}(A))$. $\square$

Definition 2.7. Let A be an $n \times m$ matrix. Define $\text{rank}(A) = \dim(\text{col}(A))$, called the **rank** of A .

Corollary 2.8. Let A be an $n \times m$ matrix. Then $\text{rank}(A) = \text{rank}(A^T)$.

3. NULL SPACE OF A MATRIX

3.1. Null space of a matrix. Let A be a $n \times m$ matrix and let b be a column vector in $\mathbb{R}^n$. Recall that a system of linear equations $Ax = b$ of n equations and m variables is said to be *homogeneous* if $b = 0_{n \times 1}$.

Theorem 3.1. (*Solution set of homogeneous system forms vector subspace*) The solution set of a homogeneous system of linear equations is a vector subspace i.e. for an $n \times m$ matrix A , $\{x \in \mathbb{R}^m : Ax = 0\}$ is a vector subspace.

Proof. Let $S = \{x \in \mathbb{R}^m : Ax = 0\}$. For $v_1, v_2 \in S$, $Av_1 = 0$ and $Av_2 = 0$. Hence $A(v_1 + v_2) = Av_1 + Av_2 = 0 + 0 = 0$. Hence, $v_1 + v_2 \in S$. For $v \in S$ and $c \in \mathbb{R}$, $A(cv) = c(Av) = c(0) = 0$ and so $cv \in S$. $\square$

Definition 3.2. Let A be an $n \times m$ matrix. Let

$$N(A) = \{v \in \mathbb{R}^m : Av = 0\}.$$

$N(A)$ is a vector subspace of $\mathbb{R}^m$, called the **null space of A** .

Exercise 3.3. Let b be a non-zero vector in $\mathbb{R}^n$ and let A be an $n \times m$ matrix. Determine if

$$\{v \in \mathbb{R}^m : Av = b\}$$

is a vector subspace of $\mathbb{R}^m$.

3.2. Basis for $N(A)$.

Theorem 3.4. (*Dimension and basis for $N(A)$*) Let A be an $n \times m$ matrix. Let R be the reduced row echelon form of A . Then

- (1) $\dim(N(A))$ is equal to the number of free variables for the system $Ax = 0$;
- (2) Suppose there are r free variables. We assign the free variables with the parameters $s_1, \dots, s_r$. Then one can rewrite the solution set for $Ax = 0$ in the form

$$\{s_1 v_1 + \dots + s_r v_r : s_1, \dots, s_r \in \mathbb{R}\}$$

for some $v_1, \dots, v_r \in \mathbb{R}^m$. Then, $\{v_1, \dots, v_r\}$ forms a basis for $N(A)$. (One sees the following example a precise way to write into such form.)

Proof. (Sketch of the proof) Note that (2) implies (1) by the definition of the dimension.

Spanning set: Recall that the solution set for $Ax = 0$ is obtained in Chapter 2, and this in particular implies $v_1, \dots, v_r$ spans the solution set.

Linear independence: Each v_i has **non-zero entry** for the **corresponding free variable** and **zero** for all **other free variables**. This implies $v_1, \dots, v_r$ are linearly independent. $\square$

Example 3.5. Let $A = (1 \ 1 \ 1 \ 1)$. Let $S = \{(x_1, x_2, x_3, x_4)^T : x_1 + x_2 + x_3 + x_4 = 0\} = N(A)$. Then,

$$\left\{ \begin{pmatrix} -s - t - u \\ s \\ t \\ u \end{pmatrix} \in \mathbb{R}^4 : s, t, u \in \mathbb{R} \right\} = \left\{ s \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} : s, t, u \in \mathbb{R} \right\}.$$

Then a basis for S is

$$\{(1, -1, 0, 0)^T, (1, 0, -1, 0)^T, (1, 0, 0, -1)^T\}.$$

Hence, $\dim(N(A)) = 3$.

Exercise 3.6. Let A be a $1 \times m$ matrix with all entries equal to 1. Find a basis for $N(A)$ and determine $\dim(N(A))$.

MATH 2101 LINEAR ALGEBRA I, FALL SEMESTER 2025
CHAPTER 4: LINEAR TRANSFORMATIONS (PART 1)

K.Y. CHAN

Key concepts in this section:

- What is a linear transformation? Examples of linear transformations?
- How to construct linear transformations?
- Concepts of null spaces, range, nullity and rank.
- Dimension formula and some applications on the formula
- Injectivity for a linear transformation

1. LINEAR TRANSFORMATIONS

1.1. Definitions. In the set theory, a function is to assign each element in the domain to an element in the codomain. Now we would like to define some functions in the content of vector spaces, so that the major properties (i.e. addition and scalar multiplication) can be reflected from functions.

Definition 1.1. Let V and W be vector spaces. Then a function $T : V \rightarrow W$ is a **linear transformation from V to W** if for any $x, y \in V$ and a scalar $c \in \mathbb{R}$,

- (1) (**Preserving addition**) $T(x + y) = T(x) + T(y)$;
- (2) (**Preserving scalar multiplication**) $T(cx) = cT(x)$.

We now give several examples:

Example 1.2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T((x, y)^T) = 3(x, y)^T,$$

Then T is a linear transformation.

Example 1.3. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$T((x_1, \dots, x_n)^T) = x_1 + \dots + x_n.$$

Example 1.4. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by

$$T((x_1, x_2, x_3)^T) = (x_1 + x_2, x_2 - 2x_3)^T.$$

To see that T is a linear transformation, let $v = (x_1, x_2, x_3)^T, w = (y_1, y_2, y_3)^T \in \mathbb{R}^3$,

$$\begin{aligned} T(v+w) &= ((x_1+y_1) + (x_2+y_2), x_2+y_2 - 2(x_3+y_3))^T \\ &= (x_1+x_2, x_2-2x_3) + (y_1+y_2, y_2-2y_3)^T \\ &= T(v) + T(w), \end{aligned}$$

where the second equality follows from addition of two vectors. And, for $c \in \mathbb{R}$ and $v \in \mathbb{R}^3$,

$$\begin{aligned} T(cv) &= (cx_1 + cx_2, cx_2 - 2cx_3)^T \\ &= c(x_1 + x_2, x_2 - x_3)^T, \end{aligned}$$

where the second equality follows from the scalar multiplication on a vector.

Example 1.5. (Matrix multiplication as a linear transformation) Let A be an $n \times m$ -matrix. Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ given by:

$$T(v) = Av,$$

where Av means the matrix multiplication between the $n \times m$ matrix A and the $m \times 1$ matrix v . Then it is a linear transformation.

Sol:

- Addition: For $v_1, v_2 \in \mathbb{R}^m$, $T(v_1 + v_2) = A(v_1 + v_2) = Av_1 + Av_2 = T(v_1) + T(v_2)$.
- Scalar multiplication: For a scalar $c \in \mathbb{R}$ and $v \in \mathbb{R}^m$, $T(cv) = A(cv) = c(Av) = cT(v)$.

1.2. Identity and zero transformations.

Definition 1.6. Let V, W be vector spaces.

- (1) The **identity transformation** $\text{Id}_V : V \rightarrow V$ is the function $\text{Id}_V(v) = v$.
- (2) The **zero transformation** $T_0 : V \rightarrow W$ is the function $T_0(v) = 0$.

Theorem 1.7. Both identity transformation and zero transformation are linear transformations.

Proof. Exercise. □

1.3. Some basic properties of linear transformations.

Theorem 1.8. (Linear transformation preserving zero and linear combinations) Let $T : V \rightarrow W$ be a linear transformation. Then

- $T(0) = 0$;
- for any $a_1, \dots, a_r \in \mathbb{R}$ and any $v_1, \dots, v_r \in V$,

$$T(a_1v_1 + \dots + a_rv_r) = a_1T(v_1) + \dots + a_rT(v_r).$$

Proof. Exercises □

Remark 1.9. The first thing to **check whether a function between vector spaces is a linear transformation** is to **check whether $T(0) = 0$** .

Example 1.10. (Non-example) Let $T : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$T(x) = 3x + 2.$$

Note that $T(0) = 2 \neq 0$, and so T is **not** a linear transformation.

Of course, even a function sends the zero vector to the zero vector, such function could still be not a linear transformation.

Example 1.11. (Non-example) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$T((x, y)^T) = xy$$

Note that $T((1, 1)^T) = 1$ and $T((2, 2)^T) = 4 \neq 2T((1, 1)^T)$. Hence, T is not a linear transformation.

1.4. Advanced examples.

Example 1.12. (Example from functions from $\mathbb{R}$ to $\mathbb{R}$) Let V be the vector space of *differentiable* functions from $\mathbb{R}$ to $\mathbb{R}$. Then the differentiation

$$f'(x) = \frac{df}{dx}(x)$$

is a linear transformation from V to V .

- Addition: for $f_1, f_2 \in V$,

$$(f_1 + f_2)'(x) = f_1'(x) + f_2'(x) = (f_1' + f_2')(x)$$

- Scalar multiplication: for $c \in \mathbb{R}$ and for $f \in V$,

$$(cf)'(x) = c(f'(x)) = (c(f'))(x).$$

Example 1.13. (Example from space of matrices) Let V be the vector space of $n \times m$ -matrices. Let P be an $n \times n$ matrix and let Q be a $m \times m$ matrix. Let

$$T : V \rightarrow V, \quad T(A) = PAQ.$$

Then T is a linear transformation.

- Addition: For $A, B \in V$, $T(A + B) = P(A + B)Q = P(AQ + BQ) = PAQ + PBQ$.
- Scalar Multiplication: For $A \in V$ and a scalar $c \in \mathbb{R}$, $T(cA) = P(cA)Q = cPAQ = cT(A)$.

Example 1.14. (non-example) Let V be the vector space of $n \times n$ -matrices. Let

$$T : V \rightarrow V, \quad T(A) = A^2.$$

Determine if T is a linear transformation.

Sol: $T(I_n) = I_n$. On the other hand, for a scalar $c \neq 0, 1$, $T(cI_n) = c^2 I_n \neq cT(I_n)$. Hence, it is not a linear transformation.

2. CONSTRUCTION OF LINEAR TRANSFORMATIONS

Theorem 2.1. (*Construct a linear transformation from a basis*) Let V and W be vector spaces. Let $\{v_1, \dots, v_n\}$ be a basis for V and let $w_1, \dots, w_n \in W$. Then there *exists a unique* linear transformation $T : V \rightarrow W$ such that $T(v_i) = w_i$ for all i .

Remark 2.2. We emphasize that $w_1, \dots, w_n$ do not necessarily form a basis.

Proof. [(Key of the proof) We shall use a result in last chapter: for any $v \in V$, v can be uniquely written as $a_1 v_1 + \dots + a_n v_n$ for some scalar $a_1, \dots, a_n \in \mathbb{R}$.]

Existence: Construction of a linear transformation

For any $v \in V$, there exists unique scalars $a_1, \dots, a_n$ such that

$$v = a_1 v_1 + \dots + a_n v_n.$$

Define

$$(*) \quad T(v) = a_1 w_1 + \dots + a_n w_n.$$

Note that $T(v_i) = w_i$ for $i = 1, \dots, n$. Checking T is linear is straightforward

Uniqueness: Let T' be another linear transformation satisfying

$$T'(v_i) = w_i$$

for all i . Now, for any $v \in V$, there exists scalars $a_1, \dots, a_n$ such that

$$v = a_1 v_1 + \dots + a_n v_n.$$

Then

$$\begin{aligned} T'(v) &= T'(a_1 v_1 + \dots + a_n v_n) \\ &= a_1 T'(v_1) + \dots + a_n T'(v_n) \\ &= a_1 w_1 + \dots + a_n w_n \\ &= a_1 T(v_1) + \dots + a_n T(v_n) \\ &= T(a_1 v_1 + \dots + a_n v_n) \\ &= T(v). \end{aligned}$$

□

Exercise 2.3. Check the map T constructed in (*) of the proof of Theorem 2.1 is linear.

Sol: Addition For $v, v' \in V$, write as:

$$v = a_1v_1 + \dots + a_nv_n, \quad v' = a'_1v_1 + \dots + a'_nv_n$$

for some unique $a_1, \dots, a_n, a'_1, \dots, a'_n$. Note that

$$v + v' = (a_1 + a'_1)v_1 + \dots + (a_n + a'_n)v_n$$

and so $T(v + v') = (a_1 + a'_1)w_1 + \dots + (a_n + a'_n)w_n$ from (*). On the other hand,

$$T(v) + T(v') = a_1v_1 + \dots + a_nv_n + a'_1v_1 + \dots + a'_nv_n = (a_1 + a'_1)v_1 + \dots + (a_n + a'_n)v_n.$$

Hence $T(v + v') = T(v) + T(v')$.

Scalar multiplication For $v \in V$ and $c \in \mathbb{R}$, write:

$$v = a_1v_1 + \dots + a_nv_n.$$

Then $cv = (ca_1)v_1 + \dots + (ca_n)v_n$ and so

$$T(cv) = (ca_1)w_1 + \dots + (ca_n)w_n.$$

On the other hand, $cT(v) = c(a_1w_1 + \dots + a_nw_n) = (ca_1)w_1 + \dots + (ca_n)w_n$. Hence,

$$T(cv) = cT(v).$$

Remark: In general, one can relax the condition of a basis in Theorem 2.1 to the condition of a linearly independent set. In such case, we have the following statement: Let V and W be vector spaces. Let $\{v_1, \dots, v_n\}$ be a linearly independent set in V and let $w_1, \dots, w_n \in W$. Then there exists a linear transformation $T : V \rightarrow W$ such that $T(v_i) = w_i$ for all i .

Example 2.4. (How is a linear transformation determined by its image on a basis?)

Let $v_1 = (1, 3)^T$ and $v_2 = (2, 4)^T$ in $\mathbb{R}^2$. Let $w_1 = (2, 3, 4)^T$ and $w_2 = (1, 0, 0)^T$. Since $\{v_1, v_2\}$ forms a basis for $\mathbb{R}^2$, there exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. Find $T((1, 1)^T)$.

Sol: Note that $(1, 1)^T = (2, 4)^T - (1, 3)^T$.

$$T((1, 1)^T) = T((2, 4)^T - (1, 3)^T) = T((2, 4)^T) - T((1, 3)^T) = (1, 0, 0)^T - (2, 3, 4)^T = -(1, 3, 4)^T.$$

Example 2.5. Let $v_1 = (1, 3)^T$ and $v_2 = (2, 6)^T$ in $\mathbb{R}^2$. Let $w_1 = (2, 3, 4)^T$ and $w_2 = (1, 0, 0)^T$. Does there exist a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$?

Sol: No. (Note that v_1 and v_2 are linearly dependent. So they do not form a basis

and we cannot use the theorem.) Suppose we have such linear transformation. Then $T((1, 3)^T) = (1, 3, 4)^T$ and by the scalar multiplication,

$$T((2, 6)^T) = T(2v_1) = 2T(v_1) = (4, 6, 8)^T \neq w_2.$$

This gives a contradiction.

Example 2.6. Determine if there is a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ satisfying the followings:

$$T((1, 0, -1)^T) = (1, 0)^T, \quad T((0, 1, 1)^T) = (0, 1)^T, \quad T((1, 1, 1)^T) = (1, 1)^T, \quad T((0, 0, -1)^T) = 0.$$

Sol: Note that

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 1 & 1 \end{pmatrix} = 1 \neq 0$$

and so the solution for the equation

$$a_1(1, 0, -1)^T + a_2(0, 1, 1)^T + a_3(1, 1, 1)^T = 0$$

has only the trivial solution. Then $\{(1, 0, -1)^T, (0, 1, 1)^T, (1, 1, 1)^T\}$ is linearly independent and so forms a basis for $\mathbb{R}^3$. Thus there exists a unique linear transformation T' satisfying

$$T'((1, 0, -1)^T) = (1, 0)^T, \quad T'((0, 1, 1)^T) = (0, 1)^T, \quad T'((1, 1, 1)^T) = (1, 1)^T.$$

Now we check if the remaining condition is also satisfied for T' .

Since

$$\begin{aligned} (0, 0, -1)^T &= (1, 0, -1)^T - (1, 1, 1)^T + (0, 1, 1)^T, \\ T'((0, 0, -1)^T) &= T'((1, 0, -1)^T - (1, 1, 1)^T + (0, 1, 1)^T) \\ &= (1, 0)^T - (1, 1)^T + (0, 1)^T \\ &= 0. \end{aligned}$$

Hence, T' satisfies the above four condition.

3. NULL SPACES AND RANGES

3.1. Null spaces and ranges.

Definition 3.1. Let $T : V \rightarrow W$ be a linear transformation.

- (1) The **null space** $N(T)$ is the set of all vectors x in V such that $T(x) = 0$. In other words, $N(T) = \{x \in V : T(x) = 0\}$.
- (2) The **range** $R(T)$ is the subset of W containing all the images of T . In other words, $R(T) = \{T(x) \in W : x \in V\}$.

Example 3.2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T((x, y)^T) = (0, 0)^T$. Then $N(T) = \mathbb{R}^2$ and $R(T) = 0$.

Example 3.3. Let V be a vector space. Then $N(\text{Id}_V) = 0$ and $R(\text{Id}_V) = V$.

Example 3.4. Let $T : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ given by

$$T(v) = \begin{pmatrix} I_n & 0_{n \times n} \\ 0_{n \times n} & 0_{n \times n} \end{pmatrix} v.$$

Then

$$\begin{aligned} N(T) &= \{(0, \dots, 0, x_{n+1}, \dots, x_{2n})^T : x_{n+1}, \dots, x_{2n} \in \mathbb{R}\}, \\ R(T) &= \{(x_1, \dots, x_n, 0, \dots, 0)^T : x_1, \dots, x_n \in \mathbb{R}\}. \end{aligned}$$

Theorem 3.5. (*Ranges and Null spaces form vector subspaces*) Let $T : V \rightarrow W$ be a linear transformation. Then $N(T)$ and $R(T)$ are linear subspaces of V and W respectively.

Proof. **$R(T)$:** Addition: For $v, w \in R(T)$, $T(v) = T(w) = 0$ and so $T(v + w) = T(v) + T(w) = 0$. Thus $v + w \in R(T)$.

Scalar Multiplication: For $c \in \mathbb{R}$ and $v \in V$, $T(cv) = cT(v) = 0$ and so $cv \in R(T)$. These check the conditions for subspaces.

$N(T)$: Addition: For $v', w' \in N(T)$, $v' = T(v)$ and $w' = T(w)$ for some $v, w \in V$. Hence, $v' + w' = T(v) + T(w) = T(v + w) \in N(T)$.

Scalar Multiplication: For $c \in \mathbb{R}$ and $v' = T(v) \in N(T)$, $cT(v) = T(cv) \in N(T)$. These check the conditions for subspaces. $\square$

We have the following result for computing the range (image) of a linear transformation:

Theorem 3.6. (*Finding a spanning set for $R(T)$*) Let $T : V \rightarrow W$ be a linear transformation. Let $\beta = \{v_1, \dots, v_m\}$ be a basis for V . Then

$$R(T) = \text{span} \{T(v_1), \dots, T(v_m)\}.$$

Proof. The inclusion $\supseteq$: For any $v' \in \text{span} \{T(v_1), \dots, T(v_m)\}$,

$$v' = a_1 T(v_1) + \dots + a_m T(v_m)$$

for some $a_1, \dots, a_m \in \mathbb{R}$. Then,

$$v' = T(a_1 v_1 + \dots + a_m v_m)$$

and so is in $R(T)$.

Another inclusion $\subseteq$: For $v' \in R(T)$, $v' = T(v)$ for some $v \in V$. By the definition of a basis, we have

$$v = b_1 v_1 + \dots + b_m v_m$$

for some $b_1, \dots, b_m \in \mathbb{R}$. Thus,

$$v' = T(b_1 v_1 + \dots + b_m v_m) = b_1 T(v_1) + \dots + b_m T(v_m) \in \text{span} \{T(v_1), \dots, T(v_m)\}.$$

This shows another inclusion. □

Example 3.7. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T((x, y, z)^T) = (x - y, y - z)^T$.

Compute $N(T)$ It is to solve the system of linear equations:

$$x - y = 0$$

$$y - z = 0$$

and so $N(T) = \{(t, t, t) \in \mathbb{R}^3 : t \in \mathbb{R}\}$.

Compute $R(T)$ Consider the standard basis $(1, 0, 0)^T, (0, 1, 0)^T, (0, 0, 1)^T$. Then, by Theorem 3.6, theorem.3.6,

$$\begin{aligned} R(T) &= \text{span}(\{T((1, 0, 0)^T), T((0, 1, 0)^T), T((0, 0, 1)^T)\}) \\ &= \text{span}(\{(1, 0)^T, (-1, 1)^T, (0, 1)^T\}) \\ &= \mathbb{R}^2. \end{aligned}$$

Exercise 3.8. (Dimension of the range bounded by dimension of domain) Let $T : V \rightarrow W$ be a linear transformation. Prove that

$$\dim(R(T)) \leq \dim(V).$$

Sol: Let $\beta = \{v_1, \dots, v_n\}$ be a basis for V . Since $R(T) = \text{span}(\{T(v_1), \dots, T(v_n)\})$, we can find a subset of linearly independent vectors in $\{T(v_1), \dots, T(v_n)\}$ to span $R(T)$. Thus $\dim(R(T)) \leq \dim(V)$.

We shall prove a stronger dimension formula in Theorem 4.2, theorem.4.2.

Remark 3.9. Let $T : V \rightarrow W$ be a linear transformation. Let $\{v_1, \dots, v_n\}$ be a basis for V . In general, $\{T(v_1), \dots, T(v_n)\}$ is **not** necessarily a basis for $R(T)$. For example:

- Let w be any vector in W . By Theorem 2.1, theorem.2.1, there exists a linear transformation $T : V \rightarrow W$ such that $T(v_1) = \dots = T(v_n) = w$. When $n \geq 2$, $\{T(v_1), \dots, T(v_n)\}$ is not a basis for $R(T)$.

4. DIMENSION FORMULA

4.1. Dimension formula. This section gives a precise connection of the dimension of the range and the null space of a linear transformation.

Definition 4.1. Let $T : V \rightarrow W$ be a linear transformation.

- (1) The **nullity** of T , denoted by **nullity**(T), is the dimension of $N(T)$; and
- (2) The **rank** of T , denoted by **rank**(T), is the dimension of $R(T)$.

Theorem 4.2. (*Dimension formula*) Let $T : V \rightarrow W$ be a linear transformation. Then

$$\text{nullity}(T) + \text{rank}(T) = \dim(V)$$

Example 4.3. (1) When $T = \text{id}_V : V \rightarrow V$ is the identity transformation, $N(T) = \{0\} = 0$ (the first zero is the zero vector and the second zero is the zero subspace), $R(T) = V$. Then $\dim(N(T)) + \dim(R(T)) = 0 + \dim V = \dim V$.

(2) $T_0 : V \rightarrow W$. In such case, $N(T_0) = V$, $R(T_0) = \{0\}$.

(3) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $T((x, y)^T) = x$. Then $N(T) = \text{span}\left(\left\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}\right)$ and $R(T) = \text{span}(\{(1)\})$, and so $\text{rank}(T) = 1$ and $\text{nullity}(T) = 1$.

Proof of Theorem 4.2

Let $r = \text{nullity}(T)$ and let $m = \dim(V)$.

Step 1: Complete a basis of V from a basis of $N(T)$:

Let $\{v_1, \dots, v_r\}$ be a basis for $\text{null}(T)$. We find vectors $w_1, \dots, w_{m-r}$ so that $v_1, \dots, v_r, w_1, \dots, w_{m-r}$ form a basis for V .

Step 2: Find a spanning set for $R(T)$ from Step 1

By Theorem 3.6

$$\begin{aligned} R(T) &= \text{span}\{T(v_1), \dots, T(v_r), T(w_1), \dots, T(w_{m-r})\} \\ &= \text{span}\{T(w_1), \dots, T(w_{m-r})\}. \end{aligned}$$

Step 3: Show the spanning set in Step 2 is linearly independent (and so forms a basis)

Suppose

$$a_1 T(w_1) + \dots + a_{m-r} T(w_{m-r}) = 0$$

and so $T(a_1 w_1 + \dots + a_{m-r} w_{m-r}) = 0$ and so $a_1 w_1 + \dots + a_{m-r} w_{m-r} \in N(T)$.

Hence,

$$a_1 w_1 + \dots + a_{m-r} w_{m-r} = b_1 v_1 + \dots + b_r v_r$$

for some $b_1, \dots, b_r \in \mathbb{R}$. Since $\{v_1, \dots, v_r, w_1, \dots, w_{m-r}\}$ forms a basis for V and so is linearly independent, we have that

$$a_1 = \dots = a_r = 0$$

This shows that $\{T(w_1), \dots, T(w_{m-r})\}$ is a set of linearly independent vectors. Hence, $\{T(w_1), \dots, T(w_{m-r})\}$ is a basis for $R(T)$.

Step 4: Finish the dimension formula

From Step 1, $\text{nullity}(T) = r$. From Step 3, $\text{rank}(T) = m - r$.

Hence $\text{nullity}(T) + \text{rank}(T) = r + (m - r) = \dim(V)$. $\square$

Example 4.4. (Find $R(T) \Rightarrow$ Compute $\text{nullity}(T)$ by dimension formula) Let $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ given by

$$T((x_1, x_2, x_3, x_4, x_5)^T) = (x_1 + x_4, x_2 + x_5, x_3 + x_4)^T.$$

Consider

$$T((1, 0, 0, 0, 0)^T) = (1, 0, 0)^T, \quad T((0, 1, 0, 0, 0)^T) = (0, 1, 0)^T$$

$$T((0, 0, 1, 0, 0)^T) = (0, 0, 1)^T, \quad T((0, 0, 0, 1, 0)^T) = (1, 0, 1)^T$$

$$T((0, 0, 0, 0, 1)^T) = (0, 1, 0)^T.$$

Hence, $R(T) = \mathbb{R}^3$. Then $\text{nullity}(T) = 5 - 3 = 2$.

Exercise 4.5. Compute $N(T)$ directly by solving a system of linear equations in Example 4.4. theorem.4.4.

Example 4.6. (Find $N(T) \Rightarrow$ Compute $\text{rank}(T)$ by dimension formula) Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ given by

$$T((x_1, x_2, x_3, x_4)^T) = (x_1 + x_2 - x_3, x_2 + x_3 + x_4)^T$$

Determine if T is surjective.

Sol: We shall approach by finding the nullity first. We first compute the space $N(T)$:

$$x_1 + x_2 - x_3 = 0$$

$$x_2 + x_3 + x_4 = 0$$

We obtain the augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array} \right).$$

The reduced row echelon form is then given by:

$$\left(\begin{array}{cccc|c} 1 & 0 & -2 & -1 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array} \right).$$

The null space $N(T)$ is given by the solution set of the system: set the free variables $x_3 = s$ and $x_4 = t$,

$$\{(2s + t, -s - t, s, t)^T : s, t \in \mathbb{R}\}.$$

Hence, a basis for $N(T)$ is given by:

$$\begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

It is explained in the previous chapter that $\text{nullity}(T) = 2$ and so by the dimension formula, $\text{rank}(T) = 2$. Then we must have $R(T) = \mathbb{R}^2$. This shows that T is surjective.

Exercise 4.7. Find $R(T)$ directly for Example 4.6 by using Theorem 3.6.

4.2. Compare with the matrix theory. For an $n \times m$ matrix A , let $L_A : \mathbb{R}^m \rightarrow \mathbb{R}^n$ defined by:

$$L_A(x) = Ax,$$

where Ax is the matrix multiplication. L_A is a linear transformation by Example 1.5.

Exercise 4.8. Show that:

- (1) $\text{rank}(A) = \text{rank}(L_A)$; and
- (2) $\dim(N(A)) = \text{nullity}(L_A)$.

It is explained that $\text{rank}(A)$ is the number of leading ones in the RREF of A ; and $\dim(N(A))$ is the number of free variables. Then **the number of leading ones plus the number of free variables = the number of columns = $\dim(\mathbb{R}^m)$** . This verifies the dimension formula.

5. INJECTIVITY

Recall that a function $f : X \rightarrow Y$ (between two sets X and Y) is said to be injective if for any $x, y \in X$, $f(x) = f(y)$ implies $x = y$.

Theorem 5.1. (*Injectivity determined by the null space*) Let $T : V \rightarrow W$ be a linear transformation. Then T is injective if and only if $N(T) = \{0\}$.

Proof. Only if direction

Suppose T is injective. Then $T(v) = 0 = T(0)$ implies $v = 0$.

If direction

Suppose $N(T) = \{0\}$. Let $v, w \in V$ such that $T(v) = T(w)$

$$\begin{aligned} T(v) - T(w) &= 0 \\ \Rightarrow T(v - w) &= 0 \\ \Rightarrow v - w &= 0 \\ \Rightarrow v &= w \end{aligned}$$

This implies T is injective. □

From the dimension formula (Theorem 4.2theorem.4.2), we also have:

Exercise 5.2. (Dimension of the range for an injective linear transformation) Let $T : V \rightarrow W$ be a linear transformation. Then the following statements are equivalent:

- T is injective;
- $N(T) = 0$;
- $\text{rank}(T) = \dim(V)$.

Example 5.3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that

$$T((x_1, x_2)^T) = (x_1 + x_2, x_1 - x_2, x_1)^T$$

Show that T is injective.

Sol: Let $(x_1, x_2)^T \in N(T)$. Then we have to solve:

$$\begin{aligned} x_1 + x_2 &= 0 \\ x_1 - x_2 &= 0 \\ x_1 &= 0 \end{aligned}$$

Solving the system of linear equations, we have $x_1 = x_2 = 0$ and so $N(T) = 0$. By Theorem 5.1theorem.5.1, T is injective.

Exercise 5.4. (Not possible to be an injective map when $\dim(V) > \dim(W)$) Let $T : V \rightarrow W$ be a linear transformation. Prove that if $\dim(V) > \dim(W)$, then T is not injective.

Sol: Since $R(T)$ is a subspace of W , $\text{rank}(T) \leq \dim(W)$. By the dimension formula, the first term is equal to $\text{nullity}(T)$ and so the inequality becomes

$$\text{nullity}(T) = \dim V - \text{rank}(T) \geq \dim V - \dim W > 0.$$

Thus $N(T) \neq 0$ and so T is not injective.

Exercise 5.5. Let $T : V \rightarrow W$ be a linear transformation. Show that T is surjective and injective if and only if $N(T) = 0$ and $\dim(V) = \dim(W)$. (A linear

transformation T is called an *isomorphism* if T is also bijective. We shall explain the relation to the invertible matrix in the next part.)

MATH 2101 LINEAR ALGEBRA I, FALL SEMESTER 2024
CHAPTER 4: LINEAR TRANSFORMATIONS (PART 2)

Key concepts:

- Matrix representations for a linear transformation
- Dictionary between operations on a matrix and operations on linear transformations
- Change of coordinate matrix and similar matrices

We shall consider vector spaces V and W to be finite-dimensional in this part.

1. MATRIX REPRESENTATIONS OF LINEAR TRANSFORMATIONS

1.1. Ordered basis.

Definition 1.1. An **ordered basis** is a basis with a given specific *order*. When we write an ordered basis $\beta = \{v_1, \dots, v_m\}$ for a vector space V , we mean v_1 to be the first one, v_2 to be the second one, v_m to be the last one, etc.

1.2. Matrix representation of a vector.

Definition 1.2. Let $\beta = \{v_1, \dots, v_m\}$ be an ordered basis for a vector space V . Recall that in Chapter 3, we have shown that for $v \in V$,

$$v = a_1 v_1 + \dots + a_m v_m$$

for some unique scalars $a_1, \dots, a_m \in \mathbb{R}$. Define the **coordinate vector of v with respect to β** as:

$$[v]_\beta = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}.$$

Example 1.3. Let $\beta = \{(1, 1)^T, (1, -1)^T\}$. Find $[(2, 4)^T]_\beta$.

Sol: $(2, 4)^T = a(1, 1)^T + b(1, -1)^T$ for some $a, b \in \mathbb{R}$. We have to solve the equation:

$$a + b = 2$$

$$a - b = 4$$

Thus, $a = 3$ and $b = -1$. Then

$$\left[\begin{pmatrix} 2 \\ 4 \end{pmatrix} \right]_\beta = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Example 1.4. Let $\beta = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be the standard ordered basis of $\mathbb{R}^n$. For $v \in \mathbb{R}^n$, $[v]_\beta = v$.

Example 1.5. Let $\mathcal{P}_3$ be the vector space of polynomials of degree at most 3. Let

$$\beta = \{1, 1 + x, x^2, x^2 + x^3\}.$$

be a basis for $\mathcal{P}_3$. Find

$$[1 + x + x^2 + x^3]_\beta.$$

Sol: $1 + x + x^2 + x^3 = 0(1) + (1)(1 + x) + (0)(x^2) + (1)(x^2 + x^3)$. Hence

$$[1 + x + x^2 + x^3]_\beta = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Exercise 1.6. Let β be an ordered basis for a vector space V . Let $n = \dim(V)$. Show that the map $T : V \rightarrow \mathbb{R}^n$ given by $T(v) = [v]_\beta$ is a linear transformation.

1.3. Matrix representation of a linear transformation. We now convert linear transformations to matrices:

Definition 1.7. Let $T : V \rightarrow W$ be a linear transformation. Let $\beta = \{v_1, \dots, v_m\}$ be an ordered basis for V and let $\gamma = \{w_1, \dots, w_n\}$ be an ordered basis for W . For each v_j ,

$$T(v_j) = a_{1j}w_1 + a_{2j}w_2 + \dots + a_{nj}w_n.$$

Define the **matrix representation of T relative to β and γ** to be the $n \times m$ -matrix given by:

$$[T]_\beta^\gamma = (a_{ij}).$$

If $\beta = \gamma$, we shall sometimes write $[T]_\beta$ for $[T]_\beta^\gamma$.

Definition 1.8. Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a linear transformation. Let β, β' be the standard basis for $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively. We shall call $[T]_\beta^{\beta'}$ the **standard matrix representation** of T .

Example 1.9. Find the standard matrix representation of the linear transformation:

$$T((x_1, x_2, x_3, x_4)^T) = (x_1 + x_3, x_2 + x_3, x_1 + x_4)^T.$$

Sol: Let β, β' be the standard basis for $\mathbb{R}^4$ and $\mathbb{R}^3$ respectively. Then

$$[T]_\beta^{\beta'} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Example 1.10. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^4$. Let

$$T((x_1, x_2)^T) = (x_1, x_1 + 2x_2, -x_1 + x_2, 3x_2)^T$$

Let β be the standard basis for $\mathbb{R}^2$ and let

$$\gamma = \{(1, 1, -1, 0)^T, (1, 3, 0, 3)^T, (0, 0, 1, 0)^T, (0, 0, 0, 1)^T\}$$

be a basis for $\mathbb{R}^4$. Find $[T]_\beta^\gamma$.

Sol: Step 1: Substitute the vectors in β to find the image under T :

$$T((1, 0)^T) = (1, 1, -1, 0)^T, \quad T((0, 1)^T) = (0, 2, 1, 3)^T.$$

Step 2: Find the linear combinations of $T(v_i)$: (sometimes it involves solving system of linear equations)

$$(1, 1, -1, 0)^T = \textcolor{red}{1}(1, 1, -1, 0)^T + \textcolor{red}{0}(1, 3, 0, 3)^T + \textcolor{red}{0}(0, 0, 1, 0)^T + \textcolor{red}{0}(0, 0, 0, 1)^T,$$

$$(0, 2, 1, 3)^T = \textcolor{orange}{-1}(1, 1, -1, 0)^T + \textcolor{orange}{1}(1, 3, 0, 3)^T + \textcolor{orange}{0}(0, 0, 1, 0)^T + \textcolor{orange}{0}(0, 0, 0, 1)^T$$

Step 3:

$$[T]_\beta^\gamma = \begin{pmatrix} \textcolor{red}{1} & \textcolor{orange}{-1} \\ \textcolor{red}{0} & \textcolor{orange}{1} \\ \textcolor{red}{0} & \textcolor{orange}{0} \\ \textcolor{red}{0} & \textcolor{orange}{0} \end{pmatrix}.$$

1.4. Relation to matrix multiplicatin. The matrix representation uses *matrix multiplication* to compute a linear transformation:

Theorem 1.11. (*Computing linear transformation via matrix multiplication*) Let $T : V \rightarrow W$ be a linear transformation. Let β, γ be ordered bases for V and W respectively. Then

$$[T(v)]_\gamma = [T]_\beta^\gamma [v]_\beta.$$

Proof. Let $\beta = \{v_1, \dots, v_m\}$ and $\gamma = \{w_1, \dots, w_n\}$.

Step 1: Working out matrix multiplication for the given basis

- $[T(v_i)]_\gamma$ is the i -th column of $[T]_\beta^\gamma$;
- $[v_i]_\beta = \mathbf{e}_i$ in $\mathbb{R}^m$.
- By matrix multiplication, $[T]_\beta^\gamma \mathbf{e}_i$ is the i -th column of $[T]_\beta^\gamma$.

Hence, $[T(v_i)]_\gamma = [T]_\beta^\gamma [v_i]_\beta$.

Step 2: Working out multiplications

Now, for $v \in V$, we write

$$v = a_1 v_1 + \dots + a_m v_m.$$

Hence,

$$\begin{aligned}
 [T(v)]_\gamma &= [T(a_1v_1 + \dots + a_mv_m)]_\gamma \\
 &= [a_1T(v_1) + \dots + a_mT(v_m)]_\gamma \\
 &= a_1[T(v_1)]_\gamma + \dots + a_m[T(v_m)]_\gamma \\
 &= a_1[T]_\beta^\gamma \mathbf{e}_1 + \dots + a_m[T]_\beta^\gamma \mathbf{e}_m \\
 &= [T]_\beta^\gamma (a_1\mathbf{e}_1 + \dots + a_m\mathbf{e}_m) \\
 &= [T]_\beta^\gamma \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}.
 \end{aligned}$$

□

Example 1.12. For example, in Example 1.10, if we do the multiplication:

$$[T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)]_\gamma = [T]_\beta^\gamma \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 - x_2 \\ x_2 \\ 0 \\ 0 \end{pmatrix}.$$

Note that $T((x_1, x_2)^T) = (x_1 - x_2)(1, 1, -1, 0)^T + x_2(1, 3, 0, 3)^T$, which recovers the original formula in Example 1.10.

2. COMPARING OPERATIONS BETWEEN LT AND MATRICES

2.1. Additions of linear transformations and additions of matrices.

Definition 2.1. (**Addition**) Let $T : V \rightarrow W$ and $T' : V \rightarrow W$ be two linear transformations. Define $T + T'$ be a function from V to W given by

$$(T + T')(v) = T(v) + T'(v).$$

Example 2.2. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \end{pmatrix}$ and let $B = \begin{pmatrix} 0 & 4 & -1 \\ 1 & 0 & 2 \end{pmatrix}$. Then

$$(L_A + L_B)(v) = L_A(v) + L_B(v) = Av + Bv = (A + B)v = L_{A+B}(v).$$

(Justify all equalities!)

Exercise 2.3. Show that $T + T'$ is a linear transformation from V to W .

Sol:

- (Addition) For $v_1, v_2 \in V$, $(T + T')(v_1 + v_2) = T(v_1 + v_2) + T'(v_1 + v_2) = T(v_1) + T'(v_1) + T(v_2) + T'(v_2) = (T + T')(v_1) + (T + T')(v_2)$.
- (Scalar multiplication) For $v \in V$ and $c \in \mathbb{R}$, $(T + T')(cv) = T(cv) + T'(cv) = cT(v) + cT'(v) = c(T(v) + T'(v)) = c(T + T')(v)$.

Theorem 2.4. (*Addition of LT=Addition of matrices*) Let $T, T' : V \rightarrow W$ be two linear transformations. Let β and γ be ordered bases for V and W respectively. Then

$$[T + T']_{\beta}^{\gamma} = [T]_{\beta}^{\gamma} + [T']_{\beta}^{\gamma}$$

Proof. For any $v \in V$,

$$\begin{aligned} [T + T']_{\beta}^{\gamma}[v]_{\beta} &= [(T + T')(v)]_{\gamma} \\ &= [T(v) + T'(v)]_{\gamma} \\ &= [T(v)]_{\gamma} + [T'(v)]_{\gamma} \\ &= [T]_{\beta}^{\gamma}[v]_{\beta} + [T']_{\beta}^{\gamma}[v]_{\beta} \\ &= ([T]_{\beta}^{\gamma} + [T']_{\beta}^{\gamma})[v]_{\beta} \end{aligned}$$

The first and forth equalities follow from Theorem 1.11; the second equality follows from the definition of $T + T'$; and the last equality follows from the distributive property of addition of two matrices. $\square$

2.2. Compositions of linear transformations and matrix multiplications.

Definition 2.5. Let $T : V \rightarrow W$ and $T' : W \rightarrow X$ be linear transformations. Let β, γ and α be ordered bases for V, W and X respectively. Define the composition to be:

$$(T' \circ T)(v) = T'(T(v)).$$

Exercise 2.6. Show that $T' \circ T$ is a linear transformation.

Theorem 2.7. (*Composition of LT = Matrix multiplication*) Let $T : V \rightarrow W$ and $T' : W \rightarrow X$ be linear transformations. Let β, γ and α be ordered bases for V, W and X respectively. Then

$$[T' \circ T]_{\beta}^{\alpha} = [T']_{\gamma}^{\alpha} [T]_{\beta}^{\gamma}.$$

Proof. Let $v \in V$. By Theorem 1.11, we have

$$[T' \circ T]_{\beta}^{\alpha}[v]_{\beta} = [T' \circ T(v)]_{\alpha}.$$

We also have:

$$[T']_{\gamma}^{\alpha} [T]_{\beta}^{\gamma} [v]_{\beta} = [T']_{\gamma}^{\alpha} [T(v)]_{\gamma} = [T'(T(v))]_{\alpha} = [T' \circ T(v)]_{\alpha}.$$

Thus, the two expressions agree for any v and so the two matrices are the same. $\square$

Example 2.8. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}$ given by $T((x_1, x_2, x_3, x_4)^T) = x_1 + x_2 + x_3 + x_4$. Let $U : \mathbb{R} \rightarrow \mathbb{R}^4$ given by $U(x) = (x, 2x, 3x, 4x)^T$. Let β and γ be the standard

ordered bases for $\mathbb{R}^4$ and $\mathbb{R}$ respectively. Use matrix multiplication to find $U \circ T$.

Sol: Then

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix}, \quad [U]_{\gamma}^{\beta} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

Note that $(U \circ T)(x_1, x_2, x_3, x_4) = (x_1 + x_2 + x_3 + x_4, 2x_1 + 2x_2 + 2x_3 + 2x_4, 3x_1 + 3x_2 + 3x_3 + 3x_4, 4x_1 + 4x_2 + 4x_3 + 4x_4)$. On the other hand,

$$[U \circ T]_{\beta}^{\beta} = [U]_{\gamma}^{\beta} [T]_{\beta}^{\gamma} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{pmatrix}.$$

Hence,

$$U \circ T((x_1, x_2, x_3, x_4)^T) = (x_1 + x_2 + x_3 + x_4, 2x_1 + 2x_2 + 2x_3 + 2x_4, 3x_1 + 3x_2 + 3x_3 + 3x_4, 4x_1 + 4x_2 + 4x_3 + 4x_4)^T.$$

3. INVERTIBILITY OF A LINEAR TRANSFORMATION

3.1. Invertibility.

- Definition 3.1.** • Let $T : V \rightarrow W$ be a linear transformation. We say that T is **invertible** if there exists a function $T^{-1} : W \rightarrow V$ such that $T^{-1} \circ T = \text{Id}_V$ and $T \circ T^{-1} = \text{Id}_W$. We shall call T^{-1} to be the **inverse** of T .
- An invertible linear transformation $T : V \rightarrow W$ is sometimes called an **isomorphism**.
 - We say that V is **isomorphic** to W if there exists an invertible linear transformation $T : V \rightarrow W$.

Exercise 3.2. The inverse of a linear transformation is still a linear transformation.

Exercise 3.3. Give an example of a linear transformation $T : V \rightarrow W$ such that $S \circ T = \text{Id}_V$, but T is not invertible.

We first have some properties for an invertible linear transformation:

Theorem 3.4. (*Criteria for invertibility of LT*) Let $T : V \rightarrow W$ be an invertible linear transformation. Then

- $\dim(V) = \dim(W)$;
- T is injective;
- T is surjective.

Proof. Since T has an inverse, T is injective and surjective.

Use T is surjective: Since T is surjective, $R(T) = W$ and so $\text{rank}(T) = \dim(W)$.

Use T is injective: Since T is injective, $N(T) = \{0\}$ and so $\text{nullity}(T) = 0$.

Use the dimension formula

$$\dim(W) = \text{rank}(T) = \text{rank}(T) + \text{nullity}(T) = \dim(V). \quad \square$$

Exercise 3.5. (Converse of Theorem 3.4) Let $T : V \rightarrow W$ be a linear transformation. Show that

- (a) If T is injective and $\dim(V) = \dim(W)$, then T is invertible.
- (b) If T is surjective and $\dim(V) = \dim(W)$, then T is invertible.

Example 3.6. (First check on the invertibility of a linear transformation by using dimensions) Determine if the following linear transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad T((x, y, z)^T) = (x + y, y - z)^T$$

is invertible.

Sol: Since $\dim(\mathbb{R}^3) \neq \dim(\mathbb{R}^2)$, T cannot be invertible.

Example 3.7. Determine if the following linear transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad T((x, y)^T) = (x, y, 0)^T$$

is invertible.

Sol: Since $\dim(\mathbb{R}^2) \neq \dim(\mathbb{R}^3)$, T cannot be invertible.

Theorem 3.8. (Invertible $LT=Invertible$ matrix) Let $T : V \rightarrow W$ be an invertible linear transformation. Let β, γ be ordered bases for V and W respectively. Then T is invertible if and only if $[T]_\beta^\gamma$ is invertible.

Proof. Let $m = \dim(V)$.

Only if direction:

Suppose T is invertible. Then $m = \dim(W)$ (by the previous theorem). Then $[T^{-1}]_\gamma^\beta [T]_\beta^\gamma = [T^{-1} \circ T]_\beta^\beta = [\text{Id}_V]_\beta^\beta = I_m$. Similarly, we have $[T]_\beta^\gamma [T^{-1}]_\gamma^\gamma = [\text{Id}_W]_\gamma^\gamma = I_m$.

If direction:

Suppose $[T]_\beta^\gamma$ is invertible. Let $A = [T]_\beta^\gamma$. For $w \in W$,

$$A^{-1}[w]_\gamma = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}$$

for some $a_1, \dots, a_m \in \mathbb{R}$. Write the ordered basis $\beta = \{v_1, \dots, v_m\}$.

(Construct the inverse) Define

$$U(w) = a_1 v_1 + \dots + a_m v_m.$$

(**Check the inverse**) By the definition of matrix representation, $[U]_\gamma^\beta = A^{-1}$ and so, by

$$[U \circ T]_\beta^\beta = [U]_\gamma^\beta [T]_\beta^\gamma = A^{-1}A = I_m.$$

By Theorem 1.11, $U \circ T = \text{Id}_V$. Checking the right inverse is similar. $\square$

The above proof also gives the following:

Corollary 3.9. *Let $T : V \rightarrow W$ be an invertible linear transformation. Let β, γ be bases for V and W respectively. Then*

$$[T^{-1}]_\gamma^\beta = ([T]_\beta^\gamma)^{-1}.$$

Example 3.10. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by

$$T((x, y, z)^T) = (x + y - z, y + z, x + z)^T.$$

Determine if T is invertible by computing the determinant of its matrix representation.

Sol: Let β be the standard basis for $\mathbb{R}^3$. Then, we have to compute the matrix representation

$$[T]_\beta.$$

Note that

$$[T(\mathbf{e}_1)]_\beta = [(1, 0, 1)^T]_\beta = \mathbf{e}_1 + \mathbf{e}_3,$$

$$[T(\mathbf{e}_2)]_\beta = [(1, 1, 0)^T]_\beta = \mathbf{e}_1 + \mathbf{e}_2,$$

$$[T(\mathbf{e}_3)]_\beta = [(-1, 1, 1)^T]_\beta = -\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3.$$

Thus, $[T]_\beta = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. Now, we compute the determinant of $[T]_\beta$. Thus,

$$\begin{aligned} \det([T]_\beta) &= \det \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \\ &= (-1)^{1+1}(1)\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} + (-1)^{3+1}(1)\det \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= 1 + 2 \\ &= 3 \neq 0 \end{aligned}$$

Hence, by Theorem 3.8, T is invertible.

Exercise 3.11. Find $T^{-1}((x, y, z)^T)$ in Example 3.10.

Sol: Note that

$$[T^{-1}]_\beta = [T]_\beta^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 1 \end{pmatrix}.$$

Thus $T^{-1}((x, y, z)^T) = \frac{1}{3}(x - y + 2z, x + 2y - z, -x + y + z)^T$.

3.2. Finding rank of a linear transformation.

Theorem 3.12. (*Rank of LT = Rank of matrix*) Let T be a linear transformation from V to W . Let β and γ be ordered bases for V and W respectively. Then

$$\text{rank}(T) = \text{rank}([T]_{\beta}^{\gamma}).$$

Proof. (Sketch) One starts with a basis $\{v_1, \dots, v_r\}$ for $N(T)$. Then complete to a basis $\{v_1, \dots, v_r, v_{r+1}, \dots, v_n\}$ for V , denoted by β' .

Now $T(v_{r+1}), \dots, T(v_n)$ form a basis for $R(T)$. Complete $\{T(v_{r+1}), \dots, T(v_n), w_1, \dots, w_s\}$ to a basis for W , denoted by γ' .

One checks that

$$[T]_{\beta'}^{\gamma'} = \begin{pmatrix} 0_{(n-r) \times r} & I_{n-r} \\ 0_{s \times r} & 0_{s \times (n-r)} \end{pmatrix}.$$

To complete the proof, one needs to change back to the given bases β and γ , which one needs the change of coordinate matrices below. $\square$

Example 3.13. (*Using matrix representation to find rank and nullity*) Find the nullity and rank of T in Example 1.10 using a matrix representation.

Sol. The standard matrix representation A of T is

$$\begin{pmatrix} 1 & 0 \\ 1 & 2 \\ -1 & 1 \\ 0 & 3 \end{pmatrix}$$

and so its reduced row echelon form is:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Hence, the rank of A is 2. By Theorem 3.12, $\text{rank}(T) = 2$ and by the dimension formula, $\text{nullity}(T) = 2 - 2 = 0$.

4. CHANGE OF COORDINATE MATRIX

4.1. Motivation. Given an equation of an ellipse:

$$2x^2 + 3xy + 5y^2 = 1,$$

we can change to the new coordinate given by:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In this new coordinates, we have $x'^2 + y'^2 = 1$ and we can visualize it as a circle.

4.2. Change of coordinate matrix.

Definition 4.1. Let β, β' be two ordered bases for a linear vector space V . The matrix $[\text{Id}_V]_{\beta}^{\beta'}$ is called a **change of coordinate matrix** from β to β' .

Example 4.2. Let $\beta = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\}$ and $\beta' = \{\mathbf{e}_1, \mathbf{e}_2\}$. Then $[\text{Id}_{\mathbb{R}^2}]_{\beta}^{\beta'} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$.

Example 4.3. Let $\beta = \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$, and let $\beta' = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$. Find the change of coordinate matrix from β to β' .

Sol: We have

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Similarly,

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Then, the change of coordinate matrix from β to β' is

$$\begin{pmatrix} 0 & \frac{1}{3} \\ -1 & -\frac{1}{3} \end{pmatrix}$$

Theorem 4.4. (*Properties of change of coordinate matrix*) Let β, β' be two ordered bases for a vector space V . Let $Q = [\text{Id}_V]_{\beta}^{\beta'}$. Then

- (1) Q is invertible;
- (2) For any $v \in V$, $[v]_{\beta'} = Q[v]_{\beta}$.

Proof. (1) follows from Theorem 3.8. (2) follows from Theorem 1.11.

□

Example 4.5. From Example 4.2, we have

$$[\text{Id}_{\mathbb{R}^2}]_{\beta'}^{\beta} = ([\text{Id}_{\mathbb{R}^2}]_{\beta}^{\beta'})^{-1} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}.$$

4.3. Similar matrices. One can also change the matrix representation from one basis to another one by the following theorem:

Theorem 4.6. (*Matrix representation under change of coordinates*) Let β, β' be two ordered bases for a vector space V . Let $T : V \rightarrow V$ be a linear transformation. Let $Q = [\text{Id}_V]_{\beta}^{\beta'}$. Then

$$[T]_{\beta} = Q^{-1}[T]_{\beta'}Q.$$

Proof. Let $v \in V$. Let $Q = [\text{Id}_V]_{\beta}^{\beta'}$.

Then

$$[T]_{\beta'}[v]_{\beta'} = [T(v)]_{\beta'}.$$

On the other hand,

$$Q^{-1}[T]_{\beta'}Q[v]_{\beta'} = Q^{-1}[T]_{\beta'}[v]_{\beta'} = Q^{-1}[T(v)]_{\beta'} = [T(v)]_{\beta}.$$

Two matrices are determined by the multiplication with all column vectors, and hence,

$$Q^{-1}[T]_{\beta'}Q[v]_{\beta'} = [T]_{\beta'}.$$

□

Definition 4.7. Two $n \times n$ matrices A and B are said to be **similar** if there exists an invertible matrix Q such that $Q^{-1}AQ = B$.

Similar matrices share some common properties. One is the following:

Exercise 4.8. Let A and B be two similar $n \times n$ matrices. Then $\det(A) = \det(B)$.

Exercise 4.9. (1) Show that if A is an $n \times n$ matrix similar to I_n , then $A = I_n$.

(2) Find a 2×2 matrix A which is similar to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and is not a diagonal matrix.

(3) Let A and B be two $n \times n$ matrices. If A and B are similar, then $\text{rank}(A) = \text{rank}(B)$.

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CHAPTER 5: DIAGONALIZATION

Goals in this chapter:

- The meaning of diagonalizability, eigenvalues, eigenvectors and their relations.
- How to compute eigenvalues from characteristic polynomials?
- The eigenspaces, algebraic multiplicity and geometric multiplicity
- Some applications: taking high powers on diagonalizable matrices

1. EIGENVALUES AND EIGENVECTORS

1.1. Motivation. Let D be an $n \times n$ diagonal matrix. It defines a linear transformation $L_D : \mathbb{R}^n \rightarrow \mathbb{R}^n$. The linear transformation takes the standard vectors $\mathbf{e}_i$ to $a_i \mathbf{e}_i$.

We now consider a matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix},$$

which is not diagonal. We consider a basis

$$\beta = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

Then the matrix representation $[L_A]_\beta$ is

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix},$$

which is diagonal. The goal of this section is to find a nice basis so that its matrix representation is a diagonal matrix.

1.2. Diagonalizability.

Definition 1.1. An $n \times n$ matrix A is said to be **diagonalizable** if there exists an invertible $n \times n$ -matrix Q such that $Q^{-1}AQ$ is a diagonal matrix. (In other words, a diagonalizable matrix A is *similar* to a diagonal matrix.)

Example 1.2. Any diagonal matrix D is diagonalizable. Since $I_n^{-1}DI_n = D$. For example,

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

is diagonalizable.

Example 1.3. Let $Q = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}$. Let $D = \text{diag}(3, 4)$. Since Q is invertible, we obtain a diagonalizable matrix:

$$QDQ^{-1} = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix} = \begin{pmatrix} -4 & -2 \\ 28 & 11 \end{pmatrix}.$$

Exercise 1.4. Let A be an invertible $n \times n$ matrix. Prove that A is diagonalizable if and only if A^{-1} is diagonalizable.

Sol: Suppose A is diagonalizable. Then there exists an invertible Q such that $Q^{-1}AQ = D$ for some diagonal matrix D . Then

$$(Q^{-1}AQ)^{-1} = D^{-1}$$

and so $Q^{-1}A^{-1}Q = D^{-1}$. Then A^{-1} is diagonalizable since D^{-1} is also a diagonal matrix.

2. EIGENVALUES AND EIGENVECTORS

We shall now introduce some machinery to *determine if a matrix is diagonalizable and how to diagonalize a matrix*.

Definition 2.1. Let A be an $n \times n$ matrix. A *non-zero* vector $v \in \mathbb{R}^n$ is said to be an **eigenvector** of A if there exists a scalar $\lambda \in \mathbb{R}$ such that $Av = \lambda v$. We shall call λ to be the **eigenvalue** corresponding to the eigenvector v .

Example 2.2. Let

$$A = \begin{pmatrix} 1 & 2 \\ 6 & 5 \end{pmatrix}.$$

Then

$$A \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 21 \end{pmatrix} = 7 \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Then $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is an eigenvector of A and 7 is the eigenvalue corresponding to $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

Example 2.3. (**Eigenvectors and eigenvalues of diagonal matrices**) Let $D = \text{diag}(a_1, \dots, a_n)$ be a $n \times n$ diagonal matrix. Then the vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the eigenvectors of D since

$$D\mathbf{e}_i = a_i\mathbf{e}_i$$

Then each a_i is an eigenvalue associated to the eigenvector $\mathbf{e}_i$.

Exercise 2.4. Let A be an invertible matrix. If λ is an eigenvalue of A , then $\lambda \neq 0$.

Exercise 2.5. Let A be an invertible $n \times n$ matrix. Then v is an eigenvector of A if and only if v is also an eigenvector of A^{-1} .

Sol. Suppose v is an eigenvector of A . Then $Av = \lambda v$ for some λ in $\mathbb{R}$. By previous exercise, $\lambda \neq 0$. Then $v = A^{-1}(\lambda v)$ and so $A^{-1}v = \lambda^{-1}v$. By definition, v is an eigenvector for A^{-1} . This proves the only if direction.

For the if direction, the proof is similar to the only if direction by replacing A with A^{-1} in above arguments.

Exercise 2.6. Let A be an invertible $n \times n$ matrix. Show that λ is an eigenvalue of A if and only if λ^{-1} is an eigenvalue of A^{-1} .

We shall come back to the question how to find eigenvalues and eigenvectors. We first see how to address the diagonalization problem.

Theorem 2.7. (*Diagonalizable via eigenvectors*) Let A be an $n \times n$ matrix. Then A is diagonalizable if and only if there exists a basis $\beta = \{v_1, \dots, v_n\}$ of $\mathbb{R}^n$ such that each v_i is also an eigenvector of A .

Proof. Only if direction

Suppose A is diagonalizable. Then $A = QDQ^{-1}$ for a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and an invertible matrix Q .

Step 1: Construct eigenvectors

Since

$$A(Q\mathbf{e}_i) = (QDQ^{-1})(Q\mathbf{e}_i) = QD\mathbf{e}_i = Q(\lambda_i\mathbf{e}_i) = \lambda_i(Q\mathbf{e}_i).$$

and $Q\mathbf{e}_i \neq 0$, each $Q\mathbf{e}_i$ is an eigenvector.

Step 2: Check eigenvectors form a basis

We now check those eigenvectors form a basis. Since Q is invertible and $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is linearly independent, $\{Q\mathbf{e}_1, \dots, Q\mathbf{e}_n\}$ is also linearly independent. We have shown in Chapter 3 that any n linearly independent vectors form a basis for $\mathbb{R}^n$.

If direction

We now prove the if direction. Suppose there exists an ordered basis $\beta = \{v_1, \dots, v_n\}$ of $\mathbb{R}^n$ such that those v_i are eigenvectors. Let λ_i be the corresponding eigenvalue of v_i . Let $Q = (v_1 \ v_2 \ \dots \ v_n)$. Then

$$Q^{-1}AQ\mathbf{e}_i = Q^{-1}Av_i = Q^{-1}(\lambda_i v_i) = \lambda_i Q^{-1}v_i = \lambda_i \mathbf{e}_i$$

for all i . This implies that $Q^{-1}AQ$ is a diagonal matrix. Hence, A is diagonalizable.

□

The following theorem tells you how to diagonalize a matrix from a set of eigenvectors which forms a basis:

Theorem 2.8. (*Diagonalizable via eigenvectors*) Let $\beta = \{v_1, \dots, v_n\}$ be a set of eigenvectors of an $n \times n$ -matrix A . Let $Q = (v_1 \ v_2 \ \dots \ v_n)$ be an $n \times n$ matrix. If $\{v_1, \dots, v_n\}$ is linearly independent (or forms a basis for $\mathbb{R}^n$), then $Q^{-1}AQ$ is a diagonal matrix with the diagonal entries equal to the eigenvalues of A .

Proof. One has to investigate the if part of the proof of Theorem 2.7. □

Exercise 2.9. Let $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$. Verify that $(-3, 1)^T$ and $(1, 1)^T$ are the eigenvectors of A . Then diagonalize A .

Sol: Let $Q = \begin{pmatrix} -3 & 1 \\ 1 & 1 \end{pmatrix}$. Then

$$Q^{-1}AQ = \frac{-1}{4} \begin{pmatrix} 1 & -1 \\ -1 & -3 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} -3 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}.$$

Exercise 2.10. Let A be an $n \times n$ matrix. Suppose A has only one eigenvalue equal to λ . Then A is diagonalizable if and only if $A = \lambda I_n$.

3. CHARACTERISTICS POLYNOMIALS

3.1. Finding eigenvalues.

Definition 3.1. Let A be an $n \times n$ matrix. The polynomial $\det(A - tI_n)$ in the variable t is called the **characteristics polynomial** of A .

Example 3.2. Find the characteristics polynomial of $A = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$.

Sol: $\det(A - tI_2) = \det \begin{pmatrix} 1-t & 3 \\ 2 & 1-t \end{pmatrix} = (1-t)^2 - 6 = t^2 - 2t - 5$.

Example 3.3. Let A be a diagonal matrix. Suppose the entries in the diagonal are $a_1, \dots, a_n$. Then the characteristics polynomial of A is:

$$\det(A - tI_n) = (a_1 - t) \dots (a_n - t).$$

Exercise 3.4. (*Same characteristics polynomials for similar matrices*) Let A be an $n \times n$ matrix. Let Q be an invertible $n \times n$ matrix. Then the characteristics polynomial of A is equal to the characteristics polynomial of $Q^{-1}AQ$.

Theorem 3.5. (*Eigenvalues as roots for the characteristic polynomial*) Let A be an $n \times n$ matrix. Then a scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof. Only if direction

Suppose λ is an eigenvalue of A . Then, there exists a non-zero vector $v \in \mathbb{R}^m$ such that $(A - \lambda I_n)v = 0$. In other words, there exists a non-trivial solution for $(A - \lambda I_n)x = 0$ and so $\det(A - \lambda I_n) = 0$.

If direction

$\det(A - \lambda I_n) = 0$ and so there is a non-trivial solution $v \in \mathbb{R}^m$ such that $(A - \lambda I_n)v = 0$ and so $Av = \lambda v$. Hence, v is an eigenvector and λ is its associated eigenvalue. $\square$

Example 3.6. Let $A = \lambda I_n$. Then the characteristics polynomial of A is $(\lambda - t)^n$. So A has only one eigenvalue, which is λ .

Example 3.7. Find all the eigenvalues of the matrix

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}.$$

Sol: We first find the characteristics polynomial of A :

$$\begin{aligned} \det(A - tI_3) &= \det \begin{pmatrix} 2-t & 0 & -1 \\ 0 & 2-t & 0 \\ -1 & 0 & 2-t \end{pmatrix} \\ &= (2-t) \det \begin{pmatrix} 2-t & -1 \\ -1 & 2-t \end{pmatrix} \\ &= (2-t)[(2-t)^2 - 1] \\ &= (2-t)(t^2 - 4t + 3) \\ &= (2-t)(t-3)(t-1) \end{aligned}$$

Hence, the roots for $\det(A - tI_3)$ are 1, 2, 3. Hence, all the eigenvalues are 1, 2, 3.

Example 3.8. (c.f. Exercise 2.10) Let $A = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}$. Show that A is not diagonalizable.

Sol: The characteristics polynomial of A is

$$\det(A - tI_2) = \det \begin{pmatrix} -1-t & 1 \\ -4 & 3-t \end{pmatrix} = (1-t)^2.$$

Then the only eigenvalue of A is 1. Thus if A is diagonalizable,

$$Q^{-1}AQ = I_2$$

for some invertible matrix Q . This implies that $A = I_2$, giving a contradiction.

Here is one important property for characteristics polynomials:

Theorem 3.9. (*Number of distinct eigenvalues bounded by n*) Let A be an $n \times n$ matrix. Then

- the characteristics polynomial of A has degree n with the leading coefficient $(-1)^n$;
- A has at most n eigenvalues.

Proof. (Main idea of the first one) One uses the cofactor expansion to compute the determinant $\det(A - tI_n)$.

The second one follows from the first one and Theorem 3.5 since there are at most n roots for a polynomial of degree n . $\square$

Remark 3.10. A polynomial with coefficients in $\mathbb{R}$ may have complex numbers as solutions. In such case, one shall diagonalize to have a matrix with entries in complex numbers. We shall not go deep for complex numbers in this course. One may assume that *all roots of the characteristics polynomial of a square matrix are real in this course.*

3.2. Examples on finding eigenvalues and eigenvectors.

Example 3.11. Let $A = \begin{pmatrix} 5 & -2 \\ 2 & 0 \end{pmatrix}$. Diagonalize A .

Sol:

Step 1: Find eigenvalues by solving the characteristics polynomial

$\det(A - tI_2) = \det \begin{pmatrix} 5-t & -2 \\ 2 & -t \end{pmatrix} = -t(5-t) - 4 = t^2 - 5t + 4 = (t-4)(t-1)$. Then A has 2 distinct eigenvalues.

Step 2: Find eigenvectors for a known eigenvalues by solving a homogeneous system

To find the eigenvector corresponding to 1, we solve $(A - I_2)x = 0$, that is $\begin{pmatrix} 4 & -2 \\ 2 & -1 \end{pmatrix} x =$

0. Then an eigenvector is $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

To find the eigenvector corresponding to 4, we solve $(A - 4I_2)x = 0$, that is $\begin{pmatrix} 1 & -2 \\ 2 & -4 \end{pmatrix} x = 0$. Then an eigenvector is $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Step 3: Diagonalize by Theorem 2.8

Let $Q = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. Then $Q^{-1}AQ = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$.

Example 3.12. Let $A = \begin{pmatrix} 1 & -4 & 3 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. Then $\det(A - tI_3) = (1-t)(-1-t)(2-t)$.

Hence, there are three eigenvalues of A , namely 1, -1, 2.

To find an eigenvector corresponding to 1, we solve $(A - I_3)x = 0$ and so an eigenvector is $(1 \ 0 \ 0)^T$.

To find an eigenvector corresponding to -1 , we solve $(A + I_3)x = 0$, that is

$$\begin{pmatrix} 2 & -4 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} x = 0. \text{ Thus } \begin{pmatrix} 2 & 1 & 0 \end{pmatrix}^T \text{ is an eigenvector.}$$

To find an eigenvector corresponding to 2 , we solve $(A - 2I_3)x = 0$, that is

$$\begin{pmatrix} -1 & -4 & 3 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{pmatrix} x = 0. \text{ Then } \begin{pmatrix} 3 & 0 & 1 \end{pmatrix}^T \text{ is an eigenvector.}$$

$$\text{Let } Q = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \text{ Then}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

4. LINEAR INDEPENDENCE OF EIGENVECTORS

4.1. Linear independence of eigenvectors. In Theorem 2.8, we want to find eigenvectors to form a basis. We now address the linear independence of eigenvectors:

Theorem 4.1. (*Linear independence for eigenvectors corresponding to distinct eigenvalues*) Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of an $n \times n$ matrix A . Let $v_1, \dots, v_k$ be eigenvectors corresponding to the eigenvalues $\lambda_1, \dots, \lambda_k$ respectively. Then $v_1, \dots, v_k$ are linearly independent.

Proof. Let $v_1, \dots, v_k$ be the eigenvectors of A with the associated eigenvalues $\lambda_1, \dots, \lambda_k$ respectively. We shall show **inductively** on k that $\{v_1, \dots, v_k\}$ is linearly independent. If $k = 1$, it follows from definitions that v_1 is non-zero.

Now, we consider the equation:

$$a_1 v_1 + \dots + a_k v_k = 0.$$

Multiplying A on the left, we have:

$$a_1 \lambda_1 v_1 + \dots + a_k \lambda_k v_k = 0.$$

Then, $a_1 \lambda_1 v_1 + \dots + a_k \lambda_k v_k = 0$. Then, subtracting with the equation:

$$a_1 \lambda_k v_1 + \dots + a_k \lambda_k v_k = 0,$$

we have $a_1(\lambda_1 - \lambda_k)v_1 + \dots + a_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} = 0$. By induction, we have:

$$a_i(\lambda_i - \lambda_k) = 0$$

for $i \leq k - 1$. Since $\lambda_i \neq \lambda_k$ for all $i \leq k - 1$, we have that $a_1 = \dots = a_{k-1} = 0$. This then forces $a_k = 0$ since $v_k \neq 0$. This shows that $v_1, \dots, v_k$ are linearly independent. $\square$

Corollary 4.2. (*a diagonalizable criteria*) Let A be an $n \times n$ matrix. If A has n distinct eigenvalues, then A is diagonalizable.

Proof. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues. Then, for each eigenvalue λ_i , we have an eigenvector v_i . By Theorem 4.1, $v_1, \dots, v_n$ are linearly independent. Since $\dim(\mathbb{R}^n) = n$, $v_1, \dots, v_n$ form a basis for $\mathbb{R}^n$. By Theorem 2.7, A is diagonalizable. $\square$

5. ALGEBRAIC AND GEOMETRIC MULTIPLICITIES

5.1. Algebraic multiplicities.

Definition 5.1. Let A be an $n \times n$ matrix. Let λ be an eigenvalue of A . The **algebraic multiplicity** of λ is the largest integer k such that $(t - \lambda)^k$ divides $\det(A - tI_n)$.

Example 5.2. Find the algebraic multiplicity of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Sol: The characteristics polynomial of A is

$$\det(A - tI_t) = \det \begin{pmatrix} 1-t & 2 & 3 & 4 \\ 0 & 2-t & 0 & 3 \\ 0 & 0 & 1-t & 0 \\ 0 & 0 & 0 & -1-t \end{pmatrix} = (1-t)(2-t)(1-t)(-1-t).$$

Then

- A.M. (algebraic multiplicity) of $1 = 2$;
- A.M. of $2 = 1$;
- A.M. of $-1 = 1$.

Exercise 5.3. (*Sum of algebraic multiplicities equal to n*) Let A be an $n \times n$ matrix. Show that the sum of algebraic multiplicities of all eigenvalues is equal to n .

Exercise 5.4. Let A be an $n \times n$ matrix. Let λ be an eigenvalue of A .

- (a) What is the maximum possible algebraic multiplicity of λ in general? What is the minimum possible algebraic multiplicity of λ in general?
- (b) Suppose we know that A has exactly three distinct eigenvalues. What is the maximum possible algebraic multiplicity of λ ?

Sol: For (a), the maximum possible algebraic multiplicity of λ is n since the characteristic polynomial has degree n . For example, take $A = I_n$ and the algebraic multiplicity of 1 is n . The minimum possible algebraic multiplicity of λ is 1.

For (b), the maximum possible algebraic multiplicity of λ is $n - 2$.

5.2. Geometric multiplicities. On the other hand, we have:

Definition 5.5. Let A be an $n \times n$ matrix. Let λ be an eigenvalue of A . Define

$$E_\lambda = \{v \in \mathbb{R}^n : Av = \lambda v\}.$$

The set is called the **eigenspace** corresponding to the eigenvalue λ . Since E_λ is the solution set of a homogeneous system of linear equations, E_λ is a vector subspace. The dimension of E_λ is called the **geometric multiplicity** of λ (for A). The geometric multiplicity of an eigenvalue λ is non-zero by the definition of an eigenvector.

Example 5.6. Let

$$A = \begin{pmatrix} 3 & 0 & -2 \\ 2 & 1 & -2 \\ 2 & 0 & -1 \end{pmatrix}.$$

Suppose we know that 1 is an eigenvalue of A . Find the geometric multiplicity of 1 for A .

Sol: Note that the eigenspace

$$E_1 = \{v \in \mathbb{R}^3 : (A - I_3)v = 0\}.$$

The key is to **solve the solution set of the homogeneous system $(A - I_3)x = 0$** . That is, to consider the augmented matrix:

$$\left(\begin{array}{ccc|c} 2 & 0 & -2 & 0 \\ 2 & 0 & -2 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right)$$

Thus, the RREF of the augmented matrix is:

$$\left(\begin{array}{ccc|c} 2 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

There are two free variables and so $\dim(E_1) = 2$. In other words, the geometric multiplicity of 1 for A is 2.

Theorem 5.7. (*G.M. $\leq$ A.M.*) Let A be an $n \times n$ matrix. Then the geometric multiplicity of λ is less than or equal to the algebraic multiplicity of λ .

Proof. Step 1: Fix a basis for E_λ

Let $v_1, v_2, \dots, v_r$ form a basis for E_λ , where r is the geometric multiplicity of λ for A .

Step 2: Complete to a basis for $\mathbb{R}^n$ We find vectors $w_1, \dots, w_{n-r}$ such that

$$v_1, \dots, v_r, w_1, \dots, w_{n-r}$$

form a basis for $\mathbb{R}^n$, denoted by β .

Step 3: Determine a block form of $[L_A]_\beta$ Let $Q = [\text{Id}]_\beta^{\text{st.basis}}$ be the matrix of change coordinates from β to the standard basis. It is shown in last chapter that

$$[L_A]_\beta = Q^{-1}AQ$$

Then, for $i = 1, \dots, r$, $Q^{-1}AQ\mathbf{e}_i = [L_A(v_i)]_\beta = [Av_i]_\beta = [\lambda v_i]_\beta = \lambda \mathbf{e}_i$. Thus,

$$Q^{-1}AQ = \begin{pmatrix} \lambda I_r & * \\ 0_{(n-r) \times r} & A' \end{pmatrix}$$

for some $(n-r) \times (n-r)$ -matrix A' .

Step 4: Compute the algebraic multiplicity of λ

Now,

$$\det(Q^{-1}AQ - tI_n) = (\lambda - t)^r \det(A' - tI_{n-1}).$$

Since the characteristics polynomial of A is equal to the characteristics polynomial of $Q^{-1}AQ$, $(t - \lambda)^r$ divides $\det(A - tI_n)$. The algebraic multiplicity of λ is at least r . $\square$

5.3. A necessary and sufficient condition for determining diagonalizability.

Theorem 5.8. (*Diagonalization criteria by geometric multiplicity and algebraic multiplicity*) Let A be an $n \times n$ matrix. Then A is diagonalizable if and only if the algebraic multiplicity of λ is equal to the geometric multiplicity of λ for all eigenvalues λ of A .

Proof. (Sketch) Only if direction. Suppose A is diagonalizable. By Theorem 2.7, we can find a basis for $\mathbb{R}^n$ whose vectors are eigenvectors of A . In other words, the sum of dimensions of each eigenspaces is equal to n , and so

- sum of all geometric multiplicities are equal to n

The sum of all algebraic multiplicities are also equal to n (Exercise 5.3). Then, the inequality in Theorem 5.7 forces that the geometric multiplicity for eigenvalue λ equal to the algebraic multiplicity of λ .

If direction. The sum of all algebraic multiplicities are equal to n (Exercise 5.3). Thus, the assumption implies that the **sum of all geometric multiplicities is also equal to n** . One finds a **basis for each eigenspaces** and the **union of those bases form a basis for $\mathbb{R}^n$** . By Theorem 2.7, A is diagonalizable. $\square$

Extra exercise:

- (a) Let $\lambda_1, \dots, \lambda_k$ be eigenvalues of a matrix A . For each i , let $v_{i,1}, \dots, v_{i,r_i}$ be a set of linearly independent vectors in E_{λ_i} . Show that the vectors

$$\cup_{i=1}^k \{v_{i,1}, \dots, v_{i,r_i}\}$$

form a set of linearly independent vectors. (Hint: Modify the proof of Theorem 4.1)

- (b) Work out the details that 'the sum of dimensions of each eigenspaces is equal to n , in the proof of Theorem 5.8.

Sol for (b): Let $v_1, \dots, v_n$ be the vectors in such basis in the proof. We let $S_\lambda = E_\lambda \cap \{v_1, \dots, v_n\}$ (λ is an eigenvalue of A) i.e. the set of eigenvectors in $\{v_1, \dots, v_n\}$ corresponding to the eigenvalue λ . If S_λ does not span E_λ , then we can find an extra eigenvector v' corresponding to eigenvalue λ such that $S_\lambda \cup \{v'\}$ is still linearly independent. But, now by (a), $\{v_1, \dots, v_n\} \cup \{v'\}$ is still linearly independent and so contradicts that the dimension of $\mathbb{R}^n$ is n . Hence, S_λ also spans E_λ , and so forms a basis for E_λ . Now $\dim E_\lambda = |S_\lambda|$ and so the sum of all eigenspaces gives n .

Example 5.9. (c.f. Example 3.8) Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Show that A is not diagonalizable by Theorem 5.8.

Sol: Finding algebraic multiplicities The characteristics polynomial of A is $\det(A - tI_2) = (1 - t)^2$ and so the **A.M. (algebraic multiplicity) of 1 is 2**.

Finding geometric multiplicities

$E_1 = \{v : (A - I_2)v = 0\}$ and so we have to solve:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v = 0.$$

Hence $E_1 = \left\{ \begin{pmatrix} t \\ 0 \end{pmatrix} : t \in \mathbb{R} \right\}$. Hence, the **G.M. (geometric multiplicity) of 1 is 1**.

Conclusion

Since **A.M. of 1 $\neq$ G.M. of 1**, Theorem 5.8 then implies A is not diagonalizable.

Exercise 5.10. Explain why Corollary 4.2 is a special situation of Theorem 5.8.

Exercise 5.11. Let A be an $n \times n$ matrix. If A has distinct eigenvalues λ_1 and λ_2 . Suppose $\dim E_{\lambda_1} = p$ and $\dim E_{\lambda_2} = n - p$. Show that A is diagonalizable.

Sol: Let m_1 and m_2 be the algebraic multiplicities of λ_1 and λ_2 respectively. We have the following three equations:

$$1 \leq \dim E_{\lambda_1} \leq m_1,$$

$$1 \leq \dim E_{\lambda_2} \leq m_2,$$

and $m_1 + m_2 = n$. Hence,

$$n = \dim E_{\lambda_1} + \dim E_{\lambda_2} \leq m_1 + m_2 \leq n$$

This forces the inequalities are equalities and so

$$\dim E_{\lambda_1} = m_1, \quad \dim E_{\lambda_2} = m_2.$$

By Theorem 5.8, A is diagonalizable.

6. APPLICATION ON FINDING POWERS OF A MATRIX

6.1. Taking high powers of a matrix. Let A be an $n \times n$ -matrix. One may ask if there is an efficient way to compute A^{100} ? For example, if one has a diagonal matrix:

$$D = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix},$$

then

$$D^{100} = \begin{pmatrix} 2^{100} & 0 \\ 0 & (-1)^{100} \end{pmatrix}.$$

In general, if one has a diagonalizable matrix A , one **first diagonalizes to get a diagonal matrix** $D = Q^{-1}AQ$ for some invertible matrix Q . **Then, one takes the power** $D^{100} = Q^{-1}A^{100}Q$. This then gives $A^{100} = QD^{100}Q^{-1}$.

Example 6.1. Let

$$A = \begin{pmatrix} 8 & -6 \\ 9 & -7 \end{pmatrix}.$$

Find A^{50} .

Sol: We first find the eigenvalues of A . The characteristics polynomial of A is

$$\det(A - xI_2) = \det \begin{pmatrix} 8-x & -6 \\ 9 & -7-x \end{pmatrix} = (8-x)(-7-x) + 54 = x^2 - x - 2.$$

Hence, the eigenvalues are 2 and -1 .

We now find an eigenvector for 2: Solve

$$(A - 2I_2)x = 0$$

and so

$$\begin{pmatrix} 6 & -6 \\ 9 & -9 \end{pmatrix} x = 0.$$

Then $(1, 1)^T$ is an eigenvector.

We now find an eigenvector for -1 : Solve

$$(A + I_2)x = 0$$

and so

$$\begin{pmatrix} 9 & -6 \\ 9 & -6 \end{pmatrix} x = 0.$$

Then $(2, 3)^T$ is an eigenvector.

Let $Q = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$. Then

$$Q^{-1}AQ = \text{diag}(2, -1)$$

and

$$A = Q\text{diag}(2, -1)Q^{-1}.$$

Then $A^{50} = Q\text{diag}(2^{50}, 1)Q^{-1}$. Thus

$$A^{50} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2^{50} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 \cdot 2^{50} - 2 & -2 \cdot 2^{50} + 2 \\ 3 \cdot 2^{50} - 3 & -2 \cdot 2^{50} + 3 \end{pmatrix}$$

6.2. Exponentiating a matrix. We now consider another problem. For a $n \times n$ matrix A , define

$$e^A = I_n + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \frac{1}{4!}A^4 + \dots$$

If A is a 1×1 matrix i.e. A is a number, then e^A is the usual exponential function.

Example 6.2. Let $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$. Compute e^A .

Sol: Then $e^A = \begin{pmatrix} e^3 & 0 \\ 0 & e^{-2} \end{pmatrix}$.

Example 6.3. Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$. Compute e^A .

Sol: We first need to diagonalize A .

- Eigenvalues: Solve $\det(A - tI_2) = 0$. Note

$$\det(A - tI_2) = \det \begin{pmatrix} 1-t & 2 \\ -1 & 4-t \end{pmatrix} = (1-t)(4-t) + 2 = t^2 - 5t + 6.$$

Solving $t^2 - 5t + 6 = 0$ gives $t = 2$ or 3 , and so 2 and 3 are the eigenvalues of A .

- Eigenvectors: For $\lambda = 2$, solve the system

$$(A - 2I_2)x = 0.$$

This gives

$$\begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix} x = 0$$

and so an eigenvector is $(2, 1)^T$.

For $\lambda = 3$, solve the system

$$(A - 3I_2)x = 0.$$

This gives

$$\begin{pmatrix} -2 & 2 \\ -1 & 1 \end{pmatrix} x = 0$$

and so an eigenvector is $(1, 1)^T$.

Let $Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. Then

$$Q^{-1}AQ = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

and so

$$A = Q \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} Q^{-1}.$$

Let $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. We then have that:

$$e^A = Qe^DQ^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} e^2 & 0 \\ 0 & e^3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2e^2 - e^3 & -2e^2 + 2e^3 \\ e^2 - e^3 & -e^2 + 2e^3 \end{pmatrix}.$$

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CHAPTER 6 – ORTHOGONALIZATION

Goals:

- What is an inner product on a vector space?
- Examples of an inner product on a vector space
- Orthonormal basis and Gram-Schmidt process
- Orthogonal complements and projections

1. INNER PRODUCT SPACES

1.1. Inner product.

Definition 1.1. Let V be a vector space. An **inner product** $\langle \cdot, \cdot \rangle$ on V is a function that assigns a scalar in $\mathbb{R}$ to every pair $(x, y) \in V \times V$ satisfying the following properties:

- (a) (**Addition for inner product**) For any $x, y, z \in V$, $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$.
- (b) (**Scalar multiple for inner product**) For any $x, y \in V$ and $c \in \mathbb{R}$, $\langle cx, y \rangle = c\langle x, y \rangle$.
- (c) (**Symmetry for inner product**) For any $x, y \in V$, $\langle x, y \rangle = \langle y, x \rangle$.
- (d) (**Positiveness for inner product**) For any $x \in V$, $\langle x, x \rangle > 0$ if $x \neq 0$.

A vector space equipped with an inner product is called an **inner product space**, and we usually denote the inner product by $\langle \cdot, \cdot \rangle$.

We first deduce some immediate consequences:

Proposition 1.2. *Let V be an inner product space. Then*

- (a) *For any $x, y, z \in V$, $\langle z, x + y \rangle = \langle z, x \rangle + \langle z, y \rangle$.*
- (b) *For any $x, y \in V$ and $c \in \mathbb{R}$, $\langle x, cy \rangle = c\langle x, y \rangle$.*
- (c) *For any $x \in V$, $\langle 0, x \rangle = \langle x, 0 \rangle = 0$.*
- (d) *For any $x \in V$, $\langle x, x \rangle = 0$ if and only if $x = 0$.*

Proof. The key ideas for (a) and (b) are to transfer the problem to the one in Definition 1.1. For (a),

$$\langle z, x + y \rangle = \langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle = \langle z, x \rangle + \langle z, y \rangle$$

For (b), $\langle x, cy \rangle = \langle cy, x \rangle = c\langle y, x \rangle = c\langle x, y \rangle$. For (c), we use that $0 = 0y$ for any $y \in V$. Hence, we have:

$$\langle 0, x \rangle = \langle 0y, x \rangle = 0\langle y, x \rangle = 0.$$

For (d), if $x \neq 0$, then $\langle x, x \rangle > 0$ by Definition 1.1(d). If $x = 0$, it follows from part (c). $\square$

1.2. Examples.

Example 1.3. Let $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$ and $(y_1, \dots, y_n)^T \in \mathbb{R}^n$, define

$$\langle x, y \rangle = x_1 y_1 + \dots + x_n y_n = x^T y.$$

Check $\langle, \rangle$ defines an inner product on $\mathbb{R}^n$. This is called a **standard inner product** on $\mathbb{R}^n$. When $n = 2$ and $n = 3$, it coincides with the dot product between two vectors.

Sol: Let $x = (x_1, \dots, x_n)^T$, $y = (y_1, \dots, y_n)^T$ and $z = (z_1, \dots, z_n)^T$. Let $c \in \mathbb{R}$.

Check (a): $\langle x + y, z \rangle = (x_1 + y_1)z_1 + \dots + (x_n + y_n)z_n = (x_1 z_1 + \dots + x_n z_n) + (y_1 z_1 + \dots + y_n z_n) = \langle x, z \rangle + \langle y, z \rangle$.

Checking (b) and (c) are straightforward.

Check (d): $\langle x, x \rangle = x_1^2 + \dots + x_n^2 > 0$ for $x \neq 0$.

Example 1.4. Let $\langle, \rangle$ be an inner product on a vector space V . Let r be a *positive* real number. Define a new pairing:

$$\langle, \rangle' : V \times V \rightarrow \mathbb{R}$$

given by $\langle x, y \rangle' = r \langle x, y \rangle$. Then $\langle, \rangle'$ also defines an inner product.

Example 1.5. Fix scalars $a_1, \dots, a_n > 0$. Define $\langle, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ given by: for $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$ and $(y_1, \dots, y_n)^T \in \mathbb{R}^n$,

$$\langle x, y \rangle = a_1 x_1 y_1 + \dots + a_n x_n y_n.$$

This defines another inner product on $\mathbb{R}^n$.

We also have some non-examples:

Example 1.6. Define $\langle, \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ given by: for $(x_1, x_2)^T, (y_1, y_2)^T \in \mathbb{R}^2$,

$$\langle (x_1, x_2)^T, (y_1, y_2)^T \rangle = x_1 y_1 - x_2 y_2.$$

Determine if $\langle, \cdot \rangle$ defines an inner product on $\mathbb{R}^2$.

Sol: It is not an inner product on $\mathbb{R}^2$. Note that $\langle (0, 1)^T, (0, 1)^T \rangle = -1 < 0$ which violates Definition 1.1(d).

Exercise 1.7. Let $T : V \rightarrow W$ be an injective linear transformation. Let $\langle, \cdot \rangle'$ be an inner product on W . For $x, y \in V$, define

$$\langle x, y \rangle = \langle T(x), T(y) \rangle'.$$

Show that $\langle \cdot, \cdot \rangle$ is an inner product on V .

Sol: For $x, y, z \in V$,

$$\langle x+y, z \rangle = \langle T(x+y), T(z) \rangle' = \langle T(x)+T(y), T(z) \rangle' = \langle T(x), T(z) \rangle' + \langle T(y), T(z) \rangle' = \langle x, z \rangle + \langle y, z \rangle.$$

This checks (a).

For $x, y \in V$ and $c \in \mathbb{R}$, define

$$\langle cx, y \rangle = \langle T(cx), T(y) \rangle' = \langle cT(x), T(y) \rangle' = c\langle T(x), T(y) \rangle' = c\langle x, y \rangle.$$

This checks (b).

For $x, y \in V$,

$$\langle x, y \rangle = \langle T(x), T(y) \rangle' = \langle T(y), T(x) \rangle' = \langle y, x \rangle.$$

This checks (c).

For $x \in V$, $\langle x, x \rangle \geq 0$ by using the condition (d) of $\langle \cdot, \cdot \rangle'$. Hence, to show $\langle x, x \rangle > 0$ for $x \neq 0$, we only have to show $\langle x, x \rangle \neq 0$ if $x \neq 0$. If $\langle x, x \rangle = 0$, then $\langle T(x), T(x) \rangle' = 0$ and so $T(x) = 0$. Thus $x = 0$ since T is injective. This checks $\langle x, x \rangle \neq 0$ if $x \neq 0$ and so checks (d).

Exercise 1.8. What happens for Exercise 1.7 if one drops the condition that T is injective?

Sol: In such case, $\langle \cdot, \cdot \rangle'$ is not necessarily an inner product. This is because if T is not injective, one may find a non-zero x such that $T(x) = 0$. In such case, $\langle x, x \rangle = \langle T(x), T(x) \rangle' = 0$. This violates Definition 1.1(d).

Exercise 1.9. (Transpose arising from standard inner product) Let A be an $n \times m$ -matrix. Let $\langle \cdot, \cdot \rangle_{\mathbb{R}^n}$ and $\langle \cdot, \cdot \rangle_{\mathbb{R}^m}$ be standard inner products on $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively. Then, for $x \in \mathbb{R}^m$ and for $y \in \mathbb{R}^n$,

$$\langle Ax, y \rangle_{\mathbb{R}^n} = \langle x, A^T y \rangle_{\mathbb{R}^m}.$$

Sol: Recall that $\langle x, y \rangle = x^T y$ for any $x, y \in \mathbb{R}^m$. Then

$$\langle Ax, y \rangle = (Ax)^T y = x^T A^T y = x^T (A^T y) = \langle x, A^T y \rangle.$$

1.3. Geometric interpretation and motivations for inner products. The concept of inner products can be used to generalize the length:

Definition 1.10. Let V be an inner product space. For $x \in V$, define $\|x\| = \sqrt{\langle x, x \rangle}$, which is called the **length** of x .

Example 1.11. Let $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$. Then

$$\|x\| = \sqrt{x_1^2 + \dots + x_n^2}.$$

Exercise 1.12. Let V be an inner product space. Let $T : V \rightarrow V$ be a linear transformation. Suppose $\|T(x)\| = \|x\|$ for all $x \in V$. Show that T is injective.

Sol. Suppose $T(x) = 0$. Then $\|x\| = \|T(x)\| = \|0\| = 0$. This implies that $x = 0$ by Proposition 1.2(d). Hence, T is injective.

We have some more properties of the inner product

Theorem 1.13. Let V be an inner product space. Then, for all $x, y \in V$ and $c \in \mathbb{R}$,

- (a) (*Length under scaling*) $\|cx\| = |c|\|x\|$;
- (b) (*Zero as only element with length zero*) $\|x\| = 0$ if and only if $x = 0$;
- (c) (*Cauchy-Schwartz inequality*) $\langle x, y \rangle \leq \|x\|\|y\|$ for any $x, y \in V$;
- (d) (*Triangle inequality*) $\|x + y\| \leq \|x\| + \|y\|$.

Proof. For (a),

$$\begin{aligned}\|cx\| &= \sqrt{\langle cx, cx \rangle} \\ &= \sqrt{c^2 \langle x, x \rangle} \\ &= |c| \sqrt{\langle x, x \rangle} \\ &= |c| \|x\|\end{aligned}$$

For (b), if $x = 0$, by Proposition 1.2, $\langle x, x \rangle = 0$ and so $\|x\| = \sqrt{\langle x, x \rangle} = 0$.

If $\|x\| = 0$, $\sqrt{\langle x, x \rangle} = 0$ and $\langle x, x \rangle = 0$ and so by Proposition 1.2, $x = 0$.

For (c), we consider

$$0 \leq \langle x - cy, x - cy \rangle = \langle x, x \rangle - c\langle x, y \rangle - c\langle y, x \rangle + c^2\langle y, y \rangle.$$

Let $c = \frac{\langle x, y \rangle}{\|y\|^2}$. Then, the inequality can be simplified as:

$$0 \leq \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2}.$$

Then $\langle x, y \rangle^2 \leq \|x\|^2 \cdot \|y\|^2$. Thus, $\langle x, y \rangle \leq \|x\| \cdot \|y\|$.

For (d),

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2.\end{aligned}$$

□

Exercise 1.14. Interpret geometrically the Cauchy-Schwartz inequality for the standard inner product for $\mathbb{R}^2$.

2. ORTHONORMAL SUBSETS

2.1. Orthogonal and orthonormal.

Definition 2.1. Let V be an inner product space.

- For $x, y \in V$, we say that x and y are **orthogonal** if $\langle x, y \rangle = 0$.
- A subset S of V is **orthogonal** if for any two *distinct* $x, y \in S$, $\langle x, y \rangle = 0$.
- A vector $v \in V$ is said to be a **unit** if $\|v\| = 1$.
- A subset S of V is said to be **orthonormal** if S is orthogonal and any vector in S is a unit vector.

Example 2.2. The vectors $\mathbf{e}_i$ are units in $\mathbb{R}^n$. The set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is orthonormal in $\mathbb{R}^n$.

Example 2.3. The set $\{2\mathbf{e}_1, 3\mathbf{e}_2, \dots, (n+1)\mathbf{e}_n\}$ is orthogonal, but not orthonormal in $\mathbb{R}^n$.

Example 2.4. The set

$$S = \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

is an orthonormal set in $\mathbb{R}^2$.

Exercise 2.5. (**Normalizing vectors**) Let v be a non-zero vector in V . Then $\frac{1}{\|v\|}v$ is always a unit vector. We shall say that the vector $\frac{1}{\|v\|}v$ is obtained by normalizing v .

Sol: $\|\frac{1}{\|v\|}v\| = \langle \frac{1}{\|v\|}v, \frac{1}{\|v\|}v \rangle = \frac{1}{\|v\|^2} \langle v, v \rangle = \frac{1}{\langle v, v \rangle} \langle v, v \rangle = 1$.

Example 2.6. We normalize the vector $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ to obtain the unit vector $\frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Exercise 2.7. Let v be a unit vector in V . Show that $-v$ is also a unit vector.

Sol: $\|-v\| = |-1|\|v\| = 1$.

Exercise 2.8. Describe how to obtain an orthonormal set from an orthogonal set in an inner product space V .

Sol: Let $S = \{v_1, \dots, v_r\}$ be an orthogonal set in V . Then, by normalizing vectors, we obtain an orthonormal set:

$$\left\{ \frac{v_1}{\|v_1\|}, \dots, \frac{v_r}{\|v_r\|} \right\}.$$

2.2. Linear independence of an orthogonal set. One main property of an orthogonal set is the following:

Theorem 2.9. (*Orthogonal $\Rightarrow$ Linearly independent*) Let S be an orthogonal set of non-zero vectors in an inner product space V . Then S is linearly independent.

Proof. Let $S = \{v_1, \dots, v_r\}$. Let $a_1, \dots, a_r \in \mathbb{R}$ such that

$$a_1 v_1 + \dots + a_r v_r = 0.$$

Then

$$\langle a_1 v_1 + \dots + a_r v_r, v_i \rangle = \langle 0, v_i \rangle = 0.$$

Hence, $a_1 \langle v_1, v_i \rangle + \dots + a_r \langle v_r, v_i \rangle = 0$. Since $\langle v_j, v_i \rangle = 0$ for $i \neq j$. It leaves:

$$a_i \langle v_i, v_i \rangle = 0.$$

Since $v_i \neq 0$, we then have $\langle v_i, v_i \rangle \neq 0$ and so a_i must be 0. This shows all $a_i = 0$. $\square$

Example 2.10. Let

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}.$$

Show, using the inner product that v_1, v_2, v_3 are linearly independent.

Sol: Then, one checks:

$$\langle v_1, v_2 \rangle = (1)(2) + (2)(-1) = 0$$

$$\langle v_1, v_3 \rangle = (1)(1) + (2)(2) + (5)(-1) = 0$$

$$\langle v_2, v_3 \rangle = (2)(1) + (-1)(2) = 0.$$

Hence, $\{v_1, v_2, v_3\}$ is an orthogonal set and so is a set of linearly independent vectors.

Remark 2.11. The converse of Theorem 2.9 is not true. For example. $(1, 1)^T$ and $(0, 1)^T$ are linearly independent in $\mathbb{R}^2$, but they do not form an orthogonal set.

Exercise 2.12. Let V be an inner product space. Suppose $\dim(V) = n$. Let S be an orthogonal set of non-zero vectors in V . Show that the number of vectors in S is less than or equal to n .

Sol: By Theorem 2.9, the vectors are linearly independent. It then follows from that the number of linearly independent vectors is less than or equal to $\dim(V)$.

Exercise 2.13. Let V be an inner product space with dimension equal to n . Explain why an orthogonal set of n vectors forms a basis for V .

2.3. Linear combination from an orthogonal set.

Theorem 2.14. (*Linear combination of orthogonal set*) Let V be an inner product space. Let $S = \{v_1, \dots, v_r\}$ be an orthogonal subset of V consisting of non-zero vectors. For any $y \in \text{span}(S)$,

$$y = \frac{\langle v_1, y \rangle}{\|v_1\|^2} v_1 + \dots + \frac{\langle v_r, y \rangle}{\|v_r\|^2} v_r.$$

Proof of Theorem 2.14. Since $y \in \text{span}(S)$,

$$y = a_1 v_1 + \dots + a_r v_r$$

for some $a_1, \dots, a_r \in \mathbb{R}$. Then,

$$\langle v_i, y \rangle = \langle v_i, a_1 v_1 + \dots + a_r v_r \rangle = a_1 \langle v_i, v_1 \rangle + \dots + a_r \langle v_i, v_r \rangle.$$

Then $\langle v_i, y \rangle = a_i \langle v_i, v_i \rangle$ and so

$$a_i = \frac{\langle v_i, y \rangle}{\|v_i\|^2}$$

for all i . □

Example 2.15. Show that $\{(3, 2)^T, (2, -3)^T\}$ is an orthogonal set. Express $(\sqrt{2}, -1)^T$ in terms of a linear combination of the two orthogonal vectors.

Sol: Let $v_1 = (3, 2)^T$ and $v_2 = (-2, 3)^T$. Then

$$\langle v_1, v_2 \rangle = (3)(2) + (-2)(3) = 0.$$

Let $v = (\sqrt{2}, -1)^T$. Now we compute

$$\langle v, (3, 2)^T \rangle = 3\sqrt{2} - 2$$

and $\langle v, (2, -3)^T \rangle = 2\sqrt{2} + 3$. Moreover,

$$\|(3, 2)^T\| = \sqrt{3^2 + 2^2} = \sqrt{13}, \quad \|(2, -3)^T\| = \sqrt{2^2 + (-3)^2} = \sqrt{13}.$$

Then

$$v = \frac{3\sqrt{2} - 2}{13} (3, 2)^T + \frac{2\sqrt{2} + 3}{13} (-2, 3)^T.$$

3. GRAM-SCHMIDT PROCESS

We now show that an orthonormal basis always exists and there is a systematic way to find that.

3.1. Orthonormal basis and Gram-Schmidt process. We have the following result to find an orthonormal basis for an inner product space.

Theorem 3.1. (*Construct orthogonal basis*) Let V be an inner product space and let $S = \{w_1, \dots, w_n\}$ be a basis for V . Let $v_1 = w_1$ and, for $2 \leq j \leq n$, let

$$v_i = w_i - \sum_{j=1}^{i-1} \frac{\langle w_i, v_j \rangle}{\|v_j\|^2} v_j.$$

Then, $\{v_1, \dots, v_k\}$ is a set of orthogonal vectors which forms a basis for V .

A consequence is the following existence result for an orthonormal basis:

Corollary 3.2. (*Existence of orthonormal basis*) Let V be an inner product space. There always exists an orthonormal basis for V .

Proof. There always exists a basis for V . Then, by Theorem 3.1, one has a basis which is orthogonal to V . Now one normalizes each vector in the basis (see Exercise 2.8), which gives an orthonormal basis. $\square$

Proof of Theorem 3.1. Let $S' = \{v_1, \dots, v_n\}$.

- Spanning condition: We can rewrite

$$(3.1) \quad w_i = v_i + \sum_{j=1}^{i-1} \frac{\langle w_i, v_j \rangle}{\|v_j\|^2} v_j$$

i.e. written as a linear combinations of $v_1, \dots, v_n$. Thus, $w_i \in \text{span}(S')$. Hence, $V = \text{span}(\{w_1, \dots, w_n\}) \subset \text{span}(S')$. The inclusion $\text{span}(S') \subset V$ is clear. Thus $\text{span}(S') = V$.

- Linear independence: We consider the equation:

$$b_1 v_1 + \dots + b_n v_n = 0$$

If we rewrite each v_i into w_i in the above equation (3.1), we have:

$$b'_1 w_1 + \dots + b'_{n-1} w_{n-1} + b_n w_n = 0$$

This implies that $b_n = 0$ by using the linear independence of $w_1, \dots, w_n$. This reduces the equation to the form:

$$b_1 v_1 + \dots + b_{n-1} v_{n-1} = 0.$$

Inductively, we can show $b_{n-1} = \dots = b_1 = 0$.

- Orthogonal: For $l < k$,

$$\begin{aligned}
 \langle v_k, v_l \rangle &= \langle w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j, v_l \rangle \\
 &= \langle w_k, v_l \rangle - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} \langle v_j, v_l \rangle \\
 &= \langle w_k, v_l \rangle - \frac{\langle w_k, v_l \rangle}{\|v_l\|^2} \langle v_l, v_l \rangle \\
 &= 0
 \end{aligned}$$

where the second equality follows the property of the inner product, the third equality follows from an induction that $\langle v_j, v_l \rangle = 0$ for $j \neq l$, and the last equality follows from $\langle v_l, v_l \rangle = \|v_l\|^2$.

Definition 3.3. The orthonormal basis obtained in Theorem 3.1 and Corollary 3.2 is called an **Gram-Schmidt orthonormal basis** and the process to get that is called the **Gram-Schmidt orthogonalization process**.

3.2. Examples.

Example 3.4. Find an orthonormal basis for the vector subspace

$$\text{span}\left(\left\{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}\right\}\right).$$

Sol: Let $w_1 = (1, 0, -1)^T$ and $w_2 = (2, 1, -1)^T$. We carry out the Gram-Schmidt orthogonalization process: Set $v_1 = w_1$. Then we find:

$$\begin{aligned}
 v_2 &= w_2 - \frac{\langle v_1, w_2 \rangle}{\|v_1\|^2} v_1 \\
 &= (2, 1, -1)^T - \frac{3}{2} (1, 0, -1)^T \\
 &= \begin{pmatrix} \frac{1}{2} \\ 1 \\ \frac{1}{2} \end{pmatrix}
 \end{aligned}$$

One then normalizes to get the orthonormal set:

$$\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

4. ORTHOGONAL COMPLEMENTS

4.1. Orthogonal complements.

Definition 4.1. Let S be a non-empty subset of an inner product space V . Define

$$S^\perp = \{x \in V : \langle x, y \rangle = 0 \text{ for all } y \in S\}.$$

Literally, $S^\perp$ is the subset of V containing all the vectors orthogonal to all vectors in S . We call $S^\perp$ the **orthogonal complement** of S .

Example 4.2. Let $S = \{0\}$. Then $S^\perp = V$ by Proposition 1.2(c).

Example 4.3. Let $S = V$. Then $S^\perp = \{0\}$ since if $v \neq 0$, then $\langle v, v \rangle \neq 0$.

Example 4.4. Let $S = \{\mathbf{e}_2\} \subset \mathbb{R}^3$. Then $S^\perp = \{(x, 0, z)^T : x, z \in \mathbb{R}\}$ is the xz -plane.

Exercise 4.5. Let S be a subset for V . Show that $S^\perp = \text{span}(S)^\perp$.

Theorem 4.6. Let S be a non-empty subset of V . Then $S^\perp$ is a vector subspace of V .

Proof. • Check addition: Let $x, y \in S^\perp$. For any $z \in S$,

$$\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle = 0.$$

• Check scalar multiplication: Let $x \in S^\perp$ and let $c \in \mathbb{R}$. For any $z \in S$,

$$\langle cx, z \rangle = c\langle x, z \rangle = 0.$$

By the definition, we have that $S^\perp$ is a vector subspace of V . □

Exercise 4.7. (Zero intersection between vector space and its orthogonal complement) Let W be a vector subspace of V . Show that $W \cap W^\perp = \{0\}$.

4.2. Orthogonal projections. One importance of the orthogonal complement is the following result:

Theorem 4.8. (Orthogonal complement as vector subspace) Let V be an inner product space. Let W be a vector subspace of V . Then, for any $v \in V$, there exists unique $x \in W$ and $y \in W^\perp$ such that $v = x + y$. The vector x is called the **orthogonal projection** of v onto W , and can be found by the formula:

$$(4.2) \quad x = \langle v, v_1 \rangle v_1 + \dots + \langle v, v_r \rangle v_r,$$

where $v_1, \dots, v_r$ is an orthonormal basis for W .

Before going into the proof of Theorem 4.8, we first show some examples:

Example 4.9. Let $W = \text{span}(\{\mathbf{e}_1, \mathbf{e}_3\})$. Then $W^\perp = \text{span}(\{\mathbf{e}_2\})$. For $v = (2, 3, 4)^T$, we have:

$$v = (2, 0, 4)^T + (0, 3, 0)^T.$$

Draw a picture to visualize that.

Example 4.10. Let $W = \text{span}(\{(1, -2)^T\})$. Then $W^\perp = \text{span}(\{(2, 1)^T\})$. For example,

$$\langle (3, 4)^T, (1, -2)^T \rangle = -5$$

and so the orthogonal projection of $(3, 4)^T$, according to formula (4.2), is

$$\frac{-5}{5}(1, -2)^T = (-1, 2)^T.$$

One can write $(3, 4)^T = (-1, 2)^T + (4, 2)^T$ with $(4, 2)^T \in W^\perp$.

Proof of Theorem 4.8.

Existence:

Let $\{v_1, \dots, v_r\}$ be an orthonormal basis for W . Define

$$x = \langle v, v_1 \rangle v_1 + \dots + \langle v, v_r \rangle v_r.$$

By definitions, $x \in W$. Let $y = v - x$. For any $1 \leq i \leq r$,

$$\begin{aligned} \langle y, v_i \rangle &= \langle v - x, v_i \rangle \\ &= \langle v, v_i \rangle - \langle x, v_i \rangle \\ &= \langle v, v_i \rangle - \langle v, v_1 \rangle \langle v_1, v_i \rangle - \dots - \langle v, v_r \rangle \langle v_r, v_i \rangle \\ &= \langle v, v_i \rangle - \langle v, v_i \rangle \langle v_i, v_i \rangle \\ &= 0 \end{aligned}$$

This shows that $y \in W^\perp$.

Uniqueness Let $v = x + y = x' + y'$ for $x, x' \in W$ and $y, y' \in W^\perp$. Then,

$$x - x' = y' - y$$

Hence, $x - x', y' - y \in W \cap W^\perp$. By Exercise 4.7, $x - x' = 0$ and $y' - y = 0$, and so $x = x'$ and $y = y'$. $\square$

5. ORTHOGONAL MATRICES AND SYMMETRIC MATRICES (OPTIONAL)

5.1. Orthogonal matrices.

Definition 5.1. An $n \times n$ matrix Q is said to be **orthogonal** if $Q^T Q = I_n$.

Exercise 5.2. (1) Let A be an orthogonal matrix. Let $v_1, \dots, v_n$ be the columns of A . Show that $\{v_1, \dots, v_n\}$ forms an orthonormal basis.

(2) Let $\{v_1, \dots, v_n\}$ be an orthonormal basis for $\mathbb{R}^n$. Then the matrix $(v_1 \ v_2 \ \dots \ v_n)$ is an orthogonal matrix.

Exercise 5.3. Let P, Q be an $n \times n$ orthogonal matrix. Show that PQ is also an orthogonal matrix.

One importance of an orthogonal matrix is to provide an isometry for the standard inner product for $\mathbb{R}^n$:

Exercise 5.4. Let Q be an $n \times n$ orthogonal matrix. Prove that, for any $v, w \in \mathbb{R}^n$,

$$\langle Qv, Qw \rangle = \langle v, w \rangle.$$

5.2. Orthogonal diagonalization.

Theorem 5.5. Let A be a symmetric $n \times n$ matrix. Then there exists an orthogonal matrix Q such that $Q^T A Q = D$ for some diagonal matrix D .

Proof. (Sketch)

Step 1. Let v_1 be an eigenvector of A with corresponding eigenvalue λ_1 . We normalize v_1 so that $\|v_1\| = 1$ (Why can we do that?). By Gram-Schmidt orthogonalization, we form an orthonormal basis $\{v_1, \dots, v_n\}$ for (the standard inner product for) $\mathbb{R}^n$. Let

$$Q_0 = (v_1 \ v_2 \ \dots \ v_n).$$

Then one checks that

$$Q_0^T A Q_0 = \begin{pmatrix} \lambda_1 & B \\ 0 & C \end{pmatrix},$$

where B is a $1 \times (n-1)$ matrix and C is an $(n-1) \times (n-1)$ matrix.

Step 2. Then $(Q_0^T A Q_0)^T = Q_0^T A^T Q_0 = Q_0^T A Q_0$ as A is symmetry. This implies that B is 0, and C is symmetric.

Step 3. By induction, there exists an orthogonal $(n-1) \times (n-1)$ matrix P such that $P^T C P$ is diagonal. Let

$$Q_1 = \begin{pmatrix} 1 & 0_{1 \times (n-1)} \\ 0_{(n-1) \times 1} & P \end{pmatrix}.$$

Then Q_1 is orthogonal and $Q_1 Q_0$ is also orthogonal.

Step 4. Let $Q = Q_1 Q_0$. Now, one checks that $Q^T A Q$ is a diagonal matrix.

□

Indeed, we also have a converse of Theorem 5.5:

Exercise 5.6. Let A be an $n \times n$ matrix. If there exists an orthogonal matrix Q such that $Q^T A Q = D$ for some diagonal matrix D , then A is symmetric.

5.3. Positive definite matrix.

Definition 5.7. An $n \times n$ matrix A is said to be *positive definite* if A is symmetric and $v^T A v > 0$ for all $v \in \mathbb{R}^n$.

Exercise 5.8. Let A be an $n \times n$ matrix. Then $\langle, \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$\langle v, w \rangle = v^T A w$$

defines an inner form if and only if A is a positive definite matrix.

Exercise 5.9. Let A be a positive definite $n \times n$ matrix. Then there exists an invertible matrix Q such that $A = Q^T Q$.